Advanced Stochastic Calculus with Applications

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1 Preliminaries about Stochastic Processes

Intermission 1

The purpose of this section is to introduce the concepts of **adapted** and **progressively measurable** processes, discuss their relationship, and present key results related to stopping times.

These are foundational for our later study of **stochastic control** and **optimal stopping**.

Adaptedness is crucial because it ensurs our models respect causality. For instance, in an American option exercise strategy, a decision at time t cannot be contingent on a future price $X_s, s > t$, as that information is unknown.

Progressive Measurability is a more technical requirement that arises in continuous time. It is a prerequisite for stochastic integration, ensure the process is "well-behaved" over time intervals.

You may view it as "path-adaptedness": while adaptedness ensures we know the process' value at a specific point, progressive measurability ensures that the entire history of the process up to t is mathematically tractable as a single object.

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be the probability space on which we're going to work.

Let $\mathbb{X} = (X_t)_{t\geq 0}$ be a stochastic process which is *measurable*, *adapted*, and *progressively measurable*.

Definition 2

A stochastic process is $(t, \omega) \to X_t(\omega)$ is **$\mathcal{B}(\mathbb{R}_+) \times \mathcal{F}$-measurable** if

$$\{(t, \omega) : X_t(\omega) \in A\} \in \mathcal{B}(\mathbb{R}_+) \times \mathcal{F}, \forall A \in \mathcal{B}(\mathbb{R}^d)$$

Definition 3

A stochastic process $\mathbb{X}$ is **adapted to $(\mathcal{F}_t)_{t\geq 0}$** if $X_t \in m\mathcal{F}_t, \forall t \geq 0$.

Note that $m\mathcal{F}_t$ denotes the set of measurable functions with respect to the σ-algebra $\mathcal{F}_t$ in the filtration $(\mathcal{F}_t)_{t\geq 0}$.

Definition 4

A stochastic process $\mathbb{X}$ is **progressively measurable** if $\forall t \geq 0, \{(s, \omega) \in [0, t] \times \Omega : X_s(\omega) \in A\} \in \mathcal{B}([0, t]) \times \mathcal{F}_t, \forall A \in \mathcal{B}(\mathbb{R}^d)$.

Homework 5

Show that if $\mathbb{X}$ is progressively measurable, then it is also adapted.

Hint: use the properties of measurable functions for the product σ-algebra.

Proposition 6

If $\mathbb{X}$ is adapted and right-continuous, then it is progressively measurable.

Proof

Fix $t > 0$, and define:

$$X_s^n = \begin{cases} X_{\frac{k+1}{2^n}t}, & s \in \left[\frac{k}{2^n}t, \frac{k+1}{2^n}t\right) \\ X_t, & s = t \end{cases}$$

> **Homework 7**
> Show that $X_s(\omega) = \lim_{n\to\infty} X_s^n(\omega), \forall (s,\omega) \in [0,t] \times \Omega$.

If we show that $(s,\omega) \to X_s^n(\omega)$ is $\mathcal{B}([0,t]) \times \mathcal{F}_t$-measurable then $(s,\omega) \to X_s(\omega)$ is also $\mathcal{B}([0,t]) \times \mathcal{F}_t$-measurable by pointwise limit.

Since $t > 0$ is arbitrary, we conclude that $\mathbb{X}$ is progressively measurable.

Fix $A \in \mathcal{B}(\mathbb{R}^d) : \{(s,\omega) \in [0,t] \times \Omega : X_s^n(\omega) \in A\}$

$$= \bigcup_{k=0}^{2^n-1} \left(\left[\frac{k}{2^n}t, \frac{k+1}{2^n}t\right) \times \left\{\omega \in \Omega : X_{\frac{k+1}{2^n}t}(\omega) \in A\right\}\right) \bigcup (\{t\} \times \{\omega \in \Omega : X_t(\omega) \in A\})$$

Then:

$$\{(s,\omega) \in [0,t] \times \Omega : X_s^n(\omega) \in A\} \in \mathcal{B}([0,t]) \times \mathcal{F}_t$$

■

Remark 8. *If $\mathbb{X}$ is measurable and $\tau : \Omega \to [0,+\infty)$ is $\mathcal{F}$-measurable, then $\omega \to X_{\tau(\omega)}$ is $\mathcal{F}$-measurable as composition of measurable maps.*

Adopt the notation $X_t(\omega) = X(t,\omega)$ so that $X^{-1}(A) = \{(t,\omega) : X(t,\omega) \in A\}$. Then, $\{\omega \in \Omega : X(\tau(\omega),\omega) \in A\} = \left\{\omega \in \Omega : (\tau(\omega),\omega) \in \underbrace{X^{-1}(A)}_{\in \mathcal{B}(\mathbb{R}_+)\times\mathcal{F}}\right\}$.

Notice that $\omega \to (\tau(\omega),\omega)$ is a mapping $\Omega \to \mathbb{R}_+ \times \Omega$. For any $E \in \mathcal{B}(\mathbb{R}_+)$ and $F \in \mathcal{F}$ we have $\{\omega \in \Omega : (\tau(\omega),\omega) \in E \times F\} = \{\omega \in \Omega : \tau(\omega) \in E\} \cap F \in \mathcal{F}$.

This extends to $\{\omega : (\tau(\omega),\omega) \in B\} \in \mathcal{F}, \forall B \in \mathcal{B}(\mathbb{R}_+)\times\mathcal{F}$ because $\{E \times F, E \in \mathcal{B}(\mathbb{R}_+), F \in \mathcal{F}\}$ generates $\mathcal{B}(\mathbb{R}_+)\times\mathcal{F}$.

In the definition below, let Π be either $[0,+\infty)$ or $\mathbb{Z}^+$.

> **Definition 9**
> Given a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\in\Pi}, \mathbb{P})$ and a stopping time τ, we define
>
> $$\mathcal{F}_\tau := \{A \in \mathcal{F}_\infty : A \cap \{\tau \le t\} \in \mathcal{F}_t, \forall t \in \Pi\}$$

> **Homework 10**
> Show that $\mathcal{F}_\tau$ is a σ-algebra on Ω.

Proposition 11

Let σ, τ be stopping times. Then:

1. $\tau \in m\mathcal{F}_\tau$
2. $\tau \wedge \sigma$ and $\tau \vee \sigma$ are stopping times
3. $\sigma(\omega) \leq \tau(\omega), \forall \omega \in \Omega \implies \mathcal{F}_\sigma \subseteq \mathcal{F}_\tau$
4. $\mathcal{F}_{\sigma\wedge\tau} = \mathcal{F}_\sigma \cap \mathcal{F}_\tau$

Proof

(ii) was proven in PFF and it's left as HW.

(i) It suffices to show $\{\omega : \tau(\omega) \leq s\} \in \mathcal{F}_\tau, \forall s \in \Pi$. By def $\{\tau \leq s\} \in \mathcal{F}_s \subseteq \mathcal{F}_\infty$. Fix $t \in \Pi$. Then

$$\{\tau \leq s\} \cap \{\tau \leq t\} = \{\tau \leq s \wedge t\} \in \mathcal{F}_{s\wedge t} \subseteq \mathcal{F}_t$$

(iii) Take $F \in \mathcal{F}_\sigma$. Then $F \in \mathcal{F}_\infty$ and $F \cap \{\sigma \leq t\} \in \mathcal{F}_t, \forall t \in \Pi$.

$$F \cap \{\tau \leq t\} = F \cap \underbrace{\{\sigma \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{\tau \leq t\}}_{\in \mathcal{F}_t} \quad \text{because } \sigma \leq \tau$$

Then $F \in \mathcal{F}_\tau$.

(iv) Take $F \in \mathcal{F}_{\sigma\wedge\tau}$. Then $F \in \mathcal{F}_\infty$ and $F \cap \{\sigma \wedge \tau \leq t\} \in \mathcal{F}_t, \forall t \in \Pi$. Now, $\forall t \in \Pi$,

$$F \cap \{\tau \leq t\} = F \cap \underbrace{\{\sigma \wedge \tau \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{\tau \leq t\}}_{\in \mathcal{F}_t}$$

$\implies F \in \mathcal{F}_\tau$ and by the same argument $F \in \mathcal{F}_\sigma$. Thus, $\mathcal{F}_{\sigma\wedge\tau} \subseteq \mathcal{F}_\tau \cap \mathcal{F}_\sigma$.

Take $F \in \mathcal{F}_\tau \cap \mathcal{F}_\sigma$. Then $F \in \mathcal{F}_\infty$, $F \cap \{\tau \leq t\} \in \mathcal{F}_t$ and $F \cap \{\sigma \leq t\} \in \mathcal{F}_t, \forall t \in \Pi$. Now, for $t \in \Pi$,

$$F \cap \{\tau \wedge \sigma \leq t\} = (F \cap \underbrace{\{\tau \leq t\}}_{\in \mathcal{F}_t} \cap \{\tau \wedge \sigma \leq t\}) \cup (F \cap \{\tau \wedge \sigma \leq t\} \cap \{\tau > t\})$$

$$= \underbrace{F \cap \{\sigma \leq t\}}_{\in \mathcal{F}_t} \cap \underbrace{\{\tau > t\}}_{\in \mathcal{F}_t}$$

$\implies F \in \mathcal{F}_{\tau\wedge\sigma} \implies \mathcal{F}_\tau \cap \mathcal{F}_\sigma \subseteq \mathcal{F}_{\tau\wedge\sigma}$. ∎

Proposition 12

If τ is a stopping time and $\mathbb{X}$ is progr. meas. then $X_\tau \in m\mathcal{F}_\tau$.

Proof

We must prove that $\forall A \in \mathcal{B}(\mathbb{R}^d), \{\omega \in \Omega : X_\tau(\omega) \in A\} \in \mathcal{F}_\tau$. Clearly $\{\omega \in \Omega : X_\tau(\omega) \in A\} \in \mathcal{F}_\infty$. Now, for any $t \geq 0$,

$$\{X_\tau \in A\} \cap \{\tau \leq t\} = \{X_{\tau\wedge t} \in A\} \cap \{\tau \leq t\}$$

The random variable $\tau \wedge t : \Omega \to [0, t]$ is $\mathcal{F}_t$-measurable because, $\forall s \geq 0$,

$$\{\tau \wedge t \leq s\} = \left\{\begin{matrix} \Omega & \text{if } s \geq t \\ \{\tau \leq s\} & \text{if } s \in [0,t] \end{matrix}\right\} \in \mathcal{F}_t$$

Now, let $X^t(s, \omega) := X_{s\wedge t}(\omega)$. Then $\forall A \in \mathcal{B}(\mathbb{R}^d)$, $\{\omega \in \Omega : X_{\tau\wedge t}(\omega) \in A\} = \{\omega \in \Omega : X^t(\tau(\omega), \omega) \in A\}$

$$= \left\{\omega \in \Omega : (\tau(\omega), \omega) \in \underbrace{(X^t)^{-1}(A)}_{\in \mathcal{B}([0,t])\times\mathcal{F}_t \text{ by progr. meas.}}\right\} \in \mathcal{F}_t \text{ because:}$$

For sets of the form $[0, s] \times F, s \in [0, t], F \in \mathcal{F}_t$, we have

$$\{\omega \in \Omega : (\tau(\omega), \omega) \in [0, s] \times F\} = \{\tau \leq s\} \cap F \in \mathcal{F}_t$$

Since $\mathcal{B}([0, t]) \times \mathcal{F}_t = \sigma([0, s] \times F, s \in [0, t], F \in \mathcal{F}_t)$, the result extends to the whole σ-algebra. ■

Proposition 13

Let $\mathbb{X}$ be continuous and $A \subseteq \mathbb{R}^d$ be an open set. Then $\tau_A := \inf\{t \geq 0 : X_t \notin A\}$ is a stopping time. If A is closed and $\bigcap_{\epsilon>0} \mathcal{F}_{t+\epsilon} = \mathcal{F}_{t^+} = \mathcal{F}_t$ then τ_A is a stopping time.

Proof

If A is open:

- by continuity of $s \mapsto X_s \implies$ continuity of $s \mapsto \text{dist}(X_s, A^c)$ and A^c is closed.

Homework 14

Fill in the details of this proof.

$$\{\tau_A > t\} = \{\inf_{0\leq s\leq t} \text{dist}(X_s, A^c) > 0\} = \bigcup_{n\in\mathbb{N}} \left\{\inf_{0\leq s\leq t} \text{dist}(X_s, A^c) > \frac{1}{n}\right\}$$
$$\overset{\text{indistinguishability}}{=} \bigcup_{n\in\mathbb{N}} \bigcap_{s\in[0,t]\cap\mathbb{Q}} \left\{\text{dist}(X_s, A^c) > \frac{1}{n}\right\} \in \mathcal{F}_t$$

If A is closed, let's approximate A with open sets $A_n \downarrow A$ ($A_n \supseteq A_{n+1} \supseteq A$). Then $\{\tau_A < t\} = \bigcup_{n\in\mathbb{N}}\{\tau_{A_n} \leq t\} \in \mathcal{F}_t$.

$\omega \in \{\tau_A < t\} \implies X_{\tau_A}(\omega) \in A^c \implies X_{\tau_A}(\omega) \in A_n^c$ for large n (because A^c is open and $\mathbb{X}$ continuous) $\implies \tau_{A_n}(\omega) \leq t$ for large n.

$\omega \in \bigcup_{n\in\mathbb{N}}\{\tau_{A_n} \leq t\} \implies \exists n \in \mathbb{N}$ s.t. $\omega \in \{\tau_{A_n} \leq t\} \subseteq \{\tau_A < t\}$.

Now: $\{\tau_A \leq t\} = \bigcap_{n\in\mathbb{N}} \underbrace{\{\tau_A < t + \frac{1}{n}\}}_{\in \mathcal{F}_{t+\frac{1}{n}}} \in \mathcal{F}_{t^+} = \mathcal{F}_t$. ■

2 Basic facts about Markov processes

Intermission 15

In this section we formalize the **Markov Property** in continuous time. In simple words, this property states that the future of a process depends only on its present state, not its past history.

We start by introducing the **Markov Transition Function** which governs how a process moves between states. Via the **Chapman-Kolmogorov equation** we ensure that moving from s to t is the same as stopping at $u \in (s, t)$ and restarting. We then extend this notion via the **Strong Markov Property** which says that the "memoryless" property holds even if the time we stop to check the process is random, as long as it's a stopping time.

This set-up is then used to introduce some very useful results and concepts for our modelling endeavors:

1. The solutions of SDEs are Markov processes. Hence, we can use the current state of a system to forecast future distributions without needing the entire historical data-set.

2. By transitioning from the Markov process to the **Infinitesimal Generator** and the **Semigroup** we can turn a stochastic path problem into a deterministic PDE, which we can solve via finite-difference methods. This saves us computational time, and resources, compared to using Monte Carlo methods.

3. **Càdlàg** functions are essential for modelling "shocks" or jumps, such as sudden market crashes.

Taken from Chapter 6 of Baldi 2017.

Definition 16 (Markov transition function)
Let $(E, \mathcal{E})$ be a measurable space. A MTF is a mapping $p : [0, \infty) \times [0, \infty) \times E \times \mathcal{E} \to [0, 1]$ s.t.

1. $x \mapsto p(s, t, x, A)$ is Borel measurable.

2. $A \mapsto p(s, t, x, A)$ is a probability measure on $(E, \mathcal{E})$.

3. **Chapman-Kolmogorov**: for $s < u < t$,

$$p(s, t, x, A) = \int_E p(s, u, x, dy)p(u, t, y, A).$$

Definition 17 (Markov process)
Let μ be a probability measure on $\mathcal{E}$.

A Markov process with MTF p is a stochastic process $\mathbb{X} := (\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, (X_t)_{t\geq 0}, \mathbb{P})$ s.t.

1. $\mathbb{P}(X_0 \in dy) = \mu(dy)$.
2. For every bounded measurable $f : E \to \mathbb{R}$ it holds:
$$\mathbb{E}[f(X_t)|\mathcal{F}_s] = \int_E f(y)p(s,t,X_s,dy)$$
(notice that for $f(y) = 1_A(y), A \in \mathcal{E} \implies \mathbb{P}(X_t \in A|\mathcal{F}_s) = p(s,t,X_s,A)$).

Remark 18. *In general $(\mathcal{F}_t)_{t\geq 0}$ is a filtration that contains $(\sigma(X_s, s \leq t))_{t\geq 0}$ but w.l.o.g. we could just consider $\mathcal{F}_t = \sigma(X_s, s \leq t)$.*

Homework 19
Show that $x \mapsto \int_E f(y)p(s,t,x,dy)$ is measurable.

Hint: take simple functions $f(\cdot)$. Approximate with the **Monotone Convergence Theorem**.

Theorem 20
Given a probability measure μ on $(E, \mathcal{E})$ and a MTF, there exists a Markov process with initial distribution μ and transition function p.

We want to be able to start our process from any time $s \in [0, \infty)$ and any position $x \in E$. Then we consider a measurable space $(\Omega, \mathcal{F})$ endowed with a family of filtrations $(\mathcal{F}^s_t)_{s\leq t, t\geq 0}$ and a family of probability measures $(\mathbb{P}_{s,x})_{s\geq 0, x\in E}$. Then a realisation of the Markov process reads:

$$\mathbb{X} := (\Omega, \mathcal{F}, (\mathcal{F}^s_t)_{t\geq s, s\geq 0}, (X)_{t\geq s}, (\mathbb{P}_{s,x})_{x\in E, s\geq 0})$$

and it holds:

1. $\mathbb{P}_{s,x}(X_s = x) = 1$.
2. $\mathbb{E}_{s,x}[f(X_t)|\mathcal{F}^s_u] = \int_E f(y)p(u,t,X_u,dy), \quad s \leq u \leq t$.

(notice that $\mathbb{E}_{s,x}[f(X_t)|\mathcal{F}^s_s] = \int_E f(y)p(s,t,x,dy)$). *Note: here we lose track of $s \geq 0$ then $\int_E f(y)p(u,t,X_u,dy) = \mathbb{E}_{u,z}[f(X_t)|\mathcal{F}^s_u]$ for $s < u < t$.*

Markov process on canonical space

It's convenient to define MP on a "space of paths". The regularity of paths depends on the specific applications. Generally one considers either:

(a) Continuous paths: $\Omega = C(\mathbb{R}_+; \mathbb{R}^d) \hookrightarrow \{f : \mathbb{R}_+ \to \mathbb{R}^d \text{ s.t. } f \text{ is continuous}\}$.

(b) Right-continuous with left limits (RCLL/càdlàg): $\Omega = D(\mathbb{R}_+; \mathbb{R}^d) \hookrightarrow \{f : \mathbb{R}_+ \to \mathbb{R}^d \text{ s.t. } f(t) = f(t+) \forall t \geq 0 \text{ and } f(t-) \text{ exists } \forall t > 0\}$.

- On either such space, the canonical process is

$$\begin{aligned} X : \mathbb{R}_+ \times \Omega &\to \mathbb{R}^d \\ (t, \omega) &\mapsto \omega(t) \end{aligned}$$

  here $\omega \in \Omega$ is a function $t \mapsto \omega(t)$ and $X_t(\omega) = \omega(t)$.

- We introduce a shift-operator $(\theta_h)_{h \geq 0}$ that acts on paths $\theta_h : \Omega \to \Omega$ as $\theta_h \omega(\cdot) = \omega(h + \cdot)$.

  Notice that $\theta_h X_t(\omega) = X_t(\theta_h \omega) = \omega(h + t)$.

- The filtration is the canonical filtration $\mathcal{F}_t^s = \sigma(X_u, s \leq u \leq t)$.

- MP on canonical space:

$$\mathbb{X} = (\Omega, \mathcal{F}, (\mathcal{F}_t^s)_{t \geq s, s \geq 0}, (X_t), (\theta_t), (\mathbb{P}_{s,x})_{s \geq 0, x \in \mathbb{R}^d})$$

Remark 21 (Homogeneous MP)**.** *A MTF is homogeneous if* $p(s, t, x, A) = p(0, t - s, x, A)$*. In this case* $\mathbb{P}_{s,x}(X_t \in A) = \mathbb{P}_{0,x}(X_{t-s} \in A)$ *and therefore it suffices to consider a MTF* $p : \mathbb{R}_+ \times E \times \mathcal{E} \to [0, 1]$*,* $p(t, x, A)$ *and a family* $(\mathbb{P}_x)_{x \in E}$*.*

Something with a flavour of the freezing lemma holds:

Proposition 22

Let Y be bounded and $\sigma(X_s, s \geq t)$-measurable. Then $\mathbb{E}_{s,x}[Y | \mathcal{F}_t^s] = \mathbb{E}_{t,X_t}[Y] := (\mathbb{E}_{t,z}[Y])|_{z = X_t(\omega)}$.

Proof. The σ-algebra $\sigma(X_s, s \geq t)$ is generated by the π-system

$$\left(\{X_{t_1}^{-1}(A_1) \cap X_{t_2}^{-1}(A_2) \cap \cdots \cap X_{t_n}^{-1}(A_n)\}, \forall (t_i, A_i) \in [t, \infty) \times \mathcal{B}(\mathbb{R}^d), i = 1 \ldots n, n \in \mathbb{N}\right)$$

Let $\mathcal{H}$ be the class of bounded $\sigma(X_s, s \geq t)$-measurable random variables for which it holds

$$\mathbb{E}_{s,x}[Y | \mathcal{F}_t^s] = \mathbb{E}_{t,X_t}[Y]$$

Clearly, if $(Y_n)_{n \in \mathbb{N}} \subset \mathcal{H}$, $0 \leq Y_n \leq Y_{n+1}$ and $Y_n \uparrow Y$, then $Y \in \mathcal{H}$ by MON.

Homework 23

Verify the statement above.

It's obvious that $\mathcal{H}$ is a vector space and that $Y(\omega) = 1 \in \mathcal{H}$. It remains to show that $Y(\omega) = 1_A(\omega) \in \mathcal{H}$ for any A from the π-system above.

Take $Y(\omega) = 1_{\{X_{t_1}\in A_1\}}1_{\{X_{t_2}\in A_2\}}\dots 1_{\{X_{t_n}\in A_n\}}$.

$$
\begin{aligned}
\mathbb{E}_{s,x}[Y|\mathcal{F}_t^s] &= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-1} 1_{\{X_{t_i}\in A_i\}}\cdot 1_{\{X_{t_n}\in A_n\}}\right] \\
&= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-1} 1_{\{X_{t_i}\in A_i\}}\mathbb{E}_{s,x}[1_{\{X_{t_n}\in A_n\}}|\mathcal{F}_{t_{n-1}}^s]\right] \\
&= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-1} 1_{\{X_{t_i}\in A_i\}} \underbrace{\mathbb{P}_{s,x}(X_{t_n}\in A_n|\mathcal{F}_{t_{n-1}}^s)}_{=p(t_{n-1},t_n,X_{t_{n-1}},A_n)=:\tilde{f}_{n-1}(X_{t_{n-1}})}\right] \\
&= \mathbb{E}_{s,x}\left[\prod_{i=1}^{n-2} 1_{\{X_{t_i}\in A_i\}} \underbrace{\mathbb{E}_{s,x}[1_{\{X_{t_{n-1}}\in A_{n-1}\}}\tilde{f}_{n-1}(X_{t_{n-1}})|\mathcal{F}_{t_{n-2}}^s]}_{=\int_{A_{n-1}}\tilde{f}_{n-1}(y)p(t_{n-2},t_{n-1},X_{t_{n-2}},dy)=:\tilde{f}_{n-2}(X_{t_{n-2}})}\right] \\
&= \dots\text{iterating}\dots = \mathbb{E}_{s,x}[1_{\{X_{t_1}\in A_1\}}\tilde{f}_1(X_{t_1})|\mathcal{F}_t^s] \\
&= \int_{A_1}\tilde{f}_1(y)p(t,t_1,X_t,dy) = \mathbb{E}_{t,X_t}[\tilde{f}_1(X_{t_1})1_{\{X_{t_1}\in A_1\}}] \\
&= \mathbb{E}_{t,X_t}\left[1_{\{X_{t_1}\in A_1\}}\int_E 1_{A_2}(y)\tilde{f}_2(y)p(t_1,t_2,X_{t_1},dy)\right] \\
&= \mathbb{E}_{t,X_t}\left[1_{\{X_{t_1}\in A_1\}}\mathbb{E}_{t,X_t}[1_{\{X_{t_2}\in A_2\}}\tilde{f}_2(X_{t_2})|\mathcal{F}_{t_1}^t]\right] \\
&= \mathbb{E}_{t,X_t}\left[1_{\{X_{t_1}\in A_1\}}1_{\{X_{t_2}\in A_2\}}\underbrace{\mathbb{E}_{t_2,X_{t_2}}[1_{\{X_{t_3}\in A_3\}}\tilde{f}_3(X_{t_3})|\mathcal{F}_{t_2}^t]}_{=\mathbb{E}_{t,X_t}[1_{\{X_{t_3}\in A_3\}}\tilde{f}_3(X_{t_3})|\mathcal{F}_{t_2}^t]}\right] \\
&= \dots\text{iterating}\dots = \mathbb{E}_{t,X_t}\left[\prod_{i=1}^{n} 1_{\{X_{t_i}\in A_i\}}\right] = \mathbb{E}_{t,X_t}[Y].
\end{aligned}
$$

Then we can apply the MCT. □

Definition 24 (Strong Markov property)

A MP is Strong MP if for any $(\mathcal{F}_t^s)_{t\ge s}$-stopping time τ $(\tau<\infty)$

$$\mathbb{P}_{s,x}(X_{\tau+h}\in A|\mathcal{F}_\tau^s) = p(\tau,\tau+h,X_\tau,A) \quad \mathbb{P}_{s,x}\text{-a.s.}$$

Remark 25. *On the canonical space, if $Y\in m\mathcal{F}_\infty^s$ then $Y = f((X_t)_{t\ge s})$. So, given a stopping time τ we have $Y\circ\theta_\tau = f((X_t\circ\theta_\tau)_{t\ge s}) = f((X_{t+\tau})_{t\ge 0})\in\mathcal{F}_\infty^\tau$.*

Proposition 26

If $Y\in m\mathcal{F}_\infty^s$ bounded or positive and $\mathbb{X}$ is Strong Markov, then

$$\mathbb{E}_{s,x}[Y\circ\theta_\tau|\mathcal{F}_\tau^s] = \mathbb{E}_{\tau,X_\tau}[Y] := (\mathbb{E}_{t,z}[Y])|_{t=\tau(\omega),z=X_\tau(\omega)}.$$

Proof. (HW) using the same tricks as before. □

Given a time-homogeneous MTF, the expectation is a linear operator that satisfies the semi-group property: $\forall$ bounded measurable $f : \mathbb{R}^d \to \mathbb{R}$

$$(P_t f)(x) := \int_E f(y) p(t, x, dy) = \mathbb{E}_x[f(X_t)]$$

satisfies $[P_s(P_t f)](x) = \int_E p(s, x, dy) \int_E f(z) p(t, y, dz)$

$$= \int_E \int_E f(z) p(t, y, dz) p(s, x, dy) = \mathbb{E}_x[\mathbb{E}_{X_s}[f(X_t)]]$$

$$\overset{\text{Chapman-Kolmogorov eq.}}{=} \int_E f(z) p(t+s, x, dz) = \mathbb{E}_x[f(X_{t+s})]$$

$$= (P_{t+s} f)(x)$$

Semigroup property: $P_t P_s = P_{t+s}$.

Infinitesimal generator:

$$(\mathcal{L} f)(x) := \lim_{t \downarrow 0} \frac{(P_t f)(x) - f(x)}{t} \quad \text{if the limit exists}$$

- Domain of $\mathcal{L}$, $D(\mathcal{L}) \ldots$
- Unbounded operator ...
- Limit in suitable norms ...

We will only be concerned with SDEs and their infinitesimal generators.

An important class of Strong MPs is the solutions of SDEs.

Stability estimates for SDEs

Here we consider $(X_t)_{t \geq 0} \in \mathbb{R}^d$ solution of

$$X_t = x + \int_0^t \mu(X_s) ds + \int_0^t \sigma(X_s) dW_s$$

for Lipschitz coefficients $\mu : \mathbb{R}^d \to \mathbb{R}^d$, $\sigma : \mathbb{R}^d \to \mathbb{R}^{d \times d'}$ and $(W_t)_{t \geq 0}$ a d'-dimensional B.m.

Standard estimates via Gronwall's inequality (HW)

(i) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_t\|^p\right] \leq C(p, T)(1 + \|x\|^p)$

(ii) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_t - x\|^p\right] \leq C(p, T)(1 + \|x\|^p) \cdot T^{p/2}$

(iii) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_t^{x_1} - X_t^{x_2}\|^p\right] \leq C(p, T)\|x_1 - x_2\|^p$

(iv) $\mathbb{E}\left[\sup_{0 \leq t \leq T} \|X_{t+h} - X_t\|^p\right] \leq C(p, T)(1 + \|x\|^p) h^{p/2} \quad h \geq 0$

where $p \in [2, \infty)$.

From (i) we deduce an important fact: $t \mapsto \int_0^t \sigma(X_s) dW_s$ is a martingale with

$$\mathbb{E}\left[\sup_{0 \leq t \leq T} \left|\int_0^t \sigma(X_s) dW_s\right|^p\right] < \infty$$

Moreover:

$$\mathbb{E}[\|X_t^x - X_s^y\|^p] \leq C(p,T)(1+\|x\|^p+\|y\|^p)(|t-s|^{p/2}+|x-y|^p)$$

So, for $\frac{p}{2}>1$ we can invoke Kolmogorov's continuity theorem and deduce $(x,t)\mapsto X_t^x(\omega)$ is continuous $\forall\omega\in\Omega$. (there exists a continuous modification ...) [*for time-in-homogeneous case* $(x,s,t)\mapsto X_t^{s,x}(\omega)$ *is continuous*]

Lemma 27

Let $\mathcal{H}_t := \sigma(W_u - W_t, u\geq t)$. Assume $(Y_t)_{t\geq 0}\in L^2(0,T)$ is such that $Y_u\in m\mathcal{H}_t, \forall u\geq t$. Then, $\forall v\geq t$, $\int_t^v Y_u dW_u \in \mathcal{H}_t$ and therefore $\int_t^v Y_u dW_u \perp\!\!\!\perp \sigma(W_s, s\leq t)$.

Recall that Y *is* $\sigma(W_s, s\leq t)$ *adapted.*

Proof. When we constructed the stoch. integral we proved that $(Y_t)_{t\geq 0}$ can be approximated by simple processes $(Y_u^n)_{t\geq 0}, n\in\mathbb{N}$ s.t. $Y_u^n\in m\mathcal{H}_t, \forall u\geq t, \forall n\in\mathbb{N}$ and $\lim_{n\to\infty}\mathbb{E}[\int_0^T |Y_u^n - Y_u|^2 du] = 0$. Then, it suffices to prove the claim for simple processes $Y_u(\omega) := \sum_{i=1}^m X_{t_i}(\omega)1_{[t_i,t_{i+1})}(u)$ with $X_{t_i}\in m\mathcal{H}_t\cap m\mathcal{F}_{t_i}, \forall i = 1\dots m$ and $t\leq t_1 < t_2 < \cdots < t_m \leq v$. For such process $\int_t^v Y_u dW_u = \sum_{i=1}^m \underbrace{X_{t_i}}_{\in m\mathcal{H}_t}\underbrace{(W_{t_{i+1}} - W_{t_i})}_{\in m\mathcal{H}_t} \in m\mathcal{H}_t$. ☐

Remark 28. *The next result holds in greater generality but we only prove it for Lipschitz coefficients.*

Proposition 29

Let $\mu:\mathbb{R}^d\to\mathbb{R}^d, \sigma:\mathbb{R}^d\to\mathbb{R}^{d\times d'}$ be Lipschitz and $(W_t)_{t\geq 0}\in\mathbb{R}^{d'}$ be a B.m. Let $(X_t)_{t\geq s}$ be the unique strong solution of $X_t = x + \int_s^t \mu(X_u)du + \int_s^t \sigma(X_u)dW_u$. Denote it $(X_t^{s,x})_{t\geq s}$. Then $X_t^{s,x}\in m\mathcal{H}_s, \forall t\geq s \implies X_t^{s,x}\perp\!\!\!\perp \sigma(W_u, u\leq s)$.

Proof. Recall that $(X_t^{s,x})_{t\geq s}$ is constructed by Picard iteration $X_t^0 := x, \forall t\geq 0$ and $X_t^{n+1} = x + \int_s^t \mu(X_u^n)du + \int_s^t \sigma(X_u^n)dW_u, \forall n\geq 0$. and $\lim_{n\to\infty}\mathbb{E}[\sup_{s\leq t\leq T}|X_t^n - X_t|^2] = 0, \forall T>0$. $\quad(*)$

It is clear that $X_t^0\in m\mathcal{H}_s, \forall t\geq s$. Then $\int_s^t \mu(X_u^0)du\in m\mathcal{H}_s$ and $\int_s^t\sigma(X_u^0)dW_u\in m\mathcal{H}_s$. Now, by induction, assume $(X_t^n)_{t\geq s}\subset m\mathcal{H}_s$. Then

$$X_t^{n+1} = x + \underbrace{\int_s^t \mu(X_u^n)du}_{\in m\mathcal{H}_s} + \underbrace{\int_s^t \sigma(X_u^n)dW_u}_{\in m\mathcal{H}_s} \in m\mathcal{H}_s, \quad \forall t\geq s.$$

From $(*)$ we know $\exists (X_t^{n_j})_{t\geq s}$ s.t. $\mathbb{P}(\lim_{j\to\infty}\sup_{s\leq t\leq T}|X_t^{n_j} - X_t| = 0) = 1$ so $(X_t^{s,x})_{t\geq s}\in m\mathcal{H}_s$ because we assume $\mathcal{H}_s$ completed with null sets. ☐

Let's go back to considering the unique strong solution of the SDE. Since we are working with strong solutions, then $\sigma(X_r, r\leq t)\subseteq\sigma(W_r, r\leq t)$ and it suffices for our aims to consider $\mathcal{F}_t := \sigma(W_r, r\leq t)$ (it would be equivalent to consider $\sigma(X_r, r\leq t)$).

$$
\begin{aligned}
X_t^{0,x} &= x + \int_0^t \mu(X_u)du + \int_0^t \sigma(X_u)dW_u \\
&= \underbrace{x + \int_0^s \mu(X_u)du + \int_0^s \sigma(X_u)dW_u}_{=X_s^{0,x}} \\
&\quad + \int_s^t \mu(X_u)du + \int_s^t \sigma(X_u)dW_u \\
&= \underbrace{X_s^{0,x}}_{\in m\mathcal{F}_s} + \underbrace{\int_s^t \mu(X_u)du + \int_s^t \sigma(X_u)dW_u}_{\in m\mathcal{H}_s \perp\!\!\!\perp \mathcal{F}_s} =: X_s^{0,x} + Z_{s,t} = X_t^{s,X_s^{0,x}}
\end{aligned}
$$

Proposition 30

For any bounded measurable function $f : \mathbb{R}^d \to \mathbb{R}$

$$
\mathbb{E}[f(X_t)|\mathcal{F}_s] = \mathbb{E}[f(X_t)|\sigma(X_s)], \quad \forall t \geq s.
$$

Proof. From the discussion preceding the proposition we have $\mathbb{E}[f(X_t)|\mathcal{F}_s] = \mathbb{E}[f(\underbrace{X_s}_{\in m\mathcal{F}_s} + \underbrace{Z_{s,t}}_{\perp\!\!\!\perp \mathcal{F}_s})|\mathcal{F}_s]$ then we can use the freezing lemma to claim: $\mathbb{E}[f(X_s + Z_{s,t})|\mathcal{F}_s] = \psi(X_s)$ where $\psi(x) := \mathbb{E}[f(x + Z_{s,t})]$. For any $A \in \sigma(X_s)$ we have

$$
\mathbb{E}[1_A f(X_t)] = \mathbb{E}[1_A \underbrace{\mathbb{E}[f(X_t)|\mathcal{F}_s]}_{\text{by tower property}}] = \mathbb{E}[\psi(X_s)1_A]
$$

and therefore, by definition of conditional expectation,

$$
\psi(X_s) = \mathbb{E}[f(X_t)|\sigma(X_s)].
$$

Then $\mathbb{E}[f(X_t)|\sigma(X_s)] = \mathbb{E}[f(X_t)|\mathcal{F}_s]$ as claimed. □

Corollary 31

The solution $(X_t^{0,x})_{t\geq 0}$ of the SDE above is a Markov process.

Proof. Given $\Gamma \in \mathbb{R}^d$, set $f(x) = 1_\Gamma(x)$. Then

$$
\begin{aligned}
\mathbb{P}(X_t \in \Gamma|\mathcal{F}_s) = \mathbb{P}(X_t \in \Gamma|\sigma(X_s)) = \psi(X_s) &= \mathbb{P}(y + Z_{s,t} \in \Gamma)|_{y=X_s} \\
&= \mathbb{P}(X_t^{s,y} \in \Gamma)|_{y=X_s} \doteq p(s,t,y,\Gamma)|_{y=X_s} = p(s,t,X_s,\Gamma).
\end{aligned}
$$

□

Remark 32. *For any bounded and continuous $f : \mathbb{R}^d \to \mathbb{R}$ we have, by continuity of $(t,s,x) \mapsto X_s^{t,x}(\omega)$ and DOM:*

$$\begin{aligned}\int_{\mathbb{R}^d} f(y)p(t,s,x,dy) &= \mathbb{E}[f(X_s^{t,x})] \\ &= \mathbb{E}[\lim_{n\to\infty} f(X_{s_n}^{t_n,x_n})] = \lim_{n\to\infty} \mathbb{E}[f(X_{s_n}^{t_n,x_n})] \\ &= \lim_{n\to\infty} \int_{\mathbb{R}^d} f(y)p(t_n,s_n,x_n,dy) \quad \text{for any } (t_n,s_n,x_n) \to (t,s,x)\end{aligned}$$

That is, **the Feller property** *holds:*

$$(t,x,s) \mapsto \int_{\mathbb{R}^d} f(y)p(t,s,x,dy) \text{ is continuous } \forall f \in C_b(\mathbb{R}^d).$$

It thus follows that $(X_t)_{t\geq 0}$ is actually Strong-Markov!

Proposition 33

The solution $(X_t)_{t\geq 0}$ of the SDE is strong Markov.

Proof. Let τ be a (bounded) stopping time and f a bounded measurable function. We must prove that for any $\Gamma \in \mathcal{F}_\tau$ it holds (notice that $t \mapsto X_t(\omega)$ is progr. meas $\implies X_\tau(\omega) \in m\mathcal{F}_\tau$)

$$\mathbb{E}[1_\Gamma 1_{\{X_{\tau+h}\in A\}}] = \mathbb{E}[1_\Gamma p(\tau, \tau+h, X_{\tau+h}, A)].$$

We can approximate τ with a decreasing sequence of stopping times $\tau_n \downarrow \tau$ as $n \to \infty$ with τ_n taking finitely many values.

$$\begin{aligned}\mathbb{E}[1_\Gamma 1_{\{X_{\tau_n+h}\in A\}}] &= \sum_{i=1}^N \mathbb{E}[1_\Gamma 1_{\{X_{t_i+h}\in A\}} 1_{\{\tau_n = t_i\}}] \\ &= \sum_{i=1}^N \mathbb{E}[\mathbb{E}[\underbrace{1_\Gamma \cap \{\tau_n = t_i\}}_{\in \mathcal{F}_{t_i} \text{ because } \{\tau_n = t_i\}\subseteq\{\tau\leq t_i\} \text{ and } \Gamma\in\mathcal{F}_\tau} 1_{\{X_{t_i+h}\in A\}} | \mathcal{F}_{t_i}]] \\ &= \sum_{i=1}^N \mathbb{E}[1_{\Gamma\cap\{\tau_n=t_i\}} \mathbb{P}(X_{t_i+h}\in A|\mathcal{F}_{t_i})] \\ &= \sum_{i=1}^N \mathbb{E}[1_{\Gamma\cap\{\tau_n=t_i\}} p(t_i, t_i+h, X_{t_i+h}, A)] \\ &= \sum_{i=1}^N \mathbb{E}[1_{\Gamma\cap\{\tau_n=t_i\}} p(\tau_n, \tau_n+h, X_{\tau_n+h}, A)] \\ &= \mathbb{E}[1_\Gamma p(\tau_n, \tau_n+h, X_{\tau_n+h}, A)]\end{aligned}$$

Now, for any continuous, bounded function,

$$\mathbb{E}[1_\Gamma f(X_{\tau_n+h})] = \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau_n, \tau_n+h, X_{\tau_n+h}, dy)]$$

Using DOM and Feller property we have

$$\begin{aligned}
\mathbb{E}[1_\Gamma f(X_{\tau+h})] &= \lim_{n\to\infty} \mathbb{E}[1_\Gamma f(X_{\tau_n+h})] \\
&= \lim_{n\to\infty} \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau_n, \tau_n + h, X_{\tau_n+h}, dy)] \\
&= \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau, \tau + h, X_{\tau+h}, dy)]
\end{aligned}$$

$$\implies \mathbb{E}[1_\Gamma f(X_{\tau+h})] = \mathbb{E}[1_\Gamma \int_{\mathbb{R}^d} f(\tau, \tau+h, X_{\tau+h}, dy)] \quad (*)$$

Any bounded measurable function can be approximated by bounded continuous functions: $f \in B_b(\mathbb{R}^d) \implies \exists (f_n)_n \subseteq C_b(\mathbb{R}^d)$ s.t. $\|f_n\|_\infty < 1 + \|f\|_\infty$ and $f_n(x) \to f(x)$ for all $x \in \mathbb{R}^d$. Then $(*)$ extends to bounded measurable functions by approximation. $\square$

3 Optimal Stopping in Discrete Time

Taken from Chapter 1 of Peskir and Shiryaev 2006.

3.1 Introduction to Optimal Stopping

- Discrete time setting: $(\Omega, \mathcal{F}, (\mathcal{F}_n)_n, \mathbb{P})$
- Gain process: $(G_n)_n$ adapted to $(\mathcal{F}_n)_n$
- Finite-time horizon: N
- Class of stopping times: $\mathcal{T} := \{\tau : \tau : \Omega \to \mathbb{Z}^+ \text{ is stopping time}\}$
- For $n \leq N$, $\mathcal{T}_{n,N} := \{\tau \in \mathcal{T} : n \leq \tau \leq N\}$

The problem of interest: $V_* = \sup_\tau \mathbb{E}[G_\tau]$. We want to:

1. Calculate V_*
2. Determine optimal τ_*

We distinguish two cases:

- **Infinite-time horizon:** $\tau \in \mathcal{T}$. In this case $N = +\infty$ and we set $G_N(\omega) \equiv 0$. We need to define $G_\tau 1_{\{\tau=+\infty\}}$. The usual choice is:
$$G_\tau 1_{\{\tau=+\infty\}} := \limsup_{n\to\infty} G_n 1_{\{\tau=+\infty\}}$$
- **Finite-time horizon:** $\tau \in \mathcal{T}_{0,N}$.

Assumption:
$$\mathbb{E}\left[\sup_{0\leq n\leq N} |G_n|\right] < \infty$$

Dynamic view of the problem: $V_n^N := \sup_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[G_\tau]$.

3.2 Method of backward induction

It works for $N < \infty$.

Set $S_N^N(\omega) := G_N(\omega)$ and
$$S_n^N(\omega) := \max\left\{G_n(\omega), \mathbb{E}[S_{n+1}^N|\mathcal{F}_n](\omega)\right\}$$
for $n \in \{0, 1, \ldots, N-1\}$. Let $\tau_n^N := \inf\{n \leq k \leq N : S_k^N = G_k\}$ and notice that $S_N^N = G_N \implies \{n \leq k \leq N : S_k^N = G_k\} \neq \emptyset$.

Theorem 34 (OS1)
$\forall 0 \le n \le N$ we have, $\mathbb{P}$-a.s.

$$\left.\begin{aligned} S_n^N &\ge \mathbb{E}[G_\tau|\mathcal{F}_n] \quad \forall \tau \in \mathcal{T}_{n,N} \\ S_n^N &= \mathbb{E}[G_{\tau_n^N}|\mathcal{F}_n] \end{aligned}\right\} \quad (A) \tag{1}$$

Moreover:

(i) τ_n^N is optimal for V_n^N, i.e., $V_n^N = \mathbb{E}[G_{\tau_n^N}]$.

(ii) If $\tau_* \in \mathcal{T}_{n,N}$ is another optimal stopping time for V_n^N, then $\mathbb{P}(\tau_* \ge \tau_n^N) = 1$ (i.e., τ_n^N is **minimal**).

(iii) $(S_k^N)_{n \le k \le N}$ is the smallest supermartingale which dominates $(G_k)_{n \le k \le N}$.

(iv) $(S_{k \wedge \tau_n^N}^N)_{n \le k \le N}$ is a martingale.

Proof. **Proof of (A):** For $n = N$ we have $S_N^N = G_N$ and $\tau_N^N = N$ so the claim holds. Now assume the claim holds for generic n. Take $\tau \in \mathcal{T}_{n-1,N}$ and set $\bar\tau := \tau \vee n$. Then $\bar\tau \in \mathcal{T}_{n,N}$ and

$$\begin{aligned} \mathbb{E}[G_\tau|\mathcal{F}_{n-1}] &= \mathbb{E}[G_\tau 1_{\{\tau=n-1\}}|\mathcal{F}_{n-1}] + \mathbb{E}[G_\tau 1_{\{\tau>n-1\}}|\mathcal{F}_{n-1}] \\ &= G_{n-1} 1_{\{\tau=n-1\}} + \mathbb{E}[G_{\bar\tau}|\mathcal{F}_{n-1}] 1_{\{\tau>n-1\}} \end{aligned}$$

By definition $S_{n-1}^N \ge G_{n-1}$. By the **induction hypothesis**:

$$\mathbb{E}[G_{\bar\tau}|\mathcal{F}_{n-1}] = \mathbb{E}[\mathbb{E}[G_{\bar\tau}|\mathcal{F}_n]|\mathcal{F}_{n-1}] \le \mathbb{E}[S_n^N|\mathcal{F}_{n-1}]$$

because $\bar\tau \in \mathcal{T}_{n,N}$. Combining the above:

$$\begin{aligned} \mathbb{E}[G_\tau|\mathcal{F}_{n-1}] &\le S_{n-1}^N 1_{\{\tau=n-1\}} + \underbrace{\mathbb{E}[S_n^N|\mathcal{F}_{n-1}]}_{\le S_{n-1}^N \text{ by def of } (S_n^N)_n} 1_{\{\tau>n-1\}} \\ &\le S_{n-1}^N \quad \checkmark \end{aligned}$$

Now we check the equality for τ_{n-1}^N:

$$\mathbb{E}[G_{\tau_{n-1}^N}|\mathcal{F}_{n-1}] = G_{n-1} 1_{\{\tau_{n-1}^N = n-1\}} + \mathbb{E}[G_{\tau_{n-1}^N} 1_{\{\tau_{n-1}^N > n-1\}}|\mathcal{F}_{n-1}]$$

On $\{\tau_{n-1}^N = n-1\}$ we have $G_{n-1} = S_{n-1}^N$. On $\{\tau_{n-1}^N > n-1\}$ we have $\tau_{n-1}^N = \tau_n^N$ and $S_{n-1}^N > G_{n-1}$. The latter implies $S_{n-1}^N = \mathbb{E}[S_n^N|\mathcal{F}_{n-1}]$ by definition of $(S_n^N)_n$.

Thus:

$$\begin{aligned}\mathbb{E}[G_{\tau^N_{n-1}}|\mathcal{F}_{n-1}] &= S^N_{n-1}1_{\{\tau^N_{n-1}=n-1\}} + \underbrace{\mathbb{E}[G_{\tau^N_n}|\mathcal{F}_{n-1}]}_{=\mathbb{E}[\mathbb{E}[G_{\tau^N_n}|\mathcal{F}_n]|\mathcal{F}_{n-1}]} 1_{\{\tau^N_{n-1}>n-1\}}\\ &= S^N_{n-1}1_{\{\tau^N_{n-1}=n-1\}} + \underbrace{\mathbb{E}[S^N_n|\mathcal{F}_{n-1}]}_{\text{by inductive hypothesis}} 1_{\{\tau^N_{n-1}>n-1\}}\\ &= S^N_{n-1}1_{\{\tau^N_{n-1}=n-1\}} + \underbrace{\mathbb{E}[S^N_n|\mathcal{F}_{n-1}]}_{S^N_{n-1}} 1_{\{\tau^N_{n-1}>n-1\}}\\ &= S^N_{n-1} \quad \checkmark\end{aligned}$$

Now we prove (i). Since $S^N_n \geq \mathbb{E}[G_\tau|\mathcal{F}_n] \quad \forall \tau \in \mathcal{T}_{n,N}$, then

$$\mathbb{E}[S^N_n] \geq \mathbb{E}[G_\tau] \quad \forall \tau \in \mathcal{T}_{n,N} \implies \mathbb{E}[S^N_n] \geq V^N_n.$$

However, $S^N_n = \mathbb{E}[G_{\tau^N_n}|\mathcal{F}_n] \implies \mathbb{E}[S^N_n] = \mathbb{E}[G_{\tau^N_n}] \leq V^N_n$. Thus $\boxed{V^N_n = \mathbb{E}[G_{\tau^N_n}] = \mathbb{E}[S^N_n]}$ and τ^N_n is **optimal**.

Now we prove (iii). By definition $S^N_n \geq \mathbb{E}[S^N_{n+1}|\mathcal{F}_n]$ and therefore $(S^N_n)_n$ is a supermartingale. Obviously $S^N_n \geq G_n$. Let $(\tilde{S}_n)_n$ be another supermartingale s.t. $\tilde{S}_n \geq G_n \forall n$. Then $\tilde{S}_N \geq G_N = S^N_N$. Assume $\tilde{S}_k \geq S^N_k$ for $n+1 \leq k \leq N$. Then

$$S^N_n = \max\{G_n, \mathbb{E}[S^N_{n+1}|\mathcal{F}_n]\} \leq \max\{\underbrace{G_n}_{\leq \tilde{S}_n}, \underbrace{\mathbb{E}[\tilde{S}_{n+1}|\mathcal{F}_n]}_{\leq \tilde{S}_n}\} \leq \tilde{S}_n. \quad \checkmark$$

Now we prove (ii). Let $\tau_* \in \mathcal{T}_{n,N}$ be optimal. Then $V^N_n = \mathbb{E}[G_{\tau_*}]$. However $S^N_k \geq G_k$, $n \leq k \leq N$ implies $S^N_{\tau_*} \geq G_{\tau_*}$. Arguing by contradiction, assume $\mathbb{P}(S^N_{\tau_*} > G_{\tau_*}) > 0$. Then $\mathbb{E}[S^N_{\tau_*} - G_{\tau_*}] > 0$ and

$$V^N_n = \mathbb{E}[G_{\tau_*}] < \mathbb{E}[S^N_{\tau_*}] = \mathbb{E}[\mathbb{E}[S^N_{\tau_*}|\mathcal{F}_n]] \leq \mathbb{E}[S^N_n] = V^N_n$$

where the inequality follows by (iii) and optional sampling (τ_* is bounded). Thus $V^N_n < V^N_n$, which is impossible. So, it must be $S^N_{\tau_*} = G_{\tau_*}$, $\mathbb{P}$-a.s. This implies $\tau_* \geq \tau^N_n$ because $\tau_* \in \mathcal{T}_{n,N}$.

Now we prove (iv). Take $n \leq k \leq N-1$.

$$\begin{aligned}\mathbb{E}[S^N_{(k+1)\wedge\tau^N_n}|\mathcal{F}_k] &= \mathbb{E}[S^N_{\tau^N_n}1_{\{\tau^N_n \leq k\}} + S^N_{k+1}1_{\{\tau^N_n \geq k+1\}}|\mathcal{F}_k]\\ &= \mathbb{E}[S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \leq k\}}|\mathcal{F}_k] + \mathbb{E}[S^N_{k+1}|\mathcal{F}_k]1_{\{\tau^N_n \geq k+1\}}\end{aligned}$$

where $\{\tau^N_n \geq k+1\} = \bigcap_{j=n}^k \{S^N_j > G_j\} \in \mathcal{F}_k$. On the event $\{\tau^N_n \geq k+1\}$ we have $S^N_j > G_j$ for $n \leq j \leq k$ and therefore $S^N_k = \mathbb{E}[S^N_{k+1}|\mathcal{F}_k]$ by definition. Then:

$$\begin{aligned}\mathbb{E}[S^N_{(k+1)\wedge\tau^N_n}|\mathcal{F}_k] &= S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \leq k\}} + S^N_k 1_{\{\tau^N_n \geq k+1\}}\\ &= S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \leq k\}} + S^N_{k\wedge\tau^N_n}1_{\{\tau^N_n \geq k+1\}}\\ &= S^N_{k\wedge\tau^N_n}. \quad \checkmark\end{aligned}$$

□

Remark 35. *Eq. (A) in Thm OS1 yields:*

$$S^N_n = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{n,N}} \mathbb{E}[G_\tau|\mathcal{F}_n]$$

and it characterises the so-called **Snell-envelope** *of* $(G_n)_n$*: the smallest supermartingale that dominates* $(G_n)_n$*. Notice that* $\mathcal{T}_{n,N}$ *is an uncountable set.*

3.3 Infinite Horizon

Infinite horizon: when $N = +\infty$ the backward induction fails.

Let $\mathcal{T}_n := \{\tau \in \mathcal{T} : \tau \geq n\}$ and $V_n := \sup_{\tau \in \mathcal{T}_n} \mathbb{E}[G_\tau]$.

To solve the problem we consider a process $(S_n)_{n \in \mathbb{Z}^+}$ defined as

$$S_n := \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_n} \mathbb{E}[G_\tau | \mathcal{F}_n]$$

and the stopping time

$$\tau_n := \inf\{k \geq n : S_k = G_k\} \quad \text{with } \inf \emptyset = +\infty$$

Homework 36

Prove that $\tau_n \in \mathcal{T}_n$.

We always assume $\mathbb{E}\left[\sup_{n \geq 0} |G_n|\right] < \infty$.

Theorem 37 (OS2)

The process $(S_n)_{n \in \mathbb{Z}^+}$ satisfies

$$S_n = \max\left(G_n, \mathbb{E}[S_{n+1} | \mathcal{F}_n]\right) \quad \forall n \in \mathbb{Z}^+. \tag{A}$$

Furthermore

$$\left.\begin{aligned} S_n &\geq \mathbb{E}[G_\tau | \mathcal{F}_n] \quad \forall \tau \in \mathcal{T}_n \\ S_n &= \mathbb{E}[G_{\tau_n} | \mathcal{F}_n] \quad \text{if } \mathbb{P}(\tau_n < \infty) = 1. \end{aligned}\right\} \quad (B) \tag{2}$$

Finally:

(i) τ_n is optimal if $\mathbb{P}(\tau_n < \infty) = 1$.

(ii) If τ_* is an optimal stopping time, then $\mathbb{P}(\tau_* \geq \tau_n) = 1$.

(iii) $(S_n)_{n \in \mathbb{Z}^+}$ is the smallest supermartingale that dominates $(G_n)_{n \in \mathbb{Z}^+}$.

(iv) $(S_{k \wedge \tau_n})_{k \geq n}$ is a martingale.

If $\mathbb{P}(\tau_n = +\infty) > 0$ then there is no optimal stopping time which is finite with probab. 1.

Proof. Let's start from (A). Take $\tau \in \mathcal{T}_n$ and set $\bar\tau := \tau \vee (n+1)$. Since $\{\tau \geq n+1\} \in \mathcal{F}_n$ we have:

$$\begin{aligned}
\mathbb{E}[G_\tau|\mathcal{F}_n] &= \mathbb{E}[G_\tau 1_{\{\tau=n\}}|\mathcal{F}_n] + \mathbb{E}[G_\tau 1_{\{\tau\geq n+1\}}|\mathcal{F}_n] \\
&= G_n 1_{\{\tau=n\}} + \mathbb{E}[G_{\bar\tau} 1_{\{\tau\geq n+1\}}|\mathcal{F}_n] \\
&= G_n 1_{\{\tau=n\}} + 1_{\{\tau\geq n+1\}}\mathbb{E}[\underbrace{\mathbb{E}[G_{\bar\tau}|\mathcal{F}_{n+1}]}_{\leq \operatorname{ess\,sup}_{\tau\in\mathcal{T}_{n+1}} \mathbb{E}[G_\tau|\mathcal{F}_{n+1}]}|\mathcal{F}_n] \\
&\leq G_n 1_{\{\tau=n\}} + \mathbb{E}[S_{n+1}|\mathcal{F}_n]1_{\{\tau\geq n+1\}} \\
&\leq \max\{G_n, \mathbb{E}[S_{n+1}|\mathcal{F}_n]\}.
\end{aligned}$$

Since $\tau \in \mathcal{T}_n$ was arbitrary, $\operatorname{ess\,sup}_{\tau\in\mathcal{T}_n} \mathbb{E}[G_\tau|\mathcal{F}_n] \leq \max(G_n, \mathbb{E}[S_{n+1}|\mathcal{F}_n])$.

Now we prove the reverse inequality. By definition $S_n \geq G_n$ and it remains to prove $S_n \geq \mathbb{E}[S_{n+1}|\mathcal{F}_n]$. Consider the family $\{\mathbb{E}[G_\tau|\mathcal{F}_{n+1}], \tau \in \mathcal{T}_{n+1}\} =: \Gamma_{n+1}$. Given $\tau_1, \tau_2 \in \mathcal{T}_{n+1}$ we set $\tau_3 := \tau_1 1_A + \tau_2 1_{A^c}$ with

$$A := \{\omega \in \Omega : \mathbb{E}[G_{\tau_1}|\mathcal{F}_{n+1}](\omega) > \mathbb{E}[G_{\tau_2}|\mathcal{F}_{n+1}](\omega)\}.$$

Since $A \in \mathcal{F}_{n+1}$ (HW) then $\tau_3 \in \mathcal{T}_{n+1}$ (HW). Moreover

$$\mathbb{E}[G_{\tau_3}|\mathcal{F}_{n+1}] \geq \max\{\mathbb{E}[G_{\tau_1}|\mathcal{F}_{n+1}], \mathbb{E}[G_{\tau_2}|\mathcal{F}_{n+1}]\} \quad \text{(HW)}.$$

That implies: the family Γ_{n+1} is upward directed and

$$\operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{n+1}} \mathbb{E}[G_\tau|\mathcal{F}_{n+1}] =\uparrow \lim_{k\to\infty} \mathbb{E}[G_{\tau_k}|\mathcal{F}_{n+1}] \quad \text{for some } (\tau_k)_{k\in\mathbb{N}} \in \mathcal{T}_{n+1}.$$

By <u>MON</u> we get

$$\begin{aligned}
\mathbb{E}[S_{n+1}|\mathcal{F}_n] &= \mathbb{E}[\uparrow \lim_{k\to\infty} \mathbb{E}[G_{\tau_k}|\mathcal{F}_{n+1}]|\mathcal{F}_n] \\
&=\uparrow \lim_{k\to\infty} \mathbb{E}[\mathbb{E}[G_{\tau_k}|\mathcal{F}_{n+1}]|\mathcal{F}_n] \\
&=\uparrow \lim_{k\to\infty} \underbrace{\mathbb{E}[G_{\tau_k}|\mathcal{F}_n]}_{\leq S_n \text{ because } \mathcal{T}_{n+1}\subset\mathcal{T}_n} \leq S_n.
\end{aligned}$$

> **Homework 38**
>
> Why can we use MON?

Then $S_n \geq \max\{G_n, \mathbb{E}[S_{n+1}|\mathcal{F}_n]\}$.

Now we prove (B). The inequality holds by definition of S_n. In order to prove $S_n = \mathbb{E}[G_{\tau_n}|\mathcal{F}_n]$ we first want to prove (iv). We know that $S_n \geq \mathbb{E}[S_{n+1}|\mathcal{F}_n]$ and therefore $(S_n)_{n\in\mathbb{N}}$ is a supermartingale. So we conclude $(S_{k\wedge\tau_n})_{k\geq n}$ is a supermartingale. It remains to show $\mathbb{E}[S_{k\wedge\tau_n}] = \mathbb{E}[S_{(k+1)\wedge\tau_n}]$.

> **Homework 39**
>
> If $(X_n)_{n\in\mathbb{Z}^+}$ is a supermg s.t. $\mathbb{E}[X_n] = \mathbb{E}[X_{n+1}] \forall n \in \mathbb{Z}^+$ then $(X_n)_{n\in\mathbb{Z}^+}$ is a martingale.
>
> *Sol:* $X_n \geq \mathbb{E}[X_{n+1}|\mathcal{F}_n]$ so $\mathbb{E}[X_n - X_{n+1}] = \mathbb{E}[X_n - \mathbb{E}[X_{n+1}|\mathcal{F}_n]] \geq 0$ but the equality implies $X_n = \mathbb{E}[X_{n+1}|\mathcal{F}_n]$.

$$
\begin{aligned}
\mathbb{E}[S_{(k+1)\wedge\tau_n}] &= \mathbb{E}[1_{\{\tau_n\le k\}}S_{\tau_n} + 1_{\{\tau_n\ge k+1\}}S_{k+1}] \\
&= \mathbb{E}[1_{\{\tau_n\le k\}}S_{\tau_n\wedge k} + 1_{\{\tau_n\ge k+1\}}\underbrace{\mathbb{E}[S_{k+1}|\mathcal{F}_k]}_{S_k}] \\
&\qquad \text{because } \{\tau_n \ge k+1\} = \bigcap_{j=n}^{k}\{S_j > G_j\} \in \mathcal{F}_k \\
&\qquad \text{and } S_k = \mathbb{E}[S_{k+1}|\mathcal{F}_k] \text{ on that event.} \\
&= \mathbb{E}[1_{\{\tau_n\le k\}}S_{\tau_n\wedge k} + 1_{\{\tau_n\ge k+1\}}S_{k\wedge\tau_n}] = \mathbb{E}[S_{\tau_n\wedge k}].
\end{aligned}
$$

This proves (iv).

Set $G^* := \sup_{k\in\mathbb{Z}^+}|G_k|$. Then

$$
|S_n| = \left|\operatorname*{ess\,sup}_{\tau\in\mathcal{T}_n}\mathbb{E}[G_\tau|\mathcal{F}_n]\right| \le \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_n}\mathbb{E}[|G_\tau||\mathcal{F}_n] \le \mathbb{E}[G^*|\mathcal{F}_n]
$$

Therefore $(S_n)_{n\in\mathbb{Z}^+}$ is bounded by the u.i. martingale $\{\mathbb{E}[G^*|\mathcal{F}_n], n\in\mathbb{Z}^+\}$. That implies $(S_n)_{n\in\mathbb{Z}^+}$ is a u.i. supermartingale. It follows that $(S_{k\wedge\tau_n})_{k\ge n}$ is a u.i. martingale, with $\lim_{k\to\infty}S_{k\wedge\tau_n} = S_{\tau_n} = G_{\tau_n}$ on $\{\tau_n < \infty\}$. Then by the proposition on u.i. martingales ($M_n = \mathbb{E}[M_\infty|\mathcal{F}_n]$)

$$
S_n = \mathbb{E}[S_{k\wedge\tau_n}|\mathcal{F}_n] = \mathbb{E}[S_{\tau_n}|\mathcal{F}_n] = \mathbb{E}[G_{\tau_n}|\mathcal{F}_n].
$$

This proves (B).

Let's prove (i).

$$
V_n = \sup_{\tau\in\mathcal{T}_n}\mathbb{E}[G_\tau] = \sup_{\tau\in\mathcal{T}_n}\mathbb{E}[\underbrace{\mathbb{E}[G_\tau|\mathcal{F}_n]}_{\le S_n}] \le \mathbb{E}[S_n]
$$

On the other hand $V_n \ge \mathbb{E}[G_{\tau_n}] = \mathbb{E}[\mathbb{E}[G_{\tau_n}|\mathcal{F}_n]] = \mathbb{E}[S_n]$ if $\mathbb{P}(\tau_n < \infty) = 1$.

Let's prove (ii). Assume τ_* is optimal, i.e. $V_n = \mathbb{E}[G_{\tau_*}]$. Since $G_k \le S_k \forall k\in\mathbb{Z}^+$ then

$$
V_n = \mathbb{E}[G_{\tau_*}] \le \mathbb{E}[S_{\tau_*}] = \mathbb{E}[\mathbb{E}[S_{\tau_*}|\mathcal{F}_n]]
$$

<u>If</u>: $\mathbb{E}[S_{\tau_*}|\mathcal{F}_n] \le S_n$ then: $\qquad (*)$

$$
V_n = \mathbb{E}[G_{\tau_*}] \le \mathbb{E}[S_{\tau_*}] \le \mathbb{E}[S_n] = V_n
$$

Therefore $\mathbb{E}[G_{\tau_*} - S_{\tau_*}] = 0 \implies S_{\tau_*} = G_{\tau_*} \implies \tau_* \ge \tau_n$.

It remains to prove (*).

$$
\begin{aligned}
\mathbb{E}[S_{\tau_*}|\mathcal{F}_n] &= \mathbb{E}[S_{\tau_*} + \mathbb{E}[G^*|\mathcal{F}_{\tau_*}] - \mathbb{E}[G^*|\mathcal{F}_{\tau_*}]|\mathcal{F}_n] \\
&= \mathbb{E}[S_{\tau_*} + G^*|\mathcal{F}_n] - \mathbb{E}[G^*|\mathcal{F}_n] \\
&= \mathbb{E}[\liminf_{k\to\infty}(\underbrace{S_{\tau_*\wedge k} + G^*}_{\ge 0})|\mathcal{F}_n] - \mathbb{E}[G^*|\mathcal{F}_n] \\
&\le \liminf_{k\to\infty}\mathbb{E}[S_{\tau_*\wedge k} + G^*|\mathcal{F}_n] - \mathbb{E}[G^*|\mathcal{F}_n] \quad \text{(Fatou's lemma)} \\
&= \liminf_{k\to\infty}\mathbb{E}[S_{\tau_*\wedge k}|\mathcal{F}_n] \le S_n, \text{ as needed.}
\end{aligned}
$$

Let's prove (iii). Assume that $(\tilde{S}_k)_{k\geq n}$ is another supermg that dominates $(G_k)_{k\geq n}$. Then

$$S_k = \mathbb{E}[G_{\tau_k}|\mathcal{F}_k] \leq \mathbb{E}[\tilde{S}_{\tau_k}|\mathcal{F}_k] \leq \tilde{S}_k.$$

Homework 40
Hint: $\tilde{S}_k \geq G_k \geq -|G_k| \geq -G^* \implies \tilde{S}_k + G^* \geq 0 \quad \forall k \geq n.$

$\square$

Remark 41. *It is clear that $V_n^N \leq V_n^{N+1} \leq V_n$ and $S_n^N \leq S_n^{N+1} \leq S_n \quad \forall N \in \mathbb{N}, n \leq N$. (HW)*

Then $V_n^\infty := \lim_{N\to\infty} V_n^N \leq V_n$ and $S_n^\infty := \lim_{N\to\infty} S_n^N \leq S_n$.

If $\mathbb{P}(\tau_n < \infty) = 1$, then

$$\begin{aligned} S_n = \mathbb{E}[G_{\tau_n}|\mathcal{F}_n] &= \mathbb{E}\left[G_{\tau_n} + \sup_{k\geq n}|G_k| \middle| \mathcal{F}_n\right] - \mathbb{E}\left[\sup_{k\geq n}|G_k|\middle|\mathcal{F}_n\right] \\ &= \mathbb{E}\left[\liminf_{N\to\infty}(G_{\tau_n\wedge N} + \underbrace{\sup_{k\geq n}|G_k|)}_{\geq 0}\middle|\mathcal{F}_n\right] - \mathbb{E}\left[\sup_{k\geq n}|G_k|\middle|\mathcal{F}_n\right] \\ &\overset{\text{Fatou's}}{\leq} \liminf_{N\to\infty}\mathbb{E}\left[G_{\tau_n\wedge N} + \sup_{k\geq n}|G_k|\middle|\mathcal{F}_n\right] - \mathbb{E}\left[\sup_{k\geq n}|G_k|\middle|\mathcal{F}_n\right] \\ &= \liminf_{N\to\infty}\mathbb{E}[G_{\tau_n\wedge N}|\mathcal{F}_n] \leq \liminf_{N\to\infty} S_n^N, \quad \text{because } \tau_n\wedge N \in \mathcal{T}_{n,N} \end{aligned}$$

Then $S_n = \lim_{N\to\infty} S_n^N$ and by analogous arguments $V_n = \lim_{N\to\infty} V_n^N$.

Notice also that $\tau_n^N = \inf\{k \geq n : S_k^N = G_k\} \leq \tau_n^{N+1} \leq \tau_n \quad \forall N \in \mathbb{N}$ because $Y_k^N := S_k^N - G_k \geq 0$ and $Y_k^N \leq Y_k^{N+1}$.

Therefore $\tau_n^\infty := \lim_{N\to\infty}\tau_n^N$ is well-defined and $\tau_n^\infty \leq \tau_n$.

Assume $\mathbb{P}(\tau_n - \tau_n^\infty > 0) > 0$. Then $\mathbb{P}(\tau_n - \tau_n^\infty \geq 1) > \delta$ for some $\delta > 0$. Let $A := \{\tau_n - \tau_n^\infty \geq 1\}$. Then for $\omega \in A$ we must have $\tau_n^\infty(\omega) \leq T_\omega$ for some $T_\omega \in \mathbb{N}$.

$S_k(\omega) - G_k(\omega) > 0, \forall n \leq k \leq \tau_n^\infty(\omega)$. That implies

$$S_k(\omega) - G_k(\omega) \geq \eta_\omega > 0, \forall n \leq k \leq \tau_n^\infty(\omega), \text{ for some } \eta_\omega > 0.$$

In other words

$$\min_{n\leq k\leq \tau_n^\infty(\omega)}(S_k(\omega) - G_k(\omega)) \geq \eta_\omega \tag{1}$$

Since $S_k^N(\omega) \nearrow S_k(\omega)$ as $N \nearrow \infty \quad \forall n \leq k \leq T_\omega$ then

$$\max_{n\leq k\leq T_\omega}(S_k(\omega) - S_k^N(\omega)) \leq \eta_\omega/2 \quad \text{for sufficiently large } N.$$

(Note: T_ω is finite! i.e., $\exists \bar{N}_{\omega,\eta,T} \in \mathbb{N}$ s.t. $\max_{n\leq k\leq T_\omega}(S_k(\omega) - S_k^N(\omega)) \leq \eta_\omega/2 \quad \forall N \geq \bar{N}_{\omega,\eta,T}$).

Therefore, from (1) *we deduce,* $\forall N \geq \bar{N}_{\omega,\eta,T}$

$$\min_{n \leq k \leq \tau_n^\infty(\omega)} (S_k^N(\omega) - G_k(\omega)) \geq \eta_\omega/2 \quad (HW)$$

which would imply the absurd $\tau_n^N(\omega) \geq \tau_n^\infty(\omega) + 1$. *Thus* $\mathbb{P}(\tau_n - \tau_n^\infty \geq 1) = 0$ *and* $\tau_n = \tau_n^\infty$ $\mathbb{P}$*-a.s.*

3.4 Markovian Setting

Let $(X_n)_{n\in\mathbb{Z}^+}$ be a Markov process taking values on some metric space $(E, \operatorname{dist}(\cdot))$ [For example $(\mathbb{R}^m, \|\cdot\|_{\mathbb{R}^m})$].

This allows us to use the Markovian setup:

$$(\Omega, \mathcal{F}, (\mathcal{F}_n^X)_{n\in\mathbb{Z}^+}, (X_n)_{n\in\mathbb{Z}^+}, (\theta_n)_{n\in\mathbb{Z}^+}, (\mathbb{P}_x)_{x\in E})$$

where θ_n is the shift operator.

Assume that $G_n = g(X_n)$ for some measurable $g : E \to \mathbb{R}$.

3.5 Finite horizon

Then $S_N^N = g(X_N)$. Arguing by induction, assume $S_{n+1}^N = u(n+1, X_{n+1})$ for some function $u(\cdot,\cdot)$. Then

$$S_n^N = \max\left\{ g(X_n), \quad \underbrace{\mathbb{E}[u(n+1, X_{n+1})|\mathcal{F}_n]}_{f(n,X_n) \text{ by Markovianity for some } f(\cdot)} \right\}$$

$\implies S_n^N = u(n, X_n)$ with

$$u(n,x) := \max\{g(x), (P_{n+1}u(n+1,\cdot))(x)\}$$

Here we use the semigroup $(P_n f)(x) := \int f(y)p_n(x,dy)$ where $p_n(x,dy) = \operatorname{Law}(X_n|X_{n-1} = x)$.

Recall that $S_n^N = \operatorname{ess\,sup}_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[G_\tau|\mathcal{F}_n]$ and therefore

$$u(n, X_n) = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[g(X_\tau)|\mathcal{F}_n]$$

Moreover we know that $\tau_n^* = \inf\{k \geq n : u(k, X_k) = g(X_k)\}$ is optimal (HW). It is not hard to see that τ_n^* is a stopping time for the filtration generated by X, i.e. $\mathcal{F}_k^X = \sigma(X_0, X_1, \ldots, X_k)$. Then, with no loss of generality we can assume $(\mathcal{F}_k)_{k\in\mathbb{Z}^+} = (\mathcal{F}_k^X)_{k\in\mathbb{Z}^+}$.

In order to pass to the infinite-time horizon we need to change our notation slightly:

$$\boxed{S_n^N = u^N(n, X_n) = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_{n,N}} \mathbb{E}[g(X_\tau)|\mathcal{F}_n]}$$

3.6 Infinite horizon

Let us now assume that $(X_n)_{n\in\mathbb{Z}^+}$ is a homogeneous Markov chain with $X_0 = x \in E$, so that $P_n = P$ and $(Pf)(x) = \int f(y)p(x,dy)$.

It follows that, for $X_0 = x \in E$,

$$u^N(n, X_n) = \operatorname*{ess\,sup}_{\tau \in \mathcal{T}_{n,N}} \underbrace{\mathbb{E}_x[g(X_{n+\tau\circ\theta_n})|\mathcal{F}_n]}_{\overset{\text{using the shift operator}}{=} \mathbb{E}_y[g(X_\tau)]\big|_{y=X_n(\omega)}}$$

$$\text{where } \tau(\omega) = n + \tau(\theta_n(\omega)) = n + \tau \circ \theta_n$$

$$= \operatorname*{ess\,sup}_{\sigma \in \mathcal{T}_{0,N-n}} \mathbb{E}_{X_n}[g(X_\tau)] = u^{N-n}(0, X_n).$$

Then

$$\begin{aligned} u^N(0,x) &= \max\{g(x), (Pu^N(1,\cdot))(x)\} \\ &= \max\{g(x), (Pu^{N-1}(0,\cdot))(x)\}. \end{aligned} \tag{1}$$

It is clear that $u^N \leq u^{N+1}$ (HW) and therefore the limit

$$u^\infty(x) := \lim_{N\to\infty} u^N(0,x) \quad \text{exists (pointwise).}$$

Taking limits in eq. (1) we get

$$u^\infty(x) = \max\left\{g(x), \lim_{N\to\infty} \mathbb{E}_x[u^{N-1}(0, X_1)]\right\}$$

Notice that

$$\begin{aligned} u^{N-1}(0,x) &= \sup_{\tau\in\mathcal{T}_{0,N-1}} \mathbb{E}_x[g(X_\tau)] \\ &= \sup_{\tau\in\mathcal{T}_{0,N-1}} \mathbb{E}_x[g(X_\tau) + \sup_{n\in\mathbb{Z}^+}|g(X_n)| - \sup_{n\in\mathbb{Z}^+}|g(X_n)|] \\ &= \sup_{\tau\in\mathcal{T}_{0,N-1}} \mathbb{E}_x\underbrace{[g(X_\tau) + \sup_{n\in\mathbb{Z}^+}|g(X_n)|]}_{\geq 0} - \underbrace{\mathbb{E}_x[\sup_{n\in\mathbb{Z}^+}|g(X_n)|]}_{=:c(x)>0} \end{aligned}$$

Then $u^{N-1}(0,x) \geq -c(x)$ (i.e., it is bounded from below). We can apply MON to obtain

$$\boxed{u^\infty(x) = \max\{g(x), \mathbb{E}_x[u^\infty(X_1)]\}.}$$

Thanks to Thm. OS2 we can identify

$$\boxed{u^\infty(x) = V_0 = \sup_{\tau\in\mathcal{T}_0} \mathbb{E}_x[g(X_\tau)] =: V(x)}$$

Moreover, by the Markovian structure

$$V(X_n) = \left(\sup_{\tau\in\mathcal{T}_0} \mathbb{E}_y[g(X_\tau)]\right)\Big|_{y=X_n} = \operatorname*{ess\,sup}_{\tau\in\mathcal{T}_n} \mathbb{E}_x[g(X_\tau)|\mathcal{F}_n]$$

Remark 42. *This is a bit imprecise but essentially true.*

4 Stochastic Control in Discrete Time

Remark. *Reference: Chapter I in Schmidli's book or Bertsekas & Shreve.*

- **Noise**: Let $(Y_n)_{n\in\mathbb{N}}$ be i.i.d. on $(E_Y, \mathcal{E}_Y)$. The noise's filtration is $\mathcal{F}_n = \sigma(Y_1, Y_2, \ldots, Y_n)$, with $\mathcal{F}_0 = \{\emptyset, \Omega\}$.
- **Control**: Let $(U_n)_{n\in\mathbb{Z}^+}$ be $(\mathcal{F}_n)$-adapted taking values on some space $(\mathbb{U}, \mathcal{U})$. The class of **admissible controls** is denoted by $\hat{\mathcal{A}}$.
- **State dynamics**: Let $(X_n)_{n\in\mathbb{Z}^+}$ be the state dynamics on $(E_X, \mathcal{E}_X)$ defined by

  $$\boxed{X_{n+1} = f(X_n, U_n, Y_{n+1})}$$

  for some measurable function $f : E_X \times \mathbb{U} \times E_Y \to E_X$.

Remark 43. *By composition of measurable functions, $(X_n)_{n\in\mathbb{Z}^+}$ is $(\mathcal{F}_n)$-adapted.*

Homework 44

Prove the adaptedness of the state dynamics process $(X_n)_{n\in\mathbb{Z}^+}$.

Remark 45. *We could generalize the dynamics to $X_{n+1} = f_{n+1}(X_n, U_n, Y_{n+1})$.*

Remark 46. *We shall consider $E_Y = \mathbb{R}^k$, $E_X = \mathbb{R}^m$, and $\mathbb{U} \subseteq \mathbb{R}^\ell$.*

Remark 47. *A priori, the dynamics is not Markovian, because $(U_n)_{n\in\mathbb{Z}^+}$ can be any adapted process. The dynamics becomes Markovian if $U_n = u_n(X_n)$ for some measurable functions $u_n : E_X \to \mathbb{U}$, $n \in \mathbb{Z}^+$. We denote*

$$\mathcal{A} := \left\{(U_n)_{n\in\mathbb{Z}^+} \in \hat{\mathcal{A}} : U_n = u_n(X_n) \text{ for measurable functions } u_n : E_X \to \mathbb{U}\right\}$$

as the class of ***Markovian controls***.

Remark. *In the next page the formulation is slightly unorthodox because I use $g(X_n, U_{n-1})$ instead of $g(X_n, U_n)$. It's for good reasons!*

Given a measurable function $g : E_X \times \mathbb{U} \to \mathbb{R}$, we consider an objective function: let $T \in \mathbb{N} \cup \{\infty\}$, $X_0 = x$.

$$\boxed{J_x^T(U) := \mathbb{E}\left[\sum_{n=1}^{T} g(X_n, U_{n-1}) e^{-rn}\right]}$$

Remark. *$r > 0$ is the discount rate. If $T = \infty \implies J_x^\infty(U) = J_x(U)$.*

The **value function** of our problem reads:

$$\boxed{V_T(x) := \sup_{U\in\hat{\mathcal{A}}} J_x^T(U)}$$

Remark. *If $T = \infty \implies V(x) := \sup_{U\in\hat{\mathcal{A}}} J_x(U)$.*

Remark 48. *Intuitively we expect $V_T \to V$ when $T \to \infty$. Now we provide a sufficient condition for this claim.*

Lemma 49

Assume $g(x,u) \geq -c$ for some constant $c > 0$. Then $V(x) =\uparrow \lim_{T\to\infty} V_T(x)$ for all $x \in E_X$.

Proof. We can rewrite $V_T(x)$ by adding and subtracting a constant term to make the running cost non-negative. Let $\tilde{g}(X_k, U_{k-1}) := g(X_k, U_{k-1}) + c$. Then:

$$\begin{aligned}
V_T(x) &= \sup_{U\in\hat{\mathcal{A}}} \mathbb{E}\left[\sum_{k=1}^{T} e^{-rk}(g(X_k, U_{k-1}) + c)\right] - \sum_{k=1}^{T} e^{-rk} c \\
&= \sup_{U\in\hat{\mathcal{A}}} \mathbb{E}\left[\sum_{k=1}^{T} e^{-rk}\tilde{g}(X_k, U_{k-1})\right] - c\left(\frac{1-e^{-r(T+1)}}{1-e^{-r}} - 1\right) \\
&= \tilde{V}_T(x) - c\frac{e^{-r} - e^{-r(T+1)}}{1-e^{-r}}
\end{aligned}$$

where $\tilde{V}_T(x) := \sup_{U\in\hat{\mathcal{A}}} \mathbb{E}\left[\sum_{k=1}^{T} e^{-rk}\tilde{g}(X_k, U_{k-1})\right]$.

Since $\tilde{g} \geq 0$, then $\tilde{V}_T(x) \leq \tilde{V}_{T+1}(x) \leq \tilde{V}(x)$ for all T, with

$$\tilde{V}(x) = \sup_{U\in\hat{\mathcal{A}}} \mathbb{E}\left[\sum_{k=1}^{\infty} e^{-rk}\tilde{g}(X_k, U_{k-1})\right].$$

Then $\tilde{V}_\infty(x) := \lim_{T\to\infty} \tilde{V}_T(x) \leq \tilde{V}(x)$.

Moreover, for any $\varepsilon > 0$, there exists $U^\varepsilon \in \hat{\mathcal{A}}$ such that $X_k^\varepsilon = f(X_{k-1}^\varepsilon, U_{k-1}^\varepsilon, Y_k)$ and

$$\begin{aligned}
\tilde{V}(x) &\leq \mathbb{E}\left[\sum_{k=1}^{\infty} e^{-rk}\tilde{g}(X_k^\varepsilon, U_{k-1}^\varepsilon)\right] + \varepsilon \\
&= \mathbb{E}\left[\lim_{T\to\infty}\sum_{k=1}^{T} e^{-rk}\tilde{g}(X_k^\varepsilon, U_{k-1}^\varepsilon)\right] + \varepsilon \\
&\overset{\text{MON}}{=} \lim_{T\to\infty} \mathbb{E}\left[\sum_{k=1}^{T} e^{-rk}\tilde{g}(X_k^\varepsilon, U_{k-1}^\varepsilon)\right] + \varepsilon \leq \tilde{V}_\infty(x) + \varepsilon
\end{aligned}$$

where the last inequality follows from the fact that $\mathbb{E}\left[\sum_{k=1}^{T} e^{-rk}\tilde{g}(X_k^\varepsilon, U_{k-1}^\varepsilon)\right] \leq \tilde{V}_T(x)$.

Thus $\tilde{V}_\infty(x) \leq \tilde{V}(x) \leq \tilde{V}_\infty(x) + \varepsilon$ for all $\varepsilon > 0$, which implies $\tilde{V}_\infty(x) = \tilde{V}(x)$.

It follows that:

$$\begin{aligned}
\lim_{T\to\infty} V_T(x) &= \lim_{T\to\infty}\left(\tilde{V}_T(x) - c\frac{e^{-r} - e^{-r(T+1)}}{1-e^{-r}}\right) \\
&= \tilde{V}(x) - c\frac{e^{-r}}{1-e^{-r}} \\
&= \sup_{U\in\hat{\mathcal{A}}} \mathbb{E}\left[\sum_{k=1}^{\infty} e^{-rk}g(X_k, U_{k-1}) + \sum_{k=1}^{\infty} ce^{-rk}\right] - c\frac{e^{-r}}{1-e^{-r}} \\
&= \sup_{U\in\hat{\mathcal{A}}} \mathbb{E}\left[\sum_{k=1}^{\infty} e^{-rk}g(X_k, U_{k-1})\right] + c\frac{e^{-r}}{1-e^{-r}} - c\frac{e^{-r}}{1-e^{-r}} = V(x)
\end{aligned}$$

where we used the fact that $\sum_{k=1}^{\infty} ce^{-rk} = c\left(\frac{1}{1-e^{-r}} - 1\right) = c\frac{e^{-r}}{1-e^{-r}}$. $\square$

Lemma 50 (**Dynamic Programming Principle**)
Suppose $|V_T(x)| < \infty$. The function $V_T(x)$ satisfies:

$$\begin{aligned} V_T(x) &= \sup_{u\in\mathbb{U}} \left\{ \mathbb{E}\left[e^{-r} g(X_1, u) + e^{-r} V_{T-1}(X_1) \right] \right\} \\ &= \sup_{u\in\mathbb{U}} \left\{ \mathbb{E}\left[e^{-r} g(f(x,u,Y_1),u) + e^{-r} V_{T-1}(f(x,u,Y_1)) \right] \right\} \end{aligned}$$

Suppose $g(x,u) \geq -c$ for $c > 0$. The function $V(x)$ satisfies:

$$\begin{aligned} V(x) &= \sup_{u\in\mathbb{U}} \left\{ \mathbb{E}\left[e^{-r} g(X_1, u) + e^{-r} V(X_1) \right] \right\} \\ &= \sup_{u\in\mathbb{U}} \left\{ \mathbb{E}\left[e^{-r} g(f(x,u,Y_1),u) + e^{-r} V(f(x,u,Y_1)) \right] \right\} \end{aligned}$$

Proof. $V_0(x) = 0$.

$$\begin{aligned} V_1(x) &= \sup_{U_0 \in m\mathcal{F}_0} \mathbb{E}[e^{-r} g(X_1, U_0)] = \sup_{U_0 \in m\mathcal{F}_0} \mathbb{E}[e^{-r} g(f(x,U_0,Y_1),U_0)] \\ &= \sup_{u\in\mathbb{U}} e^{-r}\Lambda_0(x,u) \quad \text{with } \Lambda_0(x,u) := \int g(f(x,u,y),u)\mu(dy) \text{ where } Y_1 \sim \mu \end{aligned}$$

Notice that $\Lambda_0(x,u) = \int [g(f(x,u,y),u) + V_0(f(x,u,y))]\mu(dy)$.

For any $\varepsilon > 0$ and fixed $x \in E_X$, $\exists u_0^\varepsilon(x) \in \mathbb{U}$ s.t. $e^{-r}\Lambda_0(x, u_0^\varepsilon(x)) \geq V_1(x) - \varepsilon$.

Remark 51. **A fine issue: measurable selection**: *is the mapping $x \mapsto u_0^\varepsilon(x)$ measurable $E_X \to \mathbb{U}$? Yes if the function Λ_0 is u.s.c. or l.s.c. and $\mathbb{U}$ is compact (see Bertsekas & Shreve, Prop. 7.33, Ch. 7.5, 7.34).*

$$\begin{aligned} V_2(x) &= \sup_{\substack{U_0 \in m\mathcal{F}_0 \\ U_1 \in m\mathcal{F}_1}} \mathbb{E}[e^{-r} g(X_1, U_0) + e^{-2r} g(X_2, U_1)] \\ &= \sup_{U_0, U_1} \mathbb{E}\left[e^{-r} g(X_1, U_0) + e^{-2r} \mathbb{E}[g(X_2, U_1)|\mathcal{F}_1] \right] \end{aligned}$$

By the **Freezing Lemma**, $\mathbb{E}[g(f(X_1,U_1,Y_2),U_1)|\mathcal{F}_1] = \Lambda_0(X_1, U_1)$. Thus:

$$\begin{aligned} V_2(x) &= \sup_{U_0, U_1} \mathbb{E}\left[e^{-r} g(X_1, U_0) + e^{-r}(e^{-r}\Lambda_0(X_1, U_1)) \right] \\ &\implies V_2(x) \leq \sup_{U_0} \mathbb{E}\left[e^{-r} g(X_1, U_0) + e^{-r} V_1(X_1) \right] \end{aligned}$$

$\forall \varepsilon > 0$, I can choose $U_1^\varepsilon = u_0^\varepsilon(X_1)$ so that $e^{-r}\Lambda_0(X_1, U_1^\varepsilon) \geq V_1(X_1) - \varepsilon$ $\mathbb{P}$-a.s. Then $V_2(x) \geq \sup_{U_0} \mathbb{E}\left[e^{-r} g(X_1, U_0) + e^{-r} V_1(X_1) \right] - \varepsilon$. Combining the two we obtain:

$$\begin{aligned} V_2(x) &= \sup_{U_0} \mathbb{E}\left[e^{-r} g(X_1, U_0) + e^{-r} V_1(X_1) \right] \\ &= \sup_{U_0} \mathbb{E}\left[e^{-r} g(f(x,U_0,Y_1),U_0) + e^{-r} V_1(f(x,U_0,Y_1)) \right] \\ &= \sup_{u\in\mathbb{U}} e^{-r}\Lambda_1(x,u) \end{aligned}$$

where $\Lambda_1(x,u) = \int (g(f(x,u,y),u) + V_1(f(x,u,y)))\mu(dy)$. As before we notice that $\forall \varepsilon > 0$, $\exists$ **measurable** $u_1^\varepsilon : E_X \to \mathbb{U}$ s.t. $e^{-r}\Lambda_1(x, u_1^\varepsilon(x)) \geq V_2(x) - \varepsilon \ \forall x \in E_X$.

Now we continue by induction: suppose that for $n \leq T-1$

$$\begin{aligned} V_j(x) &= \sup_{U_0} \mathbb{E}[e^{-r}g(X_1,U_0) + e^{-r}V_{j-1}(X_1)] \quad \forall j \leq n \\ &= \sup_{u\in\mathbb{U}} e^{-r}\Lambda_{j-1}(x,u) \quad \text{where} \\ \Lambda_{j-1}(x,u) &:= \int [g(f(x,u,y),u) + V_{j-1}(f(x,u,y))]\mu(dy) \end{aligned}$$

and $\forall \varepsilon > 0 \ \exists$ measurable $u_{j-1}^\varepsilon : E_X \to \mathbb{U}$ s.t. $e^{-r}\Lambda_{j-1}(x, u_{j-1}^\varepsilon(x)) \geq V_j(x) - \varepsilon \ \forall x \in E_X$. Then:

$$\begin{aligned} V_{n+1}(x) &= \sup_{\substack{U_j \in m\mathcal{F}_j \\ j=0,\dots,n}} \mathbb{E}\left[\sum_{k=1}^{n+1} e^{-kr}g(X_k,U_{k-1})\right] \\ &= \sup_{\substack{U_j \in m\mathcal{F}_j \\ j=0,\dots,n}} \mathbb{E}\left[\sum_{k=1}^{n} e^{-kr}g(X_k,U_{k-1}) + \mathbb{E}[e^{-(n+1)r}g(X_{n+1},U_n)|\mathcal{F}_n]\right] \end{aligned}$$

The inner conditional expectation is:

$$e^{-(n+1)r}\mathbb{E}[g(f(X_n,U_n,Y_{n+1}),U_n)|\mathcal{F}_n] = e^{-(n+1)r}\Lambda_0(X_n,U_n) \leq e^{-rn}V_1(X_n)$$

Moreover, choosing $U_n = U_n^\varepsilon \equiv u_0^\varepsilon(X_n)$ we have $e^{-r}\Lambda_0(X_n,U_n^\varepsilon) \geq V_1(X_n) - \varepsilon$ and therefore:

$$\begin{aligned} V_{n+1}(x) &= \sup_{\substack{U_j \in m\mathcal{F}_j \\ j=0,\dots,n-1}} \mathbb{E}\left[\sum_{k=1}^{n} e^{-rk}g(X_k,U_{k-1}) + e^{-rn}V_1(X_n)\right] \\ &= \sup_{\substack{U_j \in m\mathcal{F}_j \\ j=0,\dots,n-1}} \mathbb{E}\left[\sum_{k=1}^{n-1} e^{-rk}g(X_k,U_{k-1}) + \mathbb{E}[e^{-rn}(g(X_n,U_{n-1}) + V_1(X_n))|\mathcal{F}_{n-1}]\right] \end{aligned}$$

The inner conditional expectation is:

$$\begin{aligned} &e^{-rn}\mathbb{E}[g(f(X_{n-1},U_{n-1},Y_n),U_{n-1}) + V_1(f(X_{n-1},U_{n-1},Y_n))|\mathcal{F}_{n-1}] \\ &= e^{-rn}\Lambda_1(X_{n-1},U_{n-1}) \leq e^{-r(n-1)}V_2(X_{n-1}) \end{aligned}$$

Moreover, choosing $U_{n-1} = U_{n-1}^\varepsilon = u_1^\varepsilon(X_{n-1})$ we get $e^{-r}\Lambda_1(X_{n-1},U_{n-1}^\varepsilon) \geq V_2(X_{n-1}) - \varepsilon$. Then:

$$V_{n+1}(x) = \sup_{\substack{U_j \in m\mathcal{F}_j \\ j=0,\dots,n-2}} \mathbb{E}\left[\sum_{k=1}^{n-1} e^{-rk}g(X_k,U_{k-1}) + e^{-r(n-1)}V_2(X_{n-1})\right]$$

Iterating the procedure we get:

$$V_{n+1}(x) = \sup_{U_0 \in m\mathcal{F}_0} \mathbb{E}[e^{-r}g(X_1,U_0) + e^{-r}V_n(X_1)]$$

This proves the finite horizon case: $V_T(x) = \sup_{U_0\in m\mathcal{F}_0} \mathbb{E}[e^{-r}g(X_1,U_0) + e^{-r}V_{T-1}(X_1)]$.

For the infinite horizon case, since $g \geq -c$, then $\tilde{V}_T(x) := V_T(x) + \sum_{k=1}^T ce^{-rk} \geq 0$.

$$\implies \tilde{V}_T(x) = \sup_{U_0\in m\mathcal{F}_0} \mathbb{E}[e^{-r}\tilde{g}(X_1,U_0) + e^{-r}\tilde{V}_{T-1}(X_1)]$$

where $\tilde{V}_{T-1}(X_1) = V_{T-1}(X_1) + \sum_{k=1}^{T-1} ce^{-rk}$. Since $\tilde{V}_T(x) \leq \tilde{V}(x)$ for all $x \in E_X$, we have:

$$\tilde{V}_T(x) \leq \sup_{U_0 \in m\mathcal{F}_0} \mathbb{E}[e^{-r}\tilde{g}(X_1, U_0) + e^{-r}\tilde{V}(X_1)]$$

Letting $T \to \infty$, we obtain $\tilde{V}(x) \leq \sup_{U_0 \in m\mathcal{F}_0} \mathbb{E}[e^{-r}\tilde{g}(X_1, U_0) + e^{-r}\tilde{V}(X_1)]$. For the reverse inequality, we notice that for any $U_0 \in m\mathcal{F}_0$:

$$\tilde{V}(x) \geq \tilde{V}_T(x) \geq \mathbb{E}[e^{-r}\tilde{g}(X_1, U_0) + e^{-r}\tilde{V}_{T-1}(X_1)]$$

Letting $T \to \infty$ and using the Monotone Convergence Theorem (MON):

$$\tilde{V}(x) \geq \mathbb{E}[e^{-r}\tilde{g}(X_1, U_0) + e^{-r}\tilde{V}(X_1)]$$

By the arbitrariness of U_0, this implies $\tilde{V}(x) \geq \sup_{U_0 \in m\mathcal{F}_0} \mathbb{E}[e^{-r}\tilde{g}(X_1, U_0) + e^{-r}\tilde{V}(X_1)]$. Finally, notice that $\tilde{V}(x) = V(x) + \sum_{k=1}^{\infty} ce^{-rk}$ and conclude:

$$V(x) = \sup_{U_0 \in m\mathcal{F}_0} \mathbb{E}[e^{-r}g(X_1, U_0) + e^{-r}V(X_1)]$$

□

From DPP we have

$$\begin{aligned} V_T(x) &= \sup_{u \in \mathbb{U}} e^{-r} \int [g(f(x,u,y),u) + V_{T-1}(f(x,u,y))]\,\mu(dy) \\ &= \sup_{u \in \mathbb{U}} e^{-r} \Lambda_{T-1}(x,u). \end{aligned}$$

Analogously,

$$\begin{aligned} V(x) &= \sup_{u \in \mathbb{U}} e^{-r} \int [g(f(x,u,y),u) + V(f(x,u,y))]\,\mu(dy) \\ &= \sup_{u \in \mathbb{U}} e^{-r} \Lambda(x,u). \end{aligned}$$

Then we have a corollary:

Corollary 52 (Measurable selection)

Part I: Assume $\forall n \in \mathbb{N} \cup \{0\}$, $\exists$ measurable $u_n^* : E_X \to \mathbb{U}$ s.t. $\Lambda_n(x, u_n^*(x)) = \sup_{u \in \mathbb{U}} \Lambda_n(x,u)$. Then $U_n^* := u_{T-1-n}^*(X_n)$ is an optimal control, i.e., with $X_{n+1}^* = f(X_n^*, U_n^*, Y_{n+1})$, we have

$$V_T(x) = \mathbb{E}\left[\sum_{k=1}^{T} e^{-rk} g(X_k^*, U_{k-1}^*)\right].$$

Part II: Assume $\exists$ measurable $u^* : E_X \to \mathbb{U}$ s.t. $\Lambda(x, u^*(x)) = \sup_{u \in \mathbb{U}} \Lambda(x,u)$. Then $U_n^* = u^*(X_n)$ is an optimal control, i.e., with $X_{n+1}^* = f(X_n^*, U_n^*, Y_{n+1})$, we have

$$V(x) = \mathbb{E}\left[\sum_{k=1}^{\infty} e^{-rk} g(X_k^*, U_{k-1}^*)\right].$$

Proof. $V_T(x) = \sup_{u\in\mathbb{U}} e^{-r}\Lambda_{T-1}(x,u) = e^{-r}\Lambda_{T-1}(x, u^*_{T-1}(x))$

$$= e^{-r}\int \left[g(f(x,u^*_{T-1}(x),y),u^*_{T-1}(x)) + V_{T-1}(f(x,u^*_{T-1}(x),y))\right]\mu(dy)$$
$$= \mathbb{E}\left[e^{-r}g(X^*_1,U^*_0) + e^{-r}V_{T-1}(X^*_1)\right]$$

using $U^*_0 = u^*_{T-1}(x)$.

Now we notice:

$$V_{T-1}(X^*_1) = \sup_{u\in\mathbb{U}} e^{-r}\Lambda_{T-2}(X^*_1,u) = e^{-r}\Lambda_{T-2}(X^*_1, u^*_{T-2}(X^*_1))$$
$$= e^{-r}\mathbb{E}\left[g(X^*_2,U^*_1) + V_{T-2}(X^*_2) \mid \mathcal{F}_1\right]$$

by the **freezing lemma** because $X^*_2 = f(X^*_1,U^*_1,Y_2)$ with $X^*_1, U^*_1 \in m\mathcal{F}_1$ and $Y_2 \perp\!\!\!\perp \mathcal{F}_1$.

Combining with the first equation yields:

$$V_T(x) = \mathbb{E}\left[e^{-r}g(X^*_1,U^*_0) + e^{-2r}g(X^*_2,U^*_1) + e^{-2r}V_{T-2}(X^*_2)\right].$$

Iterating the procedure we obtain the claim in Part I. □

Homework 53

Prove the claim in Part II of the corollary.

Remark 54. *In summary:*

- *We have obtained DPP in discrete time.*
- *The argument requires* **measurable selection**. *This means that we must prove a priori that $x \mapsto V_T(x)$ is either l.s.c. or u.s.c.*
- *We obtained a procedure to construct optimal controls in Markovian form $\implies$ no loss of generality in considering only* **Markovian controls**.

5 Optimal Stopping in Continuous Time

- **Optimal stopping theory** in continuous time can be developed from scratch without using results from the discrete-time setting $\implies$ **beyond the scope of these lecture notes**.

- We deduce results for continuous time as limits of results for discrete-time.

Remark (Warning)**.** *We will use some heuristic arguments for the most technical steps. You will be able to fill all the gaps in specific examples in the second half of the course.*

Formulation for infinite-time horizon and diffusive dynamics

- Let $(X_t)_{t\geq 0}$ be the solution of
$$X_t = X_0 + \int_0^t \mu(X_s)ds + \int_0^t \sigma(X_s)dW_s$$
where $\mu : \mathbb{R}^d \to \mathbb{R}^d$, $\sigma : \mathbb{R}^d \to \mathbb{R}^{d\times d'}$, $(W_t)_{t\geq 0}$ is a d'-dimensional Brownian motion, and $X_0 \in \mathbb{R}^d$.

- $(X_t)_{t\geq 0}$ is a **Markov process** and we define the family of probability measures $(\mathbb{P}_x)_{x\in\mathbb{R}^d}$ as $\mathbb{P}_x(A) = \mathbb{P}(A \mid X_0 = x)$ for all $A \in \mathcal{F}$.

- Let $g : \mathbb{R}^d \to \mathbb{R}$ be **continuous** and such that $\mathbb{E}_x\left[\sup_{t\geq 0} e^{-rt}|g(X_t)|\right] < \infty$ for some $r \geq 0$.

  Let us set for convenience:
$$\boxed{\overline{M}(\omega) := \sup_{t\geq 0} e^{-rt}|g(X_t(\omega))|}$$

- The problem we are interested in is
$$\boxed{V(x) = \sup_{\tau\in\mathcal{T}} \mathbb{E}_x\left[e^{-r\tau}g(X_\tau)\right]}$$
where $\mathcal{T} := \{\tau : \tau \text{ is an } (\mathcal{F}_t)\text{-stopping time with } \tau(\omega) < \infty\ \forall\omega \in \Omega\}$.

Remark 55. *In the current formulation, the problem is* **time-homogeneous**.

Remark (Aims)**.** *We want to connect this problem to the theory we developed in the discrete-time setting. We do this by approximation and only in some cases, but I emphasize that the main results hold in much broader generality.*

Step 1: (time-truncation)

For $T \in \mathbb{N}$, define:
$$V_T(x) := \sup_{\tau\in\mathcal{T}_{0,T}} \mathbb{E}_x\left[e^{-r\tau}g(X_\tau)\right], \quad \text{where}$$
$\mathcal{T}_{0,T} := \{\tau \in \mathcal{T} : \tau(\omega) \in [0, T], \forall\omega \in \Omega\}$.

Clearly $V_T(x) \leq V_{T+1}(x) \leq V(x)$ for all $x \in \mathbb{R}^d$. Then $V_\infty(x) := \lim_{T\to\infty} V_T(x)$ exists pointwise and $V_\infty(x) \leq V(x)$ for $x \in \mathbb{R}^d$.

Fix $\epsilon > 0$. Then there exists $\tau_\epsilon \in \mathcal{T}$ such that $V(x) \le \mathbb{E}_x\left[e^{-r\tau_\epsilon} g(X_{\tau_\epsilon})\right] + \epsilon$ and $\tau_\epsilon(\omega) < \infty$ for all $\omega \in \Omega$. Then:

$$\begin{aligned} V(x) - \epsilon &\le \mathbb{E}_x\left[\liminf_{T\to\infty}\left(\underbrace{e^{-r(\tau_\epsilon\wedge T)} g(X_{\tau_\epsilon\wedge T}) + \overline{M}}_{\ge 0}\right)\right] - \mathbb{E}_x[\overline{M}] \\ &\le \liminf_{T\to\infty} \mathbb{E}_x\left[e^{-r(\tau_\epsilon\wedge T)} g(X_{\tau_\epsilon\wedge T}) + \overline{M}\right] - \mathbb{E}_x[\overline{M}] \end{aligned}$$

where the term inside the expectation is non-negative by the definition of $\overline{M}$, allowing the use of Fatou's Lemma.

$$= \liminf_{T\to\infty} \mathbb{E}_x\left[e^{-r(\tau_\epsilon\wedge T)} g(X_{\tau_\epsilon\wedge T})\right] \le \lim_{T\to\infty} V_T(x) = V_\infty(x).$$

Thus $V(x) - \epsilon \le V_\infty(x) \le V(x)$ and $\epsilon > 0$ is arbitrary. It follows:

Lemma 56

$\forall x \in \mathbb{R}^d, \uparrow \lim_{T\to\infty} V_T(x) = V(x)$.

Step 2: (Time discretisation)

Fix $T \in \mathbb{N}$. Let $D_N^T := \left\{\frac{k}{2^N} \cdot T, k = 0, 1, 2, \ldots, 2^N\right\}$ be a **dyadic partition** of $[0, T]$. Recall that $D_N^T \subseteq D_{N+1}^T$ and $D^T := \bigcup_{N\in\mathbb{N}} D_N^T$ is a dense subset of $[0, T]$.

Let $(X_k^N)_{k\in D_N^T}$ be defined as $X_k^N := X_{\frac{k}{2^N}T}$ and let

$$V_T^N(x) := \sup_{\tau\in\mathcal{T}_{0,T}^N} \mathbb{E}_x\left[e^{-r\tau} g(X_\tau^N)\right] \quad \text{and} \quad V_T^D(x) := \sup_{\tau\in\mathcal{T}_{0,T}^D} \mathbb{E}_x\left[e^{-r\tau} g(X_\tau)\right]$$

where $\mathcal{T}_{0,T}^N := \{\tau \in \mathcal{T} : \tau \text{ takes values in } D_N^T\}$ and $\mathcal{T}_{0,T}^D := \{\tau \in \mathcal{T} : \tau \text{ takes values in } D^T\}$.

Remark 57. *For any $\tau \in \mathcal{T}_{0,T}^N$ we have*

$$\begin{aligned} \mathbb{E}_x\left[e^{-r\tau} g(X_\tau^N)\right] &= \mathbb{E}_x\left[\sum_{k=0}^{2^N} \mathbb{1}_{\{\tau = \frac{k}{2^N}T\}} e^{-r\frac{k}{2^N}T} g(X_k^N)\right] \\ &= \mathbb{E}_x\left[\sum_{k=0}^{2^N} \mathbb{1}_{\{\tau = \frac{k}{2^N}T\}} e^{-r\frac{k}{2^N}T} g(X_{\frac{k}{2^N}T})\right] = \mathbb{E}_x\left[e^{-r\tau} g(X_\tau)\right]. \end{aligned}$$

Since $D_N^T \subseteq D_{N+1}^T \subseteq D^T$ then

$$V_T^N(x) \le V_T^{N+1}(x) \le V_T^D(x) \le V_T(x), \quad \forall x \in \mathbb{R}^d.$$

Homework 58

Prove the chain of inequalities $V_T^N(x) \le V_T^{N+1}(x) \le V_T^D(x) \le V_T(x)$.

And $V_T^\infty(x) = \lim_{N\to\infty} V_T^N(x)$ is well-defined.

Lemma 59

$V_T^\infty(x) = V_T^D(x), \quad \forall x \in \mathbb{R}^d.$

Proof. For any $\epsilon > 0, \exists \tau_\epsilon \in \mathcal{T}_{0,T}^D$ s.t. $V_T^D(x) \leq \mathbb{E}_x[e^{-r\tau_\epsilon} g(X_{\tau_\epsilon})] + \epsilon$. Let

$$\tau_\epsilon^N := \begin{cases} \frac{k}{2^N}T & \text{on } \{\omega \in \Omega : \frac{k-1}{2^N}T < \tau_\epsilon(\omega) \leq \frac{k}{2^N}T\} \\ T & \text{on } \{\omega \in \Omega : \tau_\epsilon(\omega) = T\}. \end{cases}$$

Then $\tau_\epsilon^N \in \mathcal{T}_{0,T}^N$ and $0 \leq \tau_\epsilon^N(\omega) - \tau_\epsilon(\omega) \leq \frac{1}{2^N}T \quad \forall \omega \in \Omega$.

Homework 60

Show that $\tau_\epsilon^N \in \mathcal{T}_{0,T}^N$.

Moreover $\tau_\epsilon^N \geq \tau_\epsilon^{N+1}$ so that $\downarrow \lim_{N\to\infty} \tau_\epsilon^N = \tau_\epsilon$. By continuity of $t \mapsto e^{-rt} g(X_t(\omega))$ we deduce

$$\begin{aligned} V_T^D(x) - \epsilon \leq \mathbb{E}_x\left[e^{-r\tau_\epsilon} g(X_{\tau_\epsilon})\right] &= \mathbb{E}_x\left[\liminf_{N\to\infty} e^{-r\tau_\epsilon^N} g(X_{\tau_\epsilon^N})\right] \\ &\leq \liminf_{N\to\infty} \mathbb{E}_x\left[e^{-r\tau_\epsilon^N} g(X_{\tau_\epsilon^N})\right] \leq \liminf_{N\to\infty} V_T^N(x) = V_T^\infty(x). \end{aligned}$$

where the second inequality follows from Fatou's Lemma (note that $e^{-r\tau_\epsilon^N} g(X_{\tau_\epsilon^N}) \geq -\overline{M}$). Thus $V_T^D(x) - \epsilon \leq V_T^\infty(x) \leq V_T^D(x)$ and $\epsilon > 0$ is arbitrary. $\square$

By the exact same argument we can prove:

Lemma 61

$V_T(x) = V_T^D(x), \quad \forall x \in \mathbb{R}^d.$

Homework 62

Prove that $V_T(x) = V_T^D(x)$.

Combining the above results we obtain:

Corollary 63

$\forall x \in \mathbb{R}^d, \uparrow \lim_{T\to\infty} \uparrow \lim_{N\to\infty} V_T^N(x) = V(x).$

Formal link to discrete-time OS

Since V_T^N is the value function of a discrete-time Markovian OS problem with finite-time horizon, we establish the notation:

- $V_T^N(x) = u_T(0, x; \frac{T}{2^N}) \quad$ (T is the time-horizon; $\frac{T}{2^N}$ is the mesh-size)

- $\mathcal{T}_{j,N} = \mathcal{T}^N_{\frac{j}{2^N}T,T} = \{\tau \in \mathcal{T}^N_{0,T} : \tau \geq \frac{j}{2^N}T\}$
- $\mathcal{G}^N_j := \mathcal{F}_{\frac{j}{2^N}T}$

Then $u_T(j, X^N_j; \frac{T}{2^N}) = \text{ess sup}_{\tau \in \mathcal{T}_{j,N}} \mathbb{E}_x \left[e^{-r(\tau - \frac{j}{2^N}T)} g(X^N_\tau) \mid \mathcal{G}^N_j \right]$ and we are exactly in the framework of **Thm OS1**.

In particular, $\left\{ e^{-r\frac{j}{2^N}T} u_T(j, X^N_j; \frac{T}{2^N}), j = 0, 1, \dots, 2^N \right\}$ is the smallest super-martingale that dominates $\left\{ e^{-r\frac{j}{2^N}T} g(X^N_j), j = 0, 1, \dots, 2\right.$

Moreover $\tau^{T,N}_* := \inf \left\{ 0 \leq j \leq 2^N : u_T(j, X^N_j; \frac{T}{2^N}) = g(j, X^N_j) \right\}$ is optimal if finite $\mathbb{P}$-a.s.

Remark. *With a slight abuse of notation we set*

$$\boxed{u_T(j, X^N_j; \tfrac{T}{2^N}) = u_T(t, X_t; \tfrac{T}{2^N}) \text{ for } t = \tfrac{j}{2^N}T}$$

By the Markovian structure the function $u_T(\cdot, \cdot; \frac{T}{2^N}) : D^T_N \times \mathbb{R}^d \to \mathbb{R}$ reads

$$u_T(t, x; \tfrac{T}{2^N}) = \sup_{\tau \in \mathcal{T}^N_{t,T}} \mathbb{E} \left[e^{-r(\tau - t)} g(X_\tau) \mid X_t = x \right]$$

For any fixed $t \in D^T$, there is $N_t \in \mathbb{N}$ s.t. $t \in D^T_N, \forall N \geq N_t$ and

$$u_T(t, x; \tfrac{T}{2^N}) = \sup_{\tau \in \mathcal{T}^N_{t,T}} \mathbb{E} \left[e^{-r(\tau - t)} g(X_\tau) \mid X_t = x \right]$$

letting $N \to \infty$, by the same arguments as before

$$u_T(t, x; \tfrac{T}{2^N}) \uparrow u_T(t, x) = \sup_{\tau \in \mathcal{T}^D_{t,T}} \mathbb{E} \left[e^{-r(\tau - t)} g(X_\tau) \mid X_t = x \right]$$

Notice that $u_T : D^T \times \mathbb{R}^d \to \mathbb{R}$ and letting $T \uparrow \infty$ yields

$$u_T(t, x) \uparrow \sup_{\tau \in \mathcal{T}^D_t} \mathbb{E} \left[e^{-r(\tau - t)} g(X_\tau) \mid X_t = x \right]$$

$$\text{where } \mathcal{T}^D_t := \{\tau \in \mathcal{T} : \tau \text{ takes values in } D := \bigcup_{T \in \mathbb{N}} D^T \text{ and } \tau \geq t\}$$

$$= \sup_{\tau \in \mathcal{T}^D} \mathbb{E}_x \left[e^{-r\tau} g(X_\tau) \right] \doteq V^D(x) = V(x)$$

> **Homework 64**
> Prove that $V^D(x) = V(x)$.

Remark (In summary)**.**

$$\boxed{V(x) = \uparrow \lim_{T \to \infty} \uparrow \lim_{N \to \infty} u_T(t, x; \tfrac{T}{2^N}) \quad \text{for any } t \in D \text{ and any } x \in \mathbb{R}^d}$$

Remark 65. $u_T(t, x; \frac{T}{2^N}) = u_{T-t}(0, x; \frac{T}{2^N})$

Remark 66. *Because of the peculiar way in which I'm approaching the theory we will need to use* **DOM** *a few times. For that we need bounds on $V(x)$ and continuity of $x \mapsto V(x)$. This approach is not the most general! I will make standing assumptions and if time allows I will generalise.*

Assumptions:

(i) $\exists \Theta \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ s.t. $|e^{-rt}V(X_t)| \leq \Theta \quad \forall t \geq 0$. (this allows the use of **DOM**)

(ii) $x \mapsto V(x)$, $x \mapsto V_T(x)$, $x \mapsto V_T^N(x)$ are continuous on $\mathbb{R}^d$.

Remark (Observation)**.** *Condition (i) holds easily if g is bounded. Both conditions hold if g is Lipschitz, μ, σ are Lipschitz and r is "sufficiently large".*

Proof (for simplicity I consider $d = 1$ but the proof is the same for $d > 1$).

$$\begin{aligned}\left|\mathbb{E}\left[e^{-r\tau}(g(X_\tau^1) - g(X_\tau^2))\right]\right| &\leq \mathbb{E}\left[e^{-r\tau}|g(X_\tau^1) - g(X_\tau^2)|\right] \\ &\leq L_g \mathbb{E}\left[e^{-r\tau}|X_\tau^1 - X_\tau^2|\right] \leq L_g \mathbb{E}\left[e^{-2r\tau}(X_\tau^1 - X_\tau^2)^2\right]^{\frac{1}{2}}\end{aligned}$$

□

$$\begin{aligned}e^{-2rt}(X_t^1 - X_t^2)^2 &= (x_1 - x_2)^2 - 2r\int_0^t e^{-2rs}(X_s^1 - X_s^2)^2 ds \\ &\quad + 2\int_0^t e^{-2rs}(X_s^1 - X_s^2)d(X_s^1 - X_s^2) + \int_0^t e^{-2rs} d\langle X^1 - X^2\rangle_s \\ &= (x_1 - x_2)^2 + \int_0^t e^{-2rs}\left[2(X_s^1 - X_s^2)(\mu(X_s^1) - \mu(X_s^2)) - 2r(X_s^1 - X_s^2)^2\right] ds \\ &\quad + \int_0^t e^{-2rs} 2(X_s^1 - X_s^2)(\sigma(X_s^1) - \sigma(X_s^2)) dW_s \\ &\quad + \int_0^t e^{-2rs}(\sigma(X_s^1) - \sigma(X_s^2))^2 ds\end{aligned}$$

Taking expectations and dropping the martingale (why?)

$$\begin{aligned}\mathbb{E}\left[e^{-2rt}(X_t^1 - X_t^2)^2\right] &\leq (x_1 - x_2)^2 + \mathbb{E}\left[\int_0^t e^{-2rs} 2(L + \tfrac{1}{2}L^2 - r)(X_s^1 - X_s^2)^2 ds\right] \\ &\leq (x_1 - x_2)^2 \quad \text{if } r \geq L(1 + \tfrac{1}{2}L).\end{aligned}$$

Then we have:

$$\begin{aligned}\implies \mathbb{E}\left[e^{-2r\tau}(X_\tau^1 - X_\tau^2)^2\right] &\leq \liminf_{T\to\infty} \mathbb{E}\left[e^{-2r(\tau\wedge T)}(X_{\tau\wedge T}^1 - X_{\tau\wedge T}^2)^2\right] \\ &\leq \liminf_{T\to\infty}(x_1 - x_2)^2 = (x_1 - x_2)^2.\end{aligned}$$

Combining with the previous page:

$$\left|\mathbb{E}\left[e^{-r\tau}g(X_\tau^1) - e^{-r\tau}g(X_\tau^2)\right]\right| \leq \text{const.}|x_1 - x_2|.$$

$$\implies |V(x_1) - V(x_2)| \leq \text{const.}|x_1 - x_2|.$$

Homework 67

Complete the proof that $|V(x_1) - V(x_2)| \leq \text{const.}|x_1 - x_2|$.

There is an important consequence of continuity and so-called **Dini's lemma**:

Lemma 68 (Dini's lemma)

Let K be a compact and $f : K \to \mathbb{R}$ be continuous. If $(f_n)_{n\in\mathbb{N}}$ is a sequence of **continuous functions** from K to $\mathbb{R}$ s.t. $f_n \leq f_{n+1} \quad \forall n \in \mathbb{N}$ and $\forall x \in K \quad \lim_{n\to\infty} f_n(x) = f(x)$, then

$$\lim_{n\to\infty} \sup_{x\in K} |f_n(x) - f(x)| = 0. \quad \text{(i.e., uniform convergence on compacts)}$$

Then we have for any compact $K \subset \mathbb{R}^d$:

$$\boxed{\lim_{T\to\infty} \lim_{N\to\infty} \sup_{x\in K} |V_T^N(x) - V(x)| = 0.}$$

Homework 69

It is easy to verify that for $s < t$, $s, t \in [0, T]$ it holds $V_{T-s}(x) \geq V_{T-t}(x) \quad \forall x \in \mathbb{R}^d$.

Set

$$\begin{aligned}
\tau_*^{D_N^T} &:= \inf\{t \in D_N^T : u_T(t, X_t; \tfrac{T}{2^N}) = g(X_t)\} \\
\tau_*^{D^T} &:= \inf\{t \in D^T : u_T(t, X_t) = g(X_t)\} \\
\tau_*^{D} &:= \inf\{t \in D : V(X_t) = g(X_t)\}
\end{aligned}$$

Lemma 70

$\limsup_{T\to\infty} \limsup_{N\to\infty} \tau_*^{D_N^T} \leq \tau_*^D$.

Proof. For $\omega \in \Omega$ s.t. $\tau_*^D(\omega) = +\infty$ the claim is trivial. Fix $\omega \in \Omega$ s.t. $\tau_*^D(\omega) < \infty$. By definition $\tau_*^D(\omega) \in D$ and therefore $\exists T_\omega \in \mathbb{N}$ and $N_\omega \in \mathbb{N}$, $k_\omega \in \{1, 2, \dots, 2^{N_\omega}\}$ s.t. $\tau_*^D(\omega) = \frac{k_\omega}{2^{N_\omega}} T_\omega$. The latter implies $\tau_*^D(\omega) \in D_N^T$ for all $N \geq N_\omega$ and $T \geq T_\omega$. Then

$$u_T\left(\tau_*^D(\omega), X_{\tau_*^D(\omega)}; \frac{T}{2^N}\right) \leq u_T(\tau_*^D(\omega), X_{\tau_*^D(\omega)}) \leq V(X_{\tau_*^D(\omega)}) = g(X_{\tau_*^D(\omega)})$$

$\implies \tau_*^D(\omega) \geq \tau_*^{D_N^T}(\omega), \forall N \geq N_\omega, \forall T \geq T_\omega.$ ■ □

Lemma 71

$\liminf_{T\to\infty} \liminf_{N\to\infty} \tau_*^{D_N^T} \geq \tau_*^D$.

Proof. For $\omega \in \Omega$ s.t. $\tau_*^D(\omega) = 0$ it's obvious. Fix $\omega \in \Omega$ s.t. $\tau_*^D(\omega) > 0$. For any $\delta < \tau_*^D(\omega)$ we have

$$\inf_{0 \le t \le \delta} (V(X_t(\omega)) - g(X_t(\omega))) \ge c_{\delta,\omega} > 0$$

Since $t \mapsto X_t(\omega)$ is continuous then $(X_t(\omega))_{t \in [0,\delta]} \subseteq K$ for some compact $K \subset \mathbb{R}^d$. Then, by uniform convergence of V_T^N to V we have for large enough T and N:

$$\inf_{0 \le t \le \delta} (V_T^N(X_t(\omega)) - g(X_t(\omega))) \ge \frac{c_{\delta,\omega}}{2}$$

Homework 72

Verify the above inequality using uniform convergence on compacts.

$\implies \inf_{0 \le t \le \delta}(u_T(0, X_t(\omega); \frac{T}{2^N}) - g(X_t(\omega))) \ge \frac{c_{\delta,\omega}}{2}$. There is $M \in \mathbb{N}$ s.t. $\delta \le 2^M T$ and therefore

$$u_T(0, x; \tfrac{T}{2^N}) \le u_{2^M T - t}(0, x; \tfrac{T}{2^N}) = u_{2^M T}(t, x; \tfrac{T}{2^N})$$

For $T' = 2^M T$, this is $= u_{T'}(t, x; \frac{T'}{2^{M+N}})$, $\forall t \in D_{M+N}^{T'} \cap [0, \delta]$. $\implies \inf_{t \in [0,\delta] \cap D_{M+N}^{T'}}(u_{T'}(t, X_t(\omega); \frac{T'}{2^{M+N}}) - g(X_t(\omega))) \ge \frac{c_{\delta,\omega}}{2}$. Then $\tau_*^{D_{M+N}^{T'}}(\omega) \ge \delta$ for all T', N sufficiently large. Thus $\liminf_{T \to \infty} \liminf_{N \to \infty} \tau_*^{D_N^T}(\omega) \ge \delta$. Since $\delta < \tau_*^D(\omega)$ was arbitrary, and $\omega \in \Omega$, we obtain the claim. ■ □

Corollary 73

$\lim_{T \to \infty} \lim_{N \to \infty} \tau_*^{D_N^T} = \tau_*^D$.

Remark 74. *By continuity of $t \mapsto V(X_t) - g(X_t)$ and the fact that D is dense in $[0, T]$ we also have*

$$\tau_*^D = \inf\{t \in D : V(X_t) = g(X_t)\} = \inf\{t \ge 0 : V(X_t) = g(X_t)\} \doteq \tau_*$$

Theorem 75 ((OS3)) (i) $t \mapsto e^{-rt} V(X_t)$ is the smallest **supermartingale** that dominates $e^{-rt} g(X_t)$.

(ii) For $\tau_* := \inf\{t \ge 0 : V(X_t) = g(X_t)\}$, $t \mapsto e^{-r(t \wedge \tau_*)} V(X_{t \wedge \tau_*})$ is a **martingale**.

(iii) If $\mathbb{P}_x(\tau_* < \infty) = 1$ then τ_* is **optimal**.

(iv) If $\hat{\tau}$ is an optimal stopping time, then $\hat{\tau} \ge \tau_*$.

Proof. Fix $s, t \in D^T$ with $s < t$. Then $\exists N_0, k_1, k_2 \in \mathbb{N}, k_1 < k_2$ s.t. $s = \frac{k_1}{2^{N_0}} T, t = \frac{k_2}{2^{N_0}} T$. For any $N \ge N_0$ we have $s, t \in D_N^T$ and by the discrete-time OS theory we know

$$e^{-rs} u_T^N(s, X_s; \tfrac{T}{2^N}) \ge \mathbb{E}_x \left[e^{-rt} u_T^N(t, X_t; \tfrac{T}{2^N}) \mid \mathcal{F}_s \right]$$

Letting $N \to \infty$ and then $T \to \infty$ we obtain by **MON**:

$$e^{-rs} V(X_s) \ge \mathbb{E}_x \left[e^{-rt} V(X_t) \mid \mathcal{F}_s \right], \quad s, t \in D^T, s < t$$

Then $t \mapsto e^{-rt}V(X_t)$ is a supermartingale on D^T for every T. Thus, it is a supermartingale on $D := \bigcup_{T\in\mathbb{N}} D^T$, which is a dense subset of $[0,\infty)$.

Since $x \mapsto V(x)$ is continuous and $(\mathcal{F}_t)_{t\geq 0}$ is right-continuous, then for any $s < t$ we can find sequences $(s_n)_n, (t_n)_n \subset D$ s.t. $s_n \downarrow s, t_n \downarrow t, s_n < t_n \forall n$ and $e^{-rs}V(X_s) = \lim_{n\to\infty} e^{-rs_n}V(X_{s_n})$. Taking conditional expectation:

$$
\begin{aligned}
e^{-rs}V(X_s) &= \mathbb{E}_x\left[\lim_{n\to\infty} e^{-rs_n}V(X_{s_n}) \mid \mathcal{F}_s\right] \\
&\overset{\text{DOM}}{=} \lim_{n\to\infty} \mathbb{E}_x\left[e^{-rs_n}V(X_{s_n}) \mid \mathcal{F}_s\right] \\
&\geq \lim_{n\to\infty} \mathbb{E}_x\left[\mathbb{E}_x\left[e^{-rt_n}V(X_{t_n}) \mid \mathcal{F}_{s_n}\right] \mid \mathcal{F}_s\right] \\
&= \lim_{n\to\infty} \mathbb{E}_x\left[e^{-rt_n}V(X_{t_n}) \mid \mathcal{F}_s\right] \\
&= \mathbb{E}_x\left[\lim_{n\to\infty} e^{-rt_n}V(X_{t_n}) \mid \mathcal{F}_s\right] = \mathbb{E}_x\left[e^{-rt}V(X_t) \mid \mathcal{F}_s\right]
\end{aligned}
$$

Thus $e^{-rt}V(X_t)$ is a **supermartingale** on $[0,\infty)$ and it is clear that it dominates $e^{-rt}g(X_t)$.

Let $e^{-rt}\tilde{V}(X_t)$ be a supermartingale that dominates $e^{-rt}g(X_t)$. Then for any $\tau \in \mathcal{T}$, we have

$$
\tilde{V}(x) \overset{\text{optional sampling}}{\geq} \mathbb{E}_x\left[e^{-r(\tau\wedge T)}\tilde{V}(X_{\tau\wedge T})\right] \geq \mathbb{E}_x\left[e^{-r(\tau\wedge T)}g(X_{\tau\wedge T})\right]
$$

Letting $T \to \infty$ and using **DOM**:

$$
\tilde{V}(x) \geq \mathbb{E}_x\left[e^{-r\tau}g(X_\tau)\right].
$$

Since $\tau \in \mathcal{T}$ was arbitrary, then $\tilde{V}(x) \geq \sup_{\tau\in\mathcal{T}} \mathbb{E}_x[e^{-r\tau}g(X_\tau)]$ and therefore $\tilde{V}(x) \geq V(x)$. This completes the proof of (i).

Now, from optimality of $\tau_*^{T,N}$ we have for $s < t, s, t \in D^T$ and for all $N \geq N_0$:

$$
e^{-r(s\wedge\tau_*^{T,N})}u_T\left(\tau_*^{T,N}\wedge s, X_{\tau_*^{T,N}\wedge s}; \frac{T}{2^N}\right) = \mathbb{E}_x\left[e^{-r(t\wedge\tau_*^{T,N})}u_T\left(t\wedge\tau_*^{T,N}, X_{t\wedge\tau_*^{T,N}}; \frac{T}{2^N}\right) \mid \mathcal{F}_s\right]
$$

Letting $T, N \to \infty$ and using convergence of $u_T(\cdot; \frac{T}{2^N})$ to $V(\cdot)$ and convergence of $\tau_*^{T,N}$ to τ_* we deduce:

$$
e^{-r(s\wedge\tau_*)}V(X_{s\wedge\tau_*}) = \mathbb{E}_x\left[e^{-r(t\wedge\tau_*)}V(X_{t\wedge\tau_*}) \mid \mathcal{F}_s\right]
$$

Homework 76

Make the argument precise with **DOM** and uniform convergence of $u_T(\cdot; \frac{T}{2^N})$ to $V(\cdot)$.

Then $e^{-r(t\wedge\tau_*)}V(X_{t\wedge\tau_*})$ is a **martingale** on D^T for any T. Thus on D and, from the same argument as above, on $[0,\infty)$. This completes the proof of (ii).

To prove (iii) it suffices to notice that $\forall t \geq 0$:

$$
\begin{aligned}
V(x) &= \mathbb{E}_x\left[e^{-r(t\wedge\tau_*)}V(X_{t\wedge\tau_*})\right] \\
&= \mathbb{E}_x\left[e^{-r\tau_*}V(X_{\tau_*})\mathbb{1}_{\{\tau_*\leq t\}}\right] + \mathbb{E}_x\left[e^{-rt}V(X_t)\mathbb{1}_{\{t<\tau_*\}}\right] \\
&= \mathbb{E}_x\left[e^{-r\tau_*}g(X_{\tau_*})\mathbb{1}_{\{\tau_*\leq t\}}\right] + \mathbb{E}_x\left[e^{-rt}V(X_t)\mathbb{1}_{\{t<\tau_*\}}\right].
\end{aligned}
$$

Letting $t \uparrow \infty$ we have:

$$\lim_{t\uparrow\infty} \mathbb{E}_x \left[e^{-r\tau_*} g(X_{\tau_*}) \mathbb{1}_{\{\tau_* \le t\}} \right] = \lim_{t\uparrow\infty} \mathbb{E}_x \left[(e^{-r\tau_*} g(X_{\tau_*}) + \overline{M} - \overline{M}) \mathbb{1}_{\{\tau_* \le t\}} \right]$$
$$\overset{\text{MON}}{=} \mathbb{E}_x \left[e^{-r\tau_*} g(X_{\tau_*}) \mathbb{1}_{\{\tau_* < \infty\}} \right] = \mathbb{E}_x \left[e^{-r\tau_*} g(X_{\tau_*}) \right]$$

and

$$\lim_{t\uparrow\infty} \left| \mathbb{E}_x \left[e^{-rt} V(X_t) \mathbb{1}_{\{t<\tau_*\}} \right] \right| \le \lim_{t\uparrow\infty} \mathbb{E}_x \left[\Theta \mathbb{1}_{\{t<\tau_*\}} \right]$$
$$\overset{\text{DOM}}{=} \mathbb{E}_x \left[\Theta \lim_{t\uparrow\infty} \mathbb{1}_{\{t<\tau_*\}} \right] = 0.$$

Hence $V(x) = \mathbb{E}_x[e^{-r\tau_*} g(X_{\tau_*})]$ and τ_* is **optimal**.

$\square$

> **Homework 77**
> Prove (iv).

6 The Variational Problem

We derive a PDE problem from the properties of V.

- **Obstacle constraint**: $V(x) \ge g(x) \quad \forall x \in \mathbb{R}^d$.
- **PDE**: Assume $V(x)$ is sufficiently smooth to justify some generalisation of Itô's formula. Then

$$V(x) \ge \mathbb{E}_x \left[e^{-rt} V(X_t) \right]$$
$$= V(x) + \mathbb{E}_x \left[\int_0^t e^{-rs} (\mathcal{L}_X V(X_s) - rV(X_s)) ds \right]$$

  where $\mathcal{L}_X f(x) = \langle \mu(x), \nabla f(x) \rangle + \frac{1}{2} \operatorname{tr}(\sigma\sigma^\top(x) D^2 f(x))$ is the **infinitesimal generator** of $(X_t)_{t\ge0}$.

  Then $\mathbb{E}_x \left[\int_0^t e^{-rs} (\mathcal{L}_X V(X_s) - rV(X_s)) ds \right] \le 0$. Dividing by t and letting $t \to 0$ yields

$$\lim_{t\downarrow 0} \frac{1}{t} \mathbb{E}_x \left[\int_0^t e^{-rs} (\mathcal{L}_X V(X_s) - rV(X_s)) ds \right] = \mathcal{L}_X V(x) - rV(x) \le 0, \quad \forall x \in \mathbb{R}^d.$$

- **PDE in $\mathcal{C}$**: Set $\mathcal{C} := \{x \in \mathbb{R}^d : V(x) > g(x)\}$. For $x \in \mathcal{C}$, $\mathbb{P}_x(\tau_* > 0) = 1$ and

$$V(x) = \mathbb{E}_x \left[e^{-r(t\wedge\tau_*)} V(X_{t\wedge\tau_*}) \right]$$
$$= V(x) + \mathbb{E}_x \left[\int_0^{t\wedge\tau_*} e^{-rs} (\mathcal{L}_X V(X_s) - rV(X_s)) ds \right]$$
$$\implies \mathbb{E}_x \left[\int_0^{t\wedge\tau_*} e^{-rs} (\mathcal{L}_X V(X_s) - rV(X_s)) ds \right] = 0.$$

  Since we know that $\mathcal{L}_X V - rV \le 0$, then it must be $\mathcal{L}_X V(x) - rV(x) = 0$ for $x \in \mathcal{C}$.

- **Variational inequality**: Combining the items above we have

$$\begin{cases} \mathcal{L}_X V(x) - rV(x) = 0, & x \in \mathcal{C} = \{V > g\} \\ \mathcal{L}_X V(x) - rV(x) \le 0, & x \in \mathbb{R}^d \\ V(x) \ge g(x) \end{cases}$$

$\implies \max\{\mathcal{L}_X V(x) - rV(x), g(x) - V(x)\} = 0, \quad x \in \mathbb{R}^d$. And we need some growth conditions at $+\infty$.

- **Finite-time horizon**:

$$V(t,x) := \sup_{\tau \in \mathcal{T}_{0,T-t}} \mathbb{E}_x \left[e^{-r\tau} g(X_\tau) \right]$$

$$\begin{cases} \max\{\partial_t V(t,x) + \mathcal{L}_X V(t,x) - rV(t,x), g(x) - V(t,x)\} = 0 \\ \text{for } (t,x) \in [0,T) \times \mathbb{R}^d \\ V(T,x) = g(x). \end{cases}$$

Remark 78 (Running profit)**.** $V(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x \left[\int_0^\tau e^{-rt} f(X_t) dt + e^{-r\tau} g(X_\tau) \right]$.

- $e^{-rt} V(X_t) + \int_0^t e^{-rs} f(X_s) ds$ *is a supermartingale.*
- $e^{-r(t \wedge \tau_*)} V(X_{t \wedge \tau_*}) + \int_0^{t \wedge \tau_*} e^{-rs} f(X_s) ds$ *is a martingale.*
- $\tau_* = \inf\{t \ge 0 : V(X_t) = g(X_t)\}$ *is optimal.*

Homework 79

Derive the variational inequality for the OSP with running profit.

The above derivation of the variational inequality is largely heuristic. Now we formulate a more rigorous statement:

Theorem 80 (Verification)

Assume $\exists$ a "sufficiently smooth" solution of

$$\max\{\mathcal{L}_X u(x) - ru(x) + f(x), u(x) - g(x)\} = 0, \quad \forall x \in \mathbb{R}^d$$

s.t. $\lim_{t \to \infty} \mathbb{E}_x \left[e^{-rt} u(X_t) \mathbb{1}_{\{t \le \hat{\tau}\}} \right] = 0$ (**transversality condition**) for $\hat{\tau} := \inf\{t \ge 0 : u(X_t) = g(X_t)\}$.

If $\mathbb{P}_x(\hat{\tau} < \infty) = 1$, then $u(x) = \sup_{\tau \in \mathcal{T}} \mathbb{E}_x \left[e^{-r\tau} g(X_\tau) \right] = V(x)$ and $\hat{\tau} = \tau_*$ is an optimal stopping time.

Remark 81. *Here the notion of "sufficiently smooth" is deliberately vague and it should be read as: smooth enough to guarantee that*

$$(*) \quad u(x) = \mathbb{E}_x \left[e^{-r(t \wedge \tau)} u(X_{t \wedge \tau}) - \int_0^{t \wedge \tau} e^{-rs} (\mathcal{L}_X u - ru)(X_s) ds \right]$$

holds. For the applications that you will see later it will be $d = 1$ and in that case it's enough that $u \in C^1(\mathbb{R})$ with $x \mapsto \partial_x u(x)$ locally Lipschitz. That allows the application of Tanaka's formulae.

Proof. Using $(*)$ and conditions $\mathcal{L}_X u - ru \leq 0$ and $u \geq g$ we get

$$u(x) \geq \mathbb{E}_x\left[e^{-r(t\wedge\tau)}g(X_{t\wedge\tau})\right] \quad \forall \tau \in \mathcal{T}.$$

Letting $t \uparrow \infty$ and using **DOM** we get

$$u(x) \geq \mathbb{E}_x\left[e^{-r\tau}g(X_\tau)\right] \quad \forall \tau \in \mathcal{T}.$$

$\implies u(x) \geq V(x) = \sup_{\tau\in\mathcal{T}} \mathbb{E}_x\left[e^{-r\tau}g(X_\tau)\right]$.

For the reverse inequality we choose $\tau = \hat{\tau}$ in $(*)$. That yields

$$\begin{aligned} u(x) &= \mathbb{E}_x\left[e^{-r(\hat{\tau}\wedge t)}u(X_{t\wedge\hat{\tau}})\right] \\ &= \mathbb{E}_x\left[e^{-r\hat{\tau}}u(X_{\hat{\tau}})\mathbb{1}_{\{\hat{\tau}\leq t\}} + e^{-rt}u(X_t)\mathbb{1}_{\{\hat{\tau}>t\}}\right] \\ &= \mathbb{E}_x\left[e^{-r\hat{\tau}}g(X_{\hat{\tau}})\mathbb{1}_{\{\hat{\tau}\leq t\}} + e^{-rt}u(X_t)\mathbb{1}_{\{\hat{\tau}>t\}}\right]. \end{aligned}$$

Let $t \uparrow \infty$ and use **MON** and the **transversality condition**.

$$\implies u(x) = \mathbb{E}_x\left[e^{-r\hat{\tau}}g(X_{\hat{\tau}})\mathbb{1}_{\{\hat{\tau}<\infty\}}\right] = \mathbb{E}_x\left[e^{-r\hat{\tau}}g(X_{\hat{\tau}})\right].$$

Then $u(x) \leq V(x) \implies u(x) = V(x)$ and $\hat{\tau} = \tau_*$ is **optimal**. □

Homework 82

Write down and prove a verification theorem for

$$V(x) = \sup_{\tau\in\mathcal{T}} \mathbb{E}_x\left[e^{-r\tau}g(X_\tau) + \int_0^\tau e^{-rt}f(X_t)dt\right].$$

Homework 83

Write down and prove a verification theorem for

$$V(t,x) = \sup_{\tau\in\mathcal{T}_{0,T-t}} \mathbb{E}_x\left[e^{-r\tau}g(X_\tau) + \int_0^\tau e^{-rt}f(X_t)dt\right].$$

Since $V(0) = 0$, we have

$$A_1 + A_2 = 0, \quad A_1 = -A_2$$

and the general solution is

$$V(n) = A_1 e^{-\frac{\mu}{\sigma^2}n}\left(e^{\frac{\Delta}{2}n} - e^{-\frac{\Delta}{2}n}\right)$$

Hence, we have to determine 3 constants: $C_2, A_1, \tilde{n}$.

To do this, we impose the following 3 conditions:

1. $V(\tilde{n}_-) = V(\tilde{n}_+)$ (**continuity**);
2. $V'(\tilde{n}_-) = V'(\tilde{n}_+)$ (**smooth-pasting**);
3. $V''(\tilde{n}_-) = V''(\tilde{n}_+)$ (**super-contact**) $\to$ function is $C^2 \Rightarrow$ we can apply Itô's formula.

Condition 2) implies that $V'(\tilde{n}_-) = V'(\tilde{n}_+) = 1$ $\rightarrow$ this is how $\tilde{n}$ is defined!

a solution for $0 < \tilde{n} < \infty$. It depends on K, for fixed other parameters, μ and $\rho > 0$.

In particular, it is possible to show that, setting $\frac{\mu}{\frac{\sigma^2}{2}} = \hat{\mu}$, $\frac{\rho}{\frac{\sigma^2}{2}} = \hat{\rho}$, the parameters $\hat{\rho}, \hat{\mu}, K$ should satisfy:

$$0 < \underbrace{\frac{2(\hat{\mu} + K\lambda_2)\Delta}{\hat{\rho}(\hat{\mu} + K\lambda_2) + 2\hat{\rho}\lambda_2}}_{\star} < 1$$

Call K_* the value of K s.t. $\star = 0$, and for $K > K_*$, $\tilde{n}$ is the solution of (*).

A VERIFICATION THEOREM FOR THE SOLUTION OF A CONSUMPTION/INVESTMENT PROBLEM

Consider the following general problem:

$$\sup_{\pi, C} \mathbb{E}\left[\int_0^T e^{-\rho t} w_t^{\gamma} \frac{C_t^{1-\gamma}}{1-\gamma} dt + e^{-\rho T} \Delta W_T^{\gamma} \frac{X_T^{1-\gamma}}{1-\gamma}\right]$$

where w_t^{γ} and ΔW_T^{γ} are **weights to consumption and terminal utility**. We have:

(a) $u(c) = \frac{c^{1-\gamma}}{1-\gamma}$ is the utility function;

(b) The problem considers both consumption and terminal utility from wealth;

(c) The wealth dynamics are given by:

$$dX_t = (r + \pi_t(\mu - r))X_t dt + \pi_t \sigma X_t dW_t + a_t dt - C_t dt.$$

Let $V_{t,n}$ be the associated value function.

Theorem 84 (Asmussen/Steffensen, 2020)
Assume there exists a function $\bar{V}(t, n)$ that is sufficiently integrable, such that it satisfies the HJB equation:

$$\partial_t \bar{V}(t, n) - \rho \bar{V}(t, n) + \sup_{\pi, C} \Big\{ \partial_n \bar{V}(t, n) \left[(r + \pi(\mu - r))n + a_t - C\right]$$
$$+ \frac{1}{2} \partial_{nn} \bar{V}(t, n) \pi^2 \sigma^2 n^2 + w_t^{\gamma} \frac{C^{1-\gamma}}{1-\gamma} \Big\} = 0,$$

with terminal condition:

$$\bar{V}(T, n) = \Delta W_T^{\gamma} \frac{n^{1-\gamma}}{1-\gamma}.$$

Then $V = \bar{V}$ and the optimal controls are given by:

$$C^* = \arg\sup_C \left\{ -\partial_n \bar{V}(t, n) C + w_t^{\gamma} \frac{C^{1-\gamma}}{1-\gamma} \right\};$$
$$\pi^* = \arg\sup_\pi \left\{ \partial_n \bar{V}(t, n) \pi(\mu - r) n + \frac{1}{2} \partial_{nn} \bar{V}(t, n) \pi^2 \sigma^2 n^2 \right\}.$$

Proof. Consider a strategy (π, C). Applying Itô to $\bar{V}(t, X_t)$:

$$d(e^{-\rho t}\bar{V}(t,X_t)) = e^{-\rho t}\left[-\rho\bar{V}(t,X_t)dt + \partial_t\bar{V}(t,X_t)dt\right.$$
$$\left.+\partial_n\bar{V}(t,X_t)dX_t + \frac{1}{2}\partial_{nn}\bar{V}(t,X_t)\pi_t^2\sigma^2X_t^2dt\right]$$

Integrating, we have:

$$e^{-\rho t}\bar{V}(t,X_t) = -\int_t^T d(e^{-\rho s}\bar{V}(s,X_s)) + e^{-\rho T}\bar{V}(T,X_T)$$

For every (π, C), from the HJB equation we have:

$$\partial_t\bar{V}(t,X_t) - \rho\bar{V}(t,X_t) \leq -\partial_n\bar{V}(t,X_t)\left[(r+\pi_t(\mu-r))X_t + a_t - C_t\right]$$
$$-\frac{1}{2}\partial_{nn}\bar{V}(t,X_t)\pi_t^2\sigma^2X_t^2 - w_t^\gamma\frac{C_t^{1-\gamma}}{1-\gamma}$$

and

$$\begin{aligned} e^{\rho t}e^{-\rho t}\bar{V}(t,X_t) &= -\int_t^T e^{-\rho s}\left[-\rho\bar{V}(s,X_s)ds + \partial_s\bar{V}(s,X_s)ds\right. \\ &\quad \left.+\partial_n\bar{V}(s,X_s)dX_s + \frac{1}{2}\partial_{nn}\bar{V}(s,X_s)\pi_s^2\sigma^2X_s^2ds\right] + e^{-\rho T}\bar{V}(T,X_T) \\ &\geq e^{\rho t}\int_t^T e^{-\rho s}\left[w_s^\gamma\frac{C_s^{1-\gamma}}{1-\gamma}ds - \partial_n\bar{V}(s,X_s)\pi_sX_s\sigma dW_s\right] + e^{-\rho T}\bar{V}(T,X_T)\end{aligned}$$

Taking expectations:

$$\Rightarrow \bar{V}(t,n) \geq \mathbb{E}_{t,n}\left[\int_t^T e^{-\rho(s-t)}w_s^\gamma\frac{C_s^{1-\gamma}}{1-\gamma}ds + e^{-\rho(T-t)}\Delta W_T^\gamma\frac{X_T^{1-\gamma}}{1-\gamma}\right]$$

Since (π, C) was arbitrarily chosen:

$$\bar{V}(t,n) \geq \sup_{\pi,C}\mathbb{E}\left[\dots\right]$$

Now, we have to prove the other side. Consider the strategy realizing the supremum. For that strategy:

$$\partial_t\bar{V}(t,X_t) - \rho\bar{V}(t,X_t) = -\partial_n\bar{V}(t,X_t)\left[(r+\pi_t(\mu-r))X_t + a_t - C_t\right]$$
$$-\frac{1}{2}\partial_{nn}\bar{V}(t,X_t)\pi_t^2\sigma^2X_t^2 - w_t^\gamma\frac{C_t^{1-\gamma}}{1-\gamma}$$

Using the same arguments as above:

$$\begin{aligned}\Rightarrow \bar{V}(t,n) &= \mathbb{E}_{t,n}\left[\int_t^T e^{-\rho(s-t)}w_s^\gamma\frac{C_s^{1-\gamma}}{1-\gamma}ds + e^{-\rho(T-t)}\Delta W_T^\gamma\frac{X_T^{1-\gamma}}{1-\gamma}\right] \\ &\overset{\text{by definition}}{\leq} \sup_{\pi,C}\mathbb{E}_{t,n}\left[\dots\right]\end{aligned}$$

Thus, $\bar{V}(t,n) = \sup_{\pi,C} J_{t,n}(\pi, C) = V(t,n)$. □

□

OTHER INTERESTING PROBLEMS

MINIMIZING THE RUIN PROBABILITY

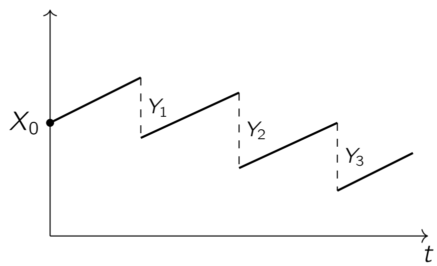
The "classical" risk theory model in insurance:

Insurance company $\rightarrow$ Receives premium in exchange for protection against events (contractually defined) that cause damage which is reimbursed by the company.

- $\uparrow \pi \rightarrow$ **premium** (constant rate)
- $\downarrow Y_i \rightarrow$ **payments** (claims)

The company has a capital endowment, which must not be depleted (go below 0).

CLASSICAL MODEL



- $X_0 \rightarrow$ **initial capital**, (invested at the rate r)
- $\rightarrow$ premiums arrive at a constant rate
- Y_1, Y_2, Y_3 are the **claims** (jumps)

Key quantity: **RUIN PROBABILITY**, i.e. defining $\tau = \inf\{t \geq 0 : X_t < 0\}$ the **TIME TO RUIN**, $\psi(n) = \mathbb{P}[\tau < \infty]$ is the **RUIN PROBABILITY**. $\delta(n) = 1 - \psi(n) \rightarrow$ **SURVIVAL PROBABILITY**

$$X_t = X_0 + \pi t - \sum_{i=1}^{N_t} Y_i$$

where:

- X_0 is the **initial capital**;
- $\pi > 0$ is the **premium rate**;
- N_t is a **counting process** (Risk process) $\rightarrow$ frequency of the claims;
- Y_i is the **size of the claims** $\rightarrow$ i.i.d., positive.

$\rightarrow$ **CRÀMER LUNDBERG MODEL**

Its diffusive approximation can be considered:

$$X_t^b = X_0 + \int_0^t \pi_s ds + \sigma \int_0^t dW_s,$$

which is easier to handle mathematically. Then, we can tackle the following problems related to the minimization of the **RUIN PROBABILITY**.

Suppose the company can:

(a) **Reinsure itself PROPORTIONALLY**, i.e. transfer to a third party a part of the risk. Its surplus process becomes:

$$X_t^b = X_0 + \int_0^t [b_s(1+\theta) - (\theta - \eta)]\lambda\mu ds + \sigma \int_0^t b_s dW_s^s,$$

where $c = (1+\eta)\lambda\mu$ is the premium rate, b is the **retention level**, and the reinsurance premium rate is $(1+\theta)(1-b)\lambda\mu$.

$$\Rightarrow c(b) = (b(1+\theta) - (\theta - \eta))\lambda\mu.$$

(b) **Invest in a risky asset**:

$$dX_t^\pi = (\eta + \pi_t m)dt + \sigma_s dW_t^s + \sigma_I \pi_t dW_t^I,$$

where W^I, W^s are independent, π_t is the **investment share**, and m is the drift of the risky asset process.

(a) $\sup_b \delta^b(n)$, where b is the **retention level**

$\downarrow$ OPTIMAL REINSURANCE

(b) $\sup_\pi \delta^\pi(n)$, where π is the **risky investment share**

$\downarrow$ OPTIMAL INVESTMENT

(c) $\sup_{\pi,b} \delta^{\pi,b}(n) \rightarrow$ OPTIMAL INVESTMENT AND REINSURANCE

$\searrow$ (a+b before must be combined)

7 Stochastic Control in Continuous Time

Remark. *A fully rigorous development of the continuous-time theory starting from our results in discrete time is a rather complex endeavour, which falls outside the scope of these lectures.*

The intuition is similar to the approach we developed for optimal stopping.

7.1 A stochastic control problem

- Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{P})$ be a filtered probability space equipped with a Brownian motion $(W_t)_{t\geq 0}$ in $\mathbb{R}^{d'}$.
- **Admissible controls**: Let $\mathcal{A}$ be the class of progressively measurable processes $(u_t)_{t\geq 0}$ taking values in a subset $\mathbb{U} \subseteq \mathbb{R}^m$, possibly compact, to keep the exposition simple. A control is **admissible** if the controlled SDE

$$X_t^u = x + \int_0^t \mu(X_s, u_s)ds + \int_0^t \sigma(X_s, u_s)dW_s \tag{3}$$

  admits a unique $(\mathcal{F}_t)_t$-adapted solution.

- If $\mu : \mathbb{R}^d \times \mathbb{U} \to \mathbb{R}^d$ and $\sigma : \mathbb{R}^d \times \mathbb{U} \to \mathbb{R}^{d\times d'}$ are such that

$$\|\mu(x,u) - \mu(y,u)\|_d + \|\sigma(x,u) - \sigma(y,u)\|_{d\times d'} \leq L\|x-y\|_d$$

  for a constant $L > 0$ independent of u, then the SDE in (1) admits a unique strong solution for each $u \in \mathcal{A}$ given and fixed.

  Remark 85. *Moreover, the same stability estimates as for standard SDEs continue to hold.*

  > **Homework** (Advanced)
  > Prove that the SDE in (1) admits a unique strong solution under the given Lipschitz condition.

- A control $u \in \mathcal{A}$ is **Markovian** if $u_t = \alpha(t, X_t^u)$ for some measurable $\alpha : [0,\infty) \times \mathbb{R}^d \to \mathbb{U}$. Notice that in the case of Markovian controls, the dynamics in (1) is a standard SDE:

$$X_t = x + \int_0^t \mu(X_s^u, \alpha(s, X_s^u))ds + \int_0^t \sigma(X_s^u, \alpha(s, X_s^u))dW_s.$$

8 Continuous-Time Stochastic Control

Remark 86. *This means that the mapping $(t,x) \mapsto \alpha(t,x)$ must be sufficiently regular to guarantee that the SDE is well posed.* **This is a major issue, because oftentimes $\alpha(t,x)$ is not even continuous!!!**

8.1 The Objective Function

Let $T \in (0,\infty)$, and let $\mathcal{O}$ be an open subset of $[0,T] \times \mathbb{R}^d$. Take continuous functions $f : [0,T] \times \mathbb{R}^d \times U \to \mathbb{R}$, $h : [0,T] \times \mathbb{R}^d \to \mathbb{R}$, and $g : \mathbb{R}^d \to \mathbb{R}$, and a discount rate $r \geq 0$.

For $(t,x) \in [0,T] \times \mathbb{R}^d$ and $u \in \mathcal{A}$, let

$$\begin{aligned} J_{t,x}(u) := \mathbb{E}_{t,x}\Big[\int_t^{T\wedge\tau_{\mathcal{O}}} & e^{-r(s-t)} f(s, X_s^u, u_s)\, ds \\ & + e^{-r(\tau_{\mathcal{O}}-t)} h(\tau_{\mathcal{O}}, X_{\tau_{\mathcal{O}}}^u) \mathbb{1}_{\{\tau_{\mathcal{O}}<T\}} \\ & + e^{-r(T-t)} g(X_T^u) \mathbb{1}_{\{\tau_{\mathcal{O}}\geq T\}} \Big], \end{aligned}$$

where $\mathbb{E}_{t,x}[\,\cdot\,] = \mathbb{E}[\,\cdot \mid X_t^u = x]$ and

$$\tau_{\mathcal{O}} := \inf\{s \geq t : (s, X_s^u) \notin \mathcal{O}\} \wedge T.$$

Remark 87. *Notice that, a priori, $(X_t^u)_{t\geq 0}$ is not Markovian.*

8.2 The Value Function

Definition 88

For $(t,x) \in [0,T] \times \mathbb{R}^d$, we define the **value function**:

$$V(t,x) := \sup_{u\in\mathcal{A}} J_{t,x}(u). \tag{2}$$

Remark 89 (Observations)**.**

- *For $t = T$, $V(T,x) = g(x)$, $x \in \mathbb{R}^d$.*
- *For $(t,x) \in \mathcal{O}^c$, $t < T$, $V(t,x) = h(t,x)$.*
- *A priori the function V is not even measurable. Much effort goes towards proving measurability directly from formula (2).*

Proposition 90

Assume $\mathcal{O} = (0,T) \times \mathbb{R}^d$ (i.e., $\tau_{\mathcal{O}} = T$). Assume f and g are Lipschitz. Then, for any $t \in [0,T]$ and $u \in \mathcal{A}$,

$$|J_{t,x}(u) - J_{t,y}(u)| \leq L\, ||x-y||_{\mathbb{R}^d} \quad \forall x, y \in \mathbb{R}^d,$$

where $L > 0$ is a given constant.

If f and g are also bounded, then, for any $x \in \mathbb{R}^d$ and $u \in \mathcal{A}$,

$$|J_{t,x}(u) - J_{s,x}(u)| \leq L(t-s)^{1/2} \quad \text{for } 0 \leq s < t \leq T,$$

where $L > 0$ is a given constant.

Homework 91 (Advanced)

Prove the Proposition above.

Corollary 92
Under the assumptions above:

$$|V(t,x) - V(s,y)| \le L\left(||x-y||_{\mathbb{R}^d} + (t-s)^{1/2}\right) \quad \text{for} \quad \begin{matrix} x, y \in \mathbb{R}^d, \\ 0 \le s < t \le T, \end{matrix}$$

where $L > 0$ is a given constant.

Homework 93
Prove the Corollary. This should be easy once you have the Proposition in place.

Remark 94. *To keep the exposition simple we shall focus on the case $\mathcal{O} = (0,T) \times \mathbb{R}^d$ so that $\tau_{\mathcal{O}} = T$ and the objective function simplifies.*

8.3 Discrete-time approximation of the dynamics via Euler scheme

Remark 95. *Fix $N \in \mathbb{N}$, set $t_j^N := \frac{j}{2^N}T$, for $j = 0, \dots, 2^N$. Let $u \in \mathcal{A}$ be given and set $\bar{X}_0^N = 0$, $\bar{u}_j^N := u_{t_j^N}$. Inductively define:*

$$\bar{X}_{j+1}^N = \bar{X}_j^N + \mu(\bar{X}_j^N, \bar{u}_j^N)\frac{T}{2^N} + \sigma(\bar{X}_j^N, \bar{u}_j^N)\Delta W_j^N$$

where $\Delta W_j^N := W_{t_{j+1}^N} - W_{t_j^N}$.

Continuous-time interpolation: $k^N(t) := \sup\{j : t_j^N \le t\}$. $X_t^N := \bar{X}_{k^N(t)}^N$ and $u_t^N := \bar{u}_{k^N(t)}^N$.

Homework 96
Show that:

$$X_t^N = x + \int_0^t \mu(X_s^N, u_s^N)\,ds + \int_0^t \sigma(X_s^N, u_s^N)\,dW_s$$

Remark 97. *Under the Lipschitz assumption on μ and σ we have:*

Lemma 98
For every $p \in [1, \infty)$ and $T > 0$, for any $u \in \mathcal{A}$,

$$\mathbb{E}\left[\sup_{0 \le t \le T} \left|\left|X_t^{N,u} - X_t^u\right|\right|_d^p\right] \le c_p \left(\frac{T}{2^N}\right)^{p/2} \quad \text{for large } N,$$

where $c_p > 0$ is a constant independent of N.

Homework 99 (Advanced)
Prove the Lemma above.

Remark 100. *For the statement of the control problem, it is convenient to also introduce $\tilde{u}_t^N := \bar{u}_{k^N(t)-1}^N$.*

8.4 Discrete-time approximation of the objective function

Remark 101.

$$J^N_{n,x}(u) := \mathbb{E}_{t^N_n,x}\left[\frac{T}{2^N}\sum_{j=n}^{2^N} e^{-r(t^N_j - t^N_n)} f(t^N_j, \bar{X}^N_j, \bar{u}^N_{j-1}) + e^{-r(T-t^N_n)} g(\bar{X}^N_{2^N})\right]$$
$$= \mathbb{E}_{t^N_n,x}\left[\int_{t^N_n}^T e^{-r(t^N_{k^N(s)} - t^N_n)} f(t^N_{k^N(s)}, X^N_s, \tilde{u}^N_s)\, ds + e^{-r(T-t^N_n)} g(X^N_T)\right]$$

The class of admissible controls simplifies to the vectors

$$\bar{u}^N := (\bar{u}_0, \bar{u}_1, \bar{u}_2, \dots, \bar{u}_{2^N-1}) \quad \textit{with } \bar{u}_j \in \mathcal{F}_{t^N_j}\ \forall j.$$

8.5 Discrete-time approximation of value function

Definition 102

$$V^N_{N-n}(x) := \sup_{\bar{u}^N} J^N_{n,x}(\bar{u}^N).$$

Remark 103. *If f and g are Lipschitz in all their variables we get*

$$|J^N_{n,x}(\bar{u}^N) - J_{t_n,x}(u)| \le \mathbb{E}_{t_n,x}\left[\int_{t_n}^T |e^{-r(t^N_{k^N(s)} - t^N_n)} f(t^N_{k^N(s)}, X^{N,u}_s, \tilde{u}^N_s) - e^{-r(s-t^N_n)} f(s, X^u_s, u_s)|\, ds\right]$$
$$+ e^{-r(T-t^N_n)}\mathbb{E}\left[|g(X^{N,u}_T) - g(X^u_T)|\right]$$
$$\le L\cdot \mathbb{E}_{t_n,x}\left[\int_{t_n}^T (||X^{N,u}_s - X^u_s||_d + ||\tilde{u}^N_s - u_s||_m)\, ds + ||X^{N,u}_T - X^u_T||_d + c\frac{T}{2^N}\right]$$
$$\le L\cdot T\cdot C'\left[\left(\frac{T}{2^N}\right)^{1/2} + \int_0^T \mathbb{E}[||\tilde{u}^N_{s\vee t_n} - u_{s\vee t_n}||_m]\, ds\right]$$

for some constants $L, c, c' > 0$ independent of N.

Moreover $\forall t \in [0,T]$, we can choose t^N_n s.t. $|t^N_n - t| \le \frac{T}{2^N}$. Then

$$|J_{t,x}(u) - J^N_{t_n,x}(u)| \le L\left(\frac{T}{2^N}\right)^{1/2}.$$

Thus, one can verify that

$$|V(t,x) - V(t_n,x)| + |V(t_n,x) - V^N_{N-n}(x)| \le C_0\left(\left(\frac{T}{2^N}\right)^{1/2} + \int_0^T \mathbb{E}[||\tilde{u}^N_s - u_s||_m]\, ds\right)$$

which means that V^N approximates V.

8.6 DPP: For the discrete-time approximation

Remark 104. *For the discrete-time approximation of the stoch. control problem we can use our previous results and obtain the DPP:*

$$V^N_{N-n}(x) = \sup_{u \in U} \left\{ e^{-r\frac{T}{2^N}} f(t_n, x, u) \frac{T}{2^N} + e^{-r\frac{T}{2^N}} \mathbb{E}_{t_n,x}[V^N_{N-(n+1)}(X^N_{t_{n+1}})] \right\}.$$

Here we take a leap of faith and we argue that DPP holds in the limit as $N \to \infty$!

8.7 DPP in continuous time

Remark 105. *Reference:* **[Yong-Zhou: Stochastic controls]**

Theorem 106

If $V(t,x)$ is continuous on $[0,T] \times \mathbb{R}^d$, then for any open set Γ, letting $\tau_\Gamma := \inf\{s \geq t : (s, X^u_s) \notin \Gamma\} \wedge T$,

$$V(t,x) = \sup_{u \in \mathcal{A}} \mathbb{E}_{t,x}\left[\int_t^{\tau_\Gamma \wedge h} e^{-r(s-t)} f(s, X^u_s, u_s)\, ds + e^{-r(\tau_\Gamma \wedge h - t)} V(\tau_\Gamma \wedge h, X^u_{\tau_\Gamma \wedge h})\right]$$

for any $h \in [t, T]$.

Proof. Quite difficult. □

9 The Hamilton-Jacobi-Bellman equation

Remark 107. *From the DPP equation we derive a PDE for V.* **The derivation is purely heuristic!**

Fix $u = const$. Then

$$V(t,x) \geq \mathbb{E}_{t,x}\left[\int_t^{\tau_\Gamma \wedge h} e^{-r(s-t)} f(s, X^u_s, u)\, ds + e^{-r(\tau_\Gamma \wedge h - t)} V(\tau_\Gamma \wedge h, X^u_{\tau_\Gamma \wedge h})\right]$$

Assume for a moment that V is sufficiently smooth to apply Itô's formula.

$$\implies V(t,x) \geq \mathbb{E}_{t,x}\Bigg[\int_t^{\tau_\Gamma \wedge h} e^{-r(s-t)} \Big(f(s, X^u_s, u) + \partial_t V(s, X^u_s) \\ + \mathcal{L}^u V(s, X^u_s) - rV(s, X^u_s)\Big)\, ds\Bigg] + V(t,x)$$

where

$$\mathcal{L}^u p(x) = \langle \mu(x,u), \nabla p(x) \rangle + \frac{1}{2} \operatorname{tr}\left(\sigma\sigma^\top(x,u) D^2 p(x)\right)$$

and we got rid of the stoch. integral

$$\mathbb{E}_{t,x}\left[\int_t^{\tau_\Gamma \wedge h} e^{-r(s-t)} \partial_x V(s, X^u_s) \sigma(X^u_s, u)\, dW_s\right] = 0.$$

Then

$$\frac{1}{h}\mathbb{E}_{t,x}\left\{\int_t^{\tau_\Gamma \wedge h} e^{-r(s-t)} \left[(\partial_t V + \mathcal{L}^u V - rV)(s, X^u_s) + f(s, X^u_s, u)\right] ds\right\} \leq 0$$

letting $h \downarrow 0$, and assuming that all functions are sufficiently smooth, we get:

$$(\partial_t V + \mathcal{L}^u V - rV)(t,x) + f(t,x,u) \leq 0$$

The result holds for any $u \in U$. Then:

$$\partial_t V(t,x) + \underbrace{\sup_{u\in U}\left(\frac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top(x,u)D^2V(t,x)\right) + \langle \mu(x,u), \nabla V(t,x)\rangle + f(t,x,u)\right)}_{=:\mathcal{H}(t,x,\nabla V, D^2 V)} \le rV(t,x)$$

If an optimal control $(u_t^*)_{t\ge 0}$ exists, then:

$$V(t,x) = \mathbb{E}_{t,x}\left[\int_t^{\tau_\Gamma\wedge h} e^{-r(s-t)} f(s, X_s^{u^*}, u_s^*)\,ds + e^{-r(\tau_\Gamma\wedge h - t)} V(\tau_\Gamma \wedge h, X^{u^*}_{\tau_\Gamma\wedge h})\right]$$

Repeating similar arguments as above (HW) we reach:

$$\begin{cases} \partial_t V(t,x) + \mathcal{H}(t,x,\nabla V, D^2V) = rV(t,x) & ,(t,x)\in[0,T)\times\mathbb{R}^d, \\ V(T,x) = g(x) & ,x\in\mathbb{R}^d. \end{cases}$$

9.1 Verification Theorem

Now we can state a verification theorem.

Theorem 108

Assume there is a "sufficiently smooth" solution $\psi(t,x)$ of the HJB equation:

$$\begin{cases} \partial_t \psi(t,x) + \mathcal{H}(t,x,\nabla \psi, D^2\psi) = r\psi(t,x) & ,(t,x)\in[0,T)\times\mathbb{R}^d, \\ \psi(T,x) = g(x) & ,x\in\mathbb{R}^d. \end{cases}$$

Then $\psi(t,x) \ge V(t,x)$.

Assume further that there exists a measurable function $\alpha^* : [0,T]\times\mathbb{R}^d \to U$ s.t. $\forall (t,x)\in[0,T]\times\mathbb{R}^d$

$$\alpha^*(t,x) = \operatorname*{argmax}_{u\in U}\left\{\frac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top(x,u)D^2\psi\right) + \langle\mu(x,u),\nabla\psi\rangle + f(t,x,u)\right\}$$

and the SDE

$$X_t^* = x + \int_0^t \mu(X_s^*, \alpha^*(s,X_s^*))\,ds + \int_0^t \sigma(X_s^*,\alpha^*(s,X_s^*))\,dW_s$$

is well-posed (i.e., $\exists!$ strong sol.). Then

$$\psi(t,x) = V(t,x) = \mathbb{E}_{t,x}\left[\int_t^T e^{-r(s-t)} f(s,X_s^*,u_s^*)\,ds + e^{-r(T-t)} g(X_T^*)\right]$$

where $u_t^* := \alpha^*(t,X_t^*)$ is an **optimal Markovian control**.

Remark 109. *Proof.* Let $(t,x)\in[0,T]\times\mathbb{R}^d$ be fixed and take an arbitrary admissible control $(u_t)_{t\in[0,T]}\in\mathcal{A}$. By smoothness of ψ:

$$\begin{aligned} &e^{-r(T-t)}\psi(T,X_T^u) + \int_t^T e^{-r(s-t)} f(s,X_s^u,u_s)\,ds \\ &\qquad = \psi(t,x) + \int_t^T e^{-r(s-t)}\left((\partial_t\psi + \mathcal{L}^u\psi - r\psi)(s,X_s^u) + f(s,X_s^u,u_s)\right)\,ds \\ &\qquad\quad + \int_t^T e^{-r(s-t)}\partial_x\psi(s,X_s^u)\sigma(X_s^u,u_s)\,dW_s \end{aligned}$$

Homework 110

Check that the stochastic integral term can be written as $\sum_{i=1}^{d}\sum_{j=1}^{d'}\partial_{x_i}\psi\cdot\sigma_{ij}\,dW_s^j$.

Taking expectation $\mathbb{E}_{t,x}$ on both sides:

$$\begin{aligned}\psi(t,x) = \mathbb{E}_{t,x}\left[e^{-r(T-t)}g(X_T^u) + \int_t^T e^{-r(s-t)}f(s,X_s^u,u_s)\,ds\right] \\ - \mathbb{E}_{t,x}\left[\int_t^T e^{-r(s-t)}\left((\partial_t\psi + \mathcal{L}^u\psi - r\psi)(s,X_s^u) + f(s,X_s^u,u_s)\right)\,ds\right]\end{aligned} \tag{1}$$

Notice that:

$$\begin{aligned}&\frac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top(X_s^u,u_s)D^2\psi(s,X_s^u)\right) + \langle\mu(X_s^u,u_s),\nabla\psi(s,X_s^u)\rangle + f(s,X_s^u,u_s) \\ &\qquad\le \sup_{\beta\in U}\left[\frac{1}{2}\operatorname{tr}\left(\sigma\sigma^\top(X_s^u,\beta)D^2\psi(s,X_s^u)\right) + \langle\mu(X_s^u,\beta),\nabla\psi(s,X_s^u)\rangle + f(s,X_s^u,\beta)\right] \\ &\qquad= \mathcal{H}(s,X_s^u,\nabla\psi,D^2\psi)\end{aligned} \tag{2}$$

Substituting this inequality into (1):

$$\begin{aligned}\psi(t,x) \ge \mathbb{E}_{t,x}\left[\int_t^T e^{-r(s-t)}f(s,X_s^u,u_s)\,ds + e^{-r(T-t)}g(X_T^u)\right] \\ - \mathbb{E}\left[\int_t^T e^{-r(s-t)}\underbrace{\left(\partial_t\psi(s,X_s^u) + \mathcal{H}(s,X_s^u,\nabla\psi,D^2\psi) - r\psi(s,X_s^u)\right)}_{=0}\,ds\right]\end{aligned}$$

Thus, $\psi(t,x) \ge \mathbb{E}_{t,x}\left[\int_t^T e^{-r(s-t)}f(s,X_s^u,u_s)\,ds + e^{-r(T-t)}g(X_T^u)\right]$ for all $u\in\mathcal{A}$, which implies $\psi(t,x)\ge V(t,x)$.

Repeating the argument with $u_s = \alpha^*(s,X_s^*)$, the inequality in Eq. (2) becomes strict equality and Eq. (1) yields:

$$\psi(t,x) = \mathbb{E}_{t,x}\left[\int_t^T e^{-r(s-t)}f(s,X_s^*,\alpha^*(s,X_s^*))\,ds + e^{-r(T-t)}g(X_T^*)\right] \le V(t,x).$$

Then $\psi = V$ and $u_t^* = \alpha^*(t,X_t^*)$ is optimal. □

Homework 111

Formulate the DPP for the problem with payoff:

$$\begin{aligned}J_{t,x}(u) := \mathbb{E}_{t,x}\Bigg[&\int_t^{T\wedge\tau_\mathcal{O}} e^{-r(s-t)}f(s,X_s^u)\,ds \\ &+ e^{-r(\tau_\mathcal{O}-t)}h(\tau_\mathcal{O},X_{\tau_\mathcal{O}}^u)\mathbb{1}_{\{\tau_\mathcal{O}<T\}} \\ &+ e^{-r(T-t)}g(X_T^u)\mathbb{1}_{\{\tau_\mathcal{O}\ge T\}}\Bigg].\end{aligned}$$

Homework 112

Obtain the HJB equation for the above problem.

10 Ito-Tanaka-Meyer formula

11 Issues and Generalisations of Itô's Formula

Remark 113. *There are at least two crucial issues with the use of Verification theorems:*

(i) existence of some kind of solution for difficult nonlinear PDEs.

(ii) sufficient regularity for the application of Itô's formula.

In the second part of this course we will consider some famous applications of the theory of stoch. control in Economics, Finance & Insurance, where explicit solutions are available. However, the vast majority of problems is not amenable to explicit solution and stoch. control is a lively area of mathematical research.

Remark 114 (Reminder)**.** *Given $\psi : [0,T] \times \mathbb{R}^d \to \mathbb{R}$, Itô's formula requires $\psi \in C^{1,2}([0,T] \times \mathbb{R}^d)$. This can be relaxed to $\psi \in C^{1,2}([0,T) \times \mathbb{R}^d) \cap C([0,T] \times \mathbb{R}^d)$ by continuity....*

In optimal stopping **we can never obtain continuity of $D^2\psi = (\partial^2_{x_i x_j}\psi)_{i,j=1\dots d}$.**

Thus, generalisations of Itô's formula are necessary. We will review a famous one, but this is only the tip of the iceberg!

Remark 115. *Reference:* **[[Revuz & Yor: Continuous martingales and Brownian motion]] chapter VI.**

Let f be a convex function.

- *left- and right- derivatives exist at all points: f'_- and f'_+*
- *There are at most countably many points $x \in \mathbb{R}$ s.t. $f'_-(x) \neq f'_+(x)$ (notice that $f'_- \leq f'_+$ by convexity)*
- *$x \mapsto f'_-(x)$ is left-cont., $x \mapsto f'_+(x)$ is right-cont.*
- *f'' exists as a positive measure on $\mathcal{B}(\mathbb{R})$.*
- *On any bounded interval (a,b) the function f can be written as*

  $$f(x) = \alpha x + \beta + \frac{1}{2}\int_a^b |x-y| f''(dy)$$

  for suitable constants $\alpha, \beta \in \mathbb{R}$ depending on the interval.

- *Let $\varphi \in C^\infty((-\infty,0])$ s.t. $\varphi \geq 0$, $\int_{-\infty}^0 \varphi(x)\,dx = 1$ and $\varphi(x) > 0 \iff x \in (-a,0]$ for some $a > 0$ (supp $\varphi = (-a,0]$). Then $\varphi_n(x) := n\varphi(nx)$ defines a sequence of mollifiers with $\int \varphi_n = 1$ and support supp $\varphi_n = (-\frac{a}{n}, 0]$.*

  Note that $\int_{-\infty}^0 n\varphi(nx)\,dx = \int_{-\infty}^0 \varphi(y)\,dy = 1$.

- *Set*

  $$f_n(x) := \int_{-\infty}^0 f(x+y)\varphi_n(y)\,dy = \int_{-\infty}^x f(z)\varphi_n(z-x)\,dz$$

Remark 116. *We have the following properties for the mollified functions f_n:*

$$|f_n(x) - f(x)| = \left| \int_{-\infty}^{0} (f(x+y) - f(x))\varphi_n(y)\, dy \right| \leq ||f(x+\cdot) - f(x)||_{L^\infty((-\frac{a}{n},0))} \xrightarrow{n\to\infty} 0.$$

Moreover, $f_n'(x) = f(x)\varphi_n(0) - \int_{-\infty}^{x} f(z)\varphi_n'(z-x)\, dz$. Since $f_-' = f_+'$ a.e., we have:

$$f_n'(x) = \int_{-\infty}^{x} f_-'(z)\varphi_n(z-x)\, dz \leq f_-'(x) \quad \text{(by convexity)}.$$

We can also write:

$$\begin{aligned} f_n'(x) &= \int_{-\infty}^{0} f_-'(x+y)\varphi_n(y)\, dy = \int_{-\frac{a}{n}}^{0} f_-'(x+y)\varphi_n(y)\, dy \\ &= \int_{-\frac{a}{n}}^{0} f_-'(x+y) n\varphi(ny)\, dy. \end{aligned}$$

Using the substitution $ny = z$, $dz = n\, dy$:

$$\begin{aligned} f_n'(x) &= \int_{-a}^{0} f_-'(x+\frac{z}{n})\varphi(z)\, dz \leq \int_{-a}^{0} f_-'(x+\frac{z}{n+1})\varphi(z)\, dz \quad \text{(by convexity)} \\ &= \int_{-\frac{a}{n+1}}^{0} f_-'(x+y)\varphi_{n+1}(y)\, dy = f_{n+1}'(x). \end{aligned}$$

Thus, $f_n' \leq f_{n+1}'$ and $f_n' \uparrow f_-'$ pointwise. Finally, $f_n'(x) = \int_{-\infty}^{x} f_-'(z)\varphi_n(z-x)\, dz$ and therefore:

$$\begin{aligned} f_n''(x) &= f_-'(x)\varphi_n(0) - \int_{-\infty}^{x} f_-'(z)\varphi_n'(z-x)\, dz \\ &= \int_{-\infty}^{x} \varphi_n(z-x) f''(dz) \geq 0 \end{aligned}$$

because $f''(dz)$ is a positive measure and $\varphi_n \geq 0$.

Theorem 117

If X is a continuous semi-martingale, there is a continuous non-decreasing process A^f s.t.

$$f(X_t) = f(X_0) + \int_0^t f_-'(X_s)\, dX_s + \frac{1}{2}A_t^f.$$

Proof. From Itô's formula:

$$f_n(X_t) = f_n(X_0) + \int_0^t f_n'(X_s)\, dX_s + \frac{1}{2}\underbrace{\int_0^t f_n''(X_s)\, d\langle X\rangle_s}_{=:A_t^{f_n}}.$$

Note that $A_t^{f_n}$ is non-decreasing and continuous since $f_n'' \geq 0$ and $d\langle X\rangle_s \geq 0$. Letting $n \to \infty$, we have $f_n(X_t) \to f(X_t)$, $f_n(X_0) \to f(X_0)$ and:

$$\begin{aligned} \lim_{n\to\infty} \mathbb{E}\left[\left| \int_0^t f_n'(X_s)\, dX_s - \int_0^t f_-'(X_s)\, dX_s \right|^2 \right] &= \lim_{n\to\infty} \mathbb{E}\left[\int_0^t |f_n'(X_s) - f_-'(X_s)|\, d\langle X\rangle_s \right] \\ &= \mathbb{E}\left[\int_0^t \lim_{n\to\infty} |f_n'(X_s) - f_-'(X_s)|\, d\langle X\rangle_s \right] = 0. \end{aligned}$$

The use of the Dominated Convergence Theorem (**DOM**) is justified because it suffices to consider X evolving on a compact set. Then, up to selecting a subsequence, we have $\mathbb{P}$-a.s.:

$$\begin{aligned}
\lim_{n\to\infty} \frac{1}{2} A_t^{f_n} &= \lim_{n\to\infty} \left(f_n(X_t) - f_n(X_0) - \int_0^t f_n'(X_s)\, dX_s \right) \\
&= f(X_t) - f(X_0) - \int_0^t f_-'(X_s)\, dX_s =: \frac{1}{2} A_t^f.
\end{aligned}$$

The process $f(X_t) - f(X_0) - \int_0^t f_-'(X_s)\, dX_s$ is continuous. Furthermore, $t \mapsto A_t^f$ is continuous and also non-decreasing as the limit of non-decreasing processes. □

12 Local Time and Tanaka's Formula

Remark 118. *We now want to characterise the process A^f in some special case. This leads to a notion of* **local time** *which will allow us to state the Itô-Tanaka-Meyer formula.*

Let us define $\operatorname{sign}(x) = \begin{cases} 1, & x > 0 \\ -1, & x \le 0 \end{cases}$ *so that if $f(x) = |x|$ then $f_-'(x) = \operatorname{sign}(x)$.*

Theorem 119 (Tanaka formula)

For any $a \in \mathbb{R}$, there exists a non-decreasing, continuous process $(L_t^a)_{t\ge 0}$, called the **local-time** of X at a, such that:

- $|X_t - a| = |X_0 - a| + \int_0^t \operatorname{sign}(X_s - a)\, dX_s + L_t^a$
- $(X_t - a)^+ = (X_0 - a)^+ + \int_0^t \nVdash_{\{X_s > a\}}\, dX_s + \frac{1}{2} L_t^a$
- $(X_t - a)^- = (X_0 - a)^- - \int_0^t \nVdash_{\{X_s \le a\}}\, dX_s + \frac{1}{2} L_t^a$

where $(y)^- = \max\{0, -y\}$. This shows that $|X_t - a|, (X_t - a)^+, (X_t - a)^-$ are semimartingales.

Proof. All functions considered are convex. Then, there exist continuous non-decreasing processes A^+ and A^- s.t.

$$\begin{aligned}
(X_t - a)^+ &= (X_0 - a)^+ + \int_0^t \nVdash_{\{X_s > a\}}\, dX_s + \frac{1}{2} A_t^+ \\
(X_t - a)^- &= (X_0 - a)^- - \int_0^t \nVdash_{\{X_s \le a\}}\, dX_s + \frac{1}{2} A_t^-
\end{aligned}$$

Then:

$$\begin{aligned}
X_t - a &= (X_t - a)^+ - (X_t - a)^- \\
&= X_0 - a + \int_0^t dX_s + \frac{1}{2}(A_t^+ - A_t^-)
\end{aligned}$$

Since $X_t - a = X_0 - a + \int_0^t dX_s$, then it must be $A_t^+ = A_t^-$ $\forall t \ge 0$ $\mathbb{P}$-a.s. We call $A_t^+ = A_t^- = L_t^a$. Moreover, $|X_t - a| = (X_t - a)^+ + (X_t - a)^-$

$$= |X_0 - a| + \int_0^t \operatorname{sign}(X_s - a)\, dX_s + L_t^a.$$

□

Remark 120. *$t \mapsto L_t^a(\omega)$ defines a non-negative measure on $[0,\infty)$. The next proposition gives us information on its support.*

Proposition 121

$dL_t^a = \mathbb{1}_{\{X_t = a\}}\, dL_t^a$, $\forall t \geq 0$, $\mathbb{P}$-a.s.

Proof. We calculate $d(X_t - a)^2$ in two ways and then equate the resulting expressions. By Itô's formula:

$$\begin{aligned} d(X_t - a)^2 &= 2(X_t - a)\, d(X_t - a) + d\langle X - a\rangle_t \\ &= 2(X_t - a)\, dX_t + d\langle X\rangle_t \end{aligned} \tag{4}$$

By Tanaka's formula:

$$d(|X_t - a|)^2 = 2|X_t - a|\, d|X_t - a| + d\langle |X - a|\rangle_t \tag{5}$$

Since $d|X_t - a| = \operatorname{sign}(X_t - a)\, dX_t + dL_t^a$, then $d\langle |X - a|\rangle_t = (\operatorname{sign}(X_t - a))^2\, d\langle X\rangle_t = d\langle X\rangle_t$. Then (5) reads:

$$d(|X_t - a|)^2 = 2\underbrace{|X_t - a| \operatorname{sign}(X_t - a)}_{=(X_t - a)}\, dX_t + 2|X_t - a|\, dL_t^a + d\langle X\rangle_t \tag{6}$$

Comparing (4) and (6) we get: $\int_0^t |X_s - a|\, dL_s^a = 0$ $\forall t \geq 0$, $\mathbb{P}$-a.s., which implies that L_s^a increases only when $X_s = a$. □

Fact 122

$(t, \omega, a) \mapsto L_t^a(\omega)$ is jointly measurable on $\mathcal{P}(\mathcal{F}_t) \otimes \mathcal{B}(\mathbb{R})$, where $\mathcal{P}(\mathcal{F}_t)$ is the σ-algebra on $\mathbb{R}_+ \times \Omega$ generated by continuous adapted processes.

Theorem 123 (Itô-Tanaka-Meyer Formula)

If $f : \mathbb{R} \to \mathbb{R}$ is the difference of two convex functions and X a continuous semi-martingale, then

$$f(X_t) = f(X_0) + \int_0^t f'_-(X_s)\, dX_s + \frac{1}{2}\int_{\mathbb{R}} L_t^a f''(da).$$

Remark 124. *This requires measurability of $a \mapsto L_t^a$.*

Remark 125. *Hence $f(X_t)$ is a semimartingale.*

Proof. Let $\tau_n := \inf\{t \geq 0 : X_t \notin (-n, n)\}$. By continuity of $t \mapsto X_t(\omega)$ we have $\tau_n(\omega) \leq \tau_{n+1}(\omega)$, $\forall n \in \mathbb{N}$, $\forall \omega \in \Omega$ and $\lim_{n\to\infty} \tau_n(\omega) = +\infty$.

Since $X_{t\wedge\tau_n} \in [-n, n]$ then we consider $f(x)$ restricted to $(-n, n)$ and we can w.l.o.g. assume that f has compact support outside of $(-n, n)$. Then

$$f(x) = \alpha_n x + \beta_n + \frac{1}{2}\int_{\mathbb{R}} |x - a| f''(da)$$

Applying this to the process:

$$f(X_{t\wedge\tau_n}) = \alpha_n X_{t\wedge\tau_n} + \beta_n + \frac{1}{2}\int_{\mathbb{R}} |X_{t\wedge\tau_n} - a| f''(da)$$

By Tanaka's formula, $|X_{t\wedge\tau_n} - a| = |X_0 - a| + \int_0^{t\wedge\tau_n} \operatorname{sign}(X_s - a)\, dX_s + L^a_{t\wedge\tau_n}$. Substituting this in:

$$\begin{aligned} f(X_{t\wedge\tau_n}) &= \alpha_n(X_{t\wedge\tau_n} - X_0) + \underbrace{\left(\alpha_n X_0 + \beta_n + \frac{1}{2}\int_{\mathbb{R}} |X_0 - a| f''(da)\right)}_{=f(X_0)} \\ &\quad + \frac{1}{2}\int_{\mathbb{R}}\left(\int_0^{t\wedge\tau_n} \operatorname{sign}(X_s - a)\, dX_s\right) f''(da) + \frac{1}{2}\int_{\mathbb{R}} L^a_{t\wedge\tau_n} f''(da) \\ &= f(X_0) + \alpha_n(X_{t\wedge\tau_n} - X_0) + \frac{1}{2}\int_0^{t\wedge\tau_n}\left(\int_{\mathbb{R}} \operatorname{sign}(X_s - a) f''(da)\right) dX_s + \frac{1}{2}\int_{\mathbb{R}} L^a_{t\wedge\tau_n} f''(da) \end{aligned}$$

The inner integral evaluates to:

$$\begin{aligned} \int_{\mathbb{R}} \operatorname{sign}(X_s - a) f''(da) &= \int_{-\infty}^{X_s} f''(da) - \int_{X_s}^{\infty} f''(da) \\ &= (f'_-(X_s) - \alpha) - (\alpha - f'_-(X_s)) = 2(f'_-(X_s) - \alpha) \end{aligned}$$

where α is the slope at infinity. Thus the stochastic integral term is:

$$\int_0^{t\wedge\tau_n} (f'_-(X_s) - \alpha)\, dX_s = \int_0^{t\wedge\tau_n} f'_-(X_s)\, dX_s - \alpha(X_{t\wedge\tau_n} - X_0)$$

Hence:

$$f(X_{t\wedge\tau_n}) = f(X_0) + \int_0^{t\wedge\tau_n} f'_-(X_s)\, dX_s + \frac{1}{2}\int_{\mathbb{R}} L^a_{t\wedge\tau_n} f''(da) \tag{1}$$

For each $\omega \in \Omega$, $\exists n_\omega \in \mathbb{N}$ large enough so that $X_s(\omega) \in (-n_\omega, n_\omega)$ $\forall s \geq 0$. Thus, letting $n \to \infty$ in (1) we obtain the claim ω by ω.

Homework 126

Fill the small gap!

□

Corollary 127 (Occupation time formula)

For any non-negative Borel function Φ,

$$\int_0^t \Phi(X_s)\, d\langle X\rangle_s = \int_{\mathbb{R}} \Phi(a) L^a_t\, da \quad \forall t \geq 0, \mathbb{P}\text{-a.s.} \tag{1}$$

(Φ may be bounded Borel).

Remark 128. *The result presented in the book is stronger: the null set does not depend on Φ.*

Proof. Let $f \in C^2(\mathbb{R})$ be s.t. $f'' = \Phi$. Then comparing Itô formula:

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\, dX_s + \frac{1}{2}\int_0^t \Phi(X_s)\, d\langle X\rangle_s$$

and Itô-Tanaka-Meyer formula:

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\, dX_s + \frac{1}{2}\int_{\mathbb{R}} L_t^a \Phi(a)\, da$$

we deduce $\int_0^t \Phi(X_s)\, d\langle X\rangle_s = \int_{\mathbb{R}} L_t^a \Phi(a)\, da$ for $\Phi \in C(\mathbb{R})$.

Let $\mathcal{H} := \{\Phi \in m\mathcal{B}(\mathbb{R}) : \Phi \text{ bounded s.t. (1) holds}\}$. $\mathcal{H}$ is a vector space, it contains $\Phi(a) = 1$ and it's closed under monotone limits by MON. Moreover $\Phi(a) = \mathbb{1}_{(-\infty,\lambda]}(a)$ can be approximated from below with continuous functions that converge pointwise $\implies \mathbb{1}_{(-\infty,\lambda]}(\cdot) \in \mathcal{H}$.

Since $\{(-\infty, \lambda], \lambda \in \mathbb{R}\}$ is a π-system that generates $\mathcal{B}(\mathbb{R})$ we can apply monotone class theorem to conclude that $\mathcal{H} \supset m\mathcal{B}(\mathbb{R})$. □

12.1 Some special cases of interest

Remark 129.
- *If $f \in W^{2,\infty}_{loc}(\mathbb{R})$, i.e., f' and f'' exist as functions that are locally bounded, then f' is absolutely continuous and by* **ITM formula***:*

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t f'(X_s)\, dX_s + \frac{1}{2}\int_{\mathbb{R}} L_t^a f''(a)\, da \\ &= f(X_0) + \int_0^t f'(X_s)\, dX_s + \frac{1}{2}\int_0^t f''(X_s)\, d\langle X\rangle_s \end{aligned}$$

  This is the formula we use in optimal stopping and stoch. control in 1-D.

- *If $f \in C(\mathbb{R})$, $f \in C^2(-\infty, 0)$ and $f \in C^2(0, \infty)$ with $f'(0-) \neq f'(0+)$ both finite and f'' loc. bounded on $\mathbb{R}$, then*

$$f''(da) = f''(a)\mathbb{1}_{(-\infty,0)}(a)\, da + f''(a)\mathbb{1}_{(0,\infty)}(a)\, da + [f'(0+) - f'(0-)]\delta_0(da)$$

  By ITM formula:

$$\begin{aligned} f(X_t) &= f(X_0) + \int_0^t f'_-(X_s)\, dX_s + \frac{1}{2}\int \mathbb{1}_{\{a\neq 0\}} L_t^a f''(a)\, da + \frac{1}{2}[f'(0+) - f'(0-)]L_t^0 \\ &= f(X_0) + \int_0^t f'_-(X_s)\, dX_s + \frac{1}{2}\int_0^t f''(X_s)\mathbb{1}_{\{X_s\neq 0\}}\, d\langle X\rangle_s + \frac{1}{2}\underbrace{[f'(0+) - f'(0-)]L_t^0}_{\text{this is precisely the effect of a lack of smooth-fit!}} \end{aligned}$$

13 American Option Pricing

13.1 Options

Remark 130. *Options are financial instruments whose payoff depends on an **underlying asset**. More generally, they are **contingent claims**.*

Definition 131

Consider a financial market with vector price process S. A **contingent claim** with maturity T is any stochastic variable $\mathcal{X} \in \mathcal{F}_T^S$. It is called **simple** if $\mathcal{X} = g(S(T))$.

The function g is the **contract function**. At time T we will know the payoff.

- **European options**: These are examples of simple contingent claims, with contract functions:
  - **Call option**: $g(x) = \max[x - K, 0] = (x - K)^+$
  - **Put option**: $g(x) = \max[K - x, 0] = (K - x)^+$
- **American options**: The option can be exercised prior to time T. The choice of the time of exercise is left to the holder of the option.

Remark 132. *At time t, if the option is exercised, the payoff is $g(S_t)$. The decision to exercise at time t can be made based on the information available up to that time, $\mathcal{F}_t^S$. From the point of view of $t = 0$, the exercise time will be a random variable τ, which will be a **stopping time**.*

*The central problem with options is to determine the **fair price** of the claim $\pi(t, g)$ for $t \leq T$.*

If $t = T$, we know $\pi(T, g) = g(S(T))$ for a simple claim. But what if $t < T$?

*Let us review pricing of contingent claims, taking a discrete-time approach first, which naturally evokes the use of **backward induction** methods.*

In the first part of the course, you learned how to address:

$$V_* = \sup_{\tau} \mathbb{E}[G_\tau].$$

How is this connected to American options pricing?

13.1.1 Preliminaries: 1-period model

Remark 133. *Key concepts: $\mathbb{P}$ measure, replicating portfolios, martingale measure, arbitrage pricing.*

*Consider the simplest set-up: a **1-period model**.*

- *S_0: current price of a stock.*
- *K: strike price.*

- $T = 1$ *period.*
- r: *risk-free interest rate.*
- $\Omega = \{\omega_1, \omega_2\}$.

The stock price at maturity is:

$$S_T(\omega) = \begin{cases} S^u & \omega = \omega_1 \\ S^d & \omega = \omega_2 \end{cases}$$

The payoff of a call option at maturity is $C_T = (S_T - K)^+$. *Specifically:*

$$C_T(\omega) = \begin{cases} S^u - K & \omega = \omega_1 \\ 0 & \omega = \omega_2 \end{cases}$$

(assuming $S^u > K$ *and* $S^d < K$*).*

What is the fair price C_0*? It is rational to define a "fair" price as the* **expected discounted (present) value** *of the terminal payoff of the option:*

$$\mathbb{E}\left[(1+r)^{-1}C_T\right].$$

This requires an evaluation of the probability that $\omega = \omega_1$ *and* $\omega = \omega_2$. *This leads to the* **choice of the measure**!

Remark 134. *One possible (natural) choice for pricing is to use the measure that reflects the investor's assessment of probabilities: the* **subjective probability** $\mathbb{P}$ *measure (actual, real-world). However, different investors have different views, which would lead to different prices* C_0. *We need a different measure, in a model that guarantees a unique price.*

Remark 135 (Replicating Portfolio)**.** *The idea is to construct a* **synthetic portfolio** *(a mix of securities) such that it replicates the payoff of the contingent claim at time* T. *Consider two assets:*

- *The stock on which the option is written.*
- *A bank account.*

Let $\phi_0 = (\alpha_0, \beta_0)$ *be the portfolio at time* 0, *and* $V_t(\phi)$ *be the value of the portfolio at time* t. *Then:*

$$V_0(\phi) = \alpha_0 S_0 + \beta_0, \quad V_T(\phi) = \alpha_0 S_T + (1+r)\beta_0.$$

We say ϕ *is a* **replicating portfolio** *for* C *if* $V_T(\phi) = C_T$. *In our 2-state model, this means:*

$$V_T(\phi) = \begin{cases} V^u(\phi) = \alpha_0 S^u + (1+r)\beta_0 = C^u & \text{if } \omega = \omega_1 \\ V^d(\phi) = \alpha_0 S^d + (1+r)\beta_0 = C^d & \text{if } \omega = \omega_2 \end{cases}$$

In this setup, it is possible to find the portfolio (α_0, β_0) *constructed at time* 0 *s.t. the payoff of the option is replicated. Then, the cost of the contingent claim can be defined as the investment needed to construct the replicating portfolio:*

$$C_0 = V_0(\phi) = \alpha_0 S_0 + \beta_0.$$

Notice that this price depends only on S_0 *and is independent of the subjective probabilities of who prices!*

Remark 136 (No Arbitrage Pricing)**.** *Given the replicating portfolio, we consider the following strategy:*

- *At $t = 0$:*
  - *sell the contingent claim: $+C_0$*
  - *buy α_0 stocks: $-\alpha_0 S_0$*
  - *lend an amount β_0: $-\beta_0$*
- *At $t = T$:*
  - *payoff of the contingent claim: $-C_T$*
  - *sell the stocks: $+\alpha_0 S_T$*
  - *receive the loan back: $\beta_0(1 + r)$*

We know that $\alpha_0 S_T + \beta_0(1 + r)$ is always equal to C_T. If $C_0 \neq \alpha_0 S_0 + \beta_0$, then it would be possible to construct an **arbitrage***, i.e., a strategy that yields a sure profit at time $t = 0$. If the model is* **arbitrage-free***, then C_0 can be considered as the fair price of the claim.*

Remark 137 (Martingale Measure)**.** *It is convenient to try to obtain the value of a contingent claim as the* **expected discounted present value** *of its cash flows:*

$$C_0 = \mathbb{E}\left[(1 + r)^{-1} C_T\right] \quad \textit{under which measure?}$$

The idea is to find a probability measure $\mathbb{P}^*$ equivalent to $\mathbb{P}$ s.t. the discounted price process is a **martingale***. For the stock, let $S_0^* = S_0$ and $S_T^* = (1 + r)^{-1} S_T$. The martingale property requires:*

$$S_0 = S_0^* = \mathbb{E}_{\mathbb{P}^*}[S_T^*] = \mathbb{E}_{\mathbb{P}^*}[(1 + r)^{-1} S_T].$$

In our 2-state model, this is:

$$S_0 = (1 + r)^{-1}[S^u p^* + S^d(1 - p^*)], \quad \textit{where } p^* = \mathbb{P}^*\{\omega = \omega_1\}.$$

Solving this equation for p^*:

$$S_0(1 + r) = S^u p^* + S^d - S^d p^* \implies p^* = \frac{S_0(1 + r) - S^d}{S^u - S^d}.$$

Now we ask: is $C_0^* = \mathbb{E}_{\mathbb{P}^*}[(1 + r)^{-1} C_T] = C_0$? From the replicating portfolio equations, we can find:

$$\alpha_0 = \frac{C^u - C^d}{S^u - S^d}, \quad \beta_0 = \frac{C^d S^u - C^u S^d}{(S^u - S^d)(1 + r)}.$$

Then:

$$\begin{aligned}
C_0^* &= (1+r)^{-1}[C^u p^* + C^d(1-p^*)] \\
&= (1+r)^{-1}\left[C^u \frac{S_0(1+r)-S^d}{S^u-S^d} + C^d\left(1-\frac{S_0(1+r)-S^d}{S^u-S^d}\right)\right] \\
&= (1+r)^{-1}\left[\frac{S_0(1+r)}{S^u-S^d}(C^u-C^d) + \frac{C^dS^u-C^uS^d}{S^u-S^d}\right] \\
&= S_0\frac{C^u-C^d}{S^u-S^d} + (1+r)^{-1}\frac{C^dS^u-C^uS^d}{S^u-S^d} \\
&= S_0\alpha_0 + \beta_0 = C_0.
\end{aligned}$$

We have now established the link between C_0 and C_0^*. However, in order to exclude arbitrage, we need to further assume that C_0 is always non-negative.

This transcription follows the style and structure of your provided preamble and previous chapter, using the defined environments and commands.

14 Arbitrage-Free Pricing and American Options

Definition 138

A pricing model M is **arbitrage-free** if there exists no portfolio Φ such that:

$$V_0(\Phi) = 0, \quad V_T(\Phi) \geq 0, \quad \mathbb{P}[V_T(\Phi) > 0] > 0$$

or if there is no portfolio such that:

$$V_0(\Phi) < 0, \quad V_T(\Phi) \geq 0 \implies \textbf{Strong Arbitrage}$$

If the market is arbitrage-free, then C_0^* is the **no-arbitrage price**.

Proposition 139

To conclude this recap, consider $S_T = \begin{cases} S^u & \omega = \omega_1 \\ S^d & \omega = \omega_2 \end{cases}$ and $B_0 = 1, B_T = 1 + r$. Let Φ be the linear space of portfolios of S and B.

The market $M = (S, B, \Phi)$ is arbitrage-free if and only if S^* admits a **martingale measure** $\mathbb{P}^*$. Then, the arbitrage-price of any contingent claim C_T that settles at T is given by:

$$C_0 = \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}C_T\right]$$

Remark 140. *Let us now consider* **American options** *in this setup. Since there are only 2 dates, the option can be exercised either at* 0 *or at* 1.

How to price an American claim? We want to find $\pi_0(X^a)$.

- **At maturity:** $X_1^a = X_T \implies$ *equivalent to a European claim.*

- **At time 0:** *The holder has the option to exercise. The decision is optimally (rationally) taken by comparing the immediate payoff to the expected payoff obtained at maturity.*

Proposition 141

Price of an American Call. Assume $S^d < K < S^u$. Assume $r \geq 0$ and $S^d < S_0(1+r) < S^u$. Then $C_0^a = C_0^e$.

Proof. Assume the contrary, i.e., $C_0^a > C_0$ at first. First, notice that:

$$\begin{aligned} C_0 &= \mathbb{E}^*\left[(1+r)^{-1}(S_T - K)^+\right] \\ &= (1+r)^{-1}p^*[S^u - K] \\ &= \frac{(1+r)S_0 - S^d}{S^u - S^d} \cdot \frac{S^u - K}{(1+r)} \end{aligned}$$

Notice that:

$$\frac{(1+r)S_0 - S^d}{S^u - S^d} \cdot \frac{S^u - K}{(1+r)} = \left(S_0 - \frac{S^d}{1+r}\right) \cdot \frac{S^u - K}{S^u - S^d}$$

Since $K > S^d$, we have $S^u - K < S^u - S^d$, which implies $\frac{S^u - K}{S^u - S^d} < 1$. However, we also observe:

$$S_0 - \frac{S^d}{1+r} > S_0 - K \quad \text{since } K > S^d \text{ and } r \geq 0$$

This leads to the conclusion that $C_0 > S_0 - K$.

Therefore, if $C_0^a > C_0$, we can sell the American call option at C_0^a and buy the European call option at C_0. □

Remark 142. *The arbitrage strategy can be summarized by the following payoff table:*

Action	**Time 0**	**If ex. at 0**	**Time 1 (if not ex.)**
Sell American	$+C_0^a$	$-(S_0 - K)$	$-(S_T - K)^+$
Buy European	$-C_0$	0	$(S_T - K)^+$
Net	$C_0^a - C_0 > 0$	$C_0^a - (S_0 - K)$	*matched*

- **At 0 if exercised:** *The profit is $C_0^a - (S_0 - K)$. Since $C_0^a \geq C_0$ and we showed $C_0 > S_0 - K$, this is strictly positive.*

- **If not exercised:** *We buy C_0 and put $C_0^a - C_0 > 0$ in the bank at time 0. At time 1, the payoffs are matched, leaving the initial profit.*

Homework 143

Describe the arbitrage when $C_0^a < C_0$.

This transcription follows the style and structure of your provided preamble and previous chapter, using the defined environments and commands.

15 American Put Pricing

Proposition 144

American Put Price. Assume $r > 0$, $S^d < S_0(1+r) < S^u$, and $S^d < K < S^u$. Then, $P_0^a = P_0$ if and only if

$$K - S_0 \leq \frac{S^u - (1+r)S_0}{S^u - S^d} \cdot \frac{K - S^d}{1+r} \tag{$*$}$$

Otherwise, $P_0^a = K - S_0 > P_0$. If $r = 0$, then $P_0^a = P_0$.

Proof. Notice that

$$\begin{aligned} P_0 &= \mathbb{E}^*\left[(1+r)^{-1}(K - S_T)^+\right] \\ &= \frac{1}{1+r}(1-p^*)(K - S^d) \\ &= \frac{K - S^d}{1+r} \cdot \frac{S^u - (1+r)S_0}{S^u - S^d} \end{aligned}$$

and indeed $(*)$ is equivalent to $P_0 \geq K - S_0$.

Suppose first the inequality holds. If $P_0^a > P_0$, one can create an arbitrage by selling the American put and buying the European put.

Homework 145

Write the payoffs of the strategy described above.

Hence, $P_0 = P_0^a$. Now suppose $(*)$ does not hold, which implies $P_0 < K - S_0$.

Assume $P_0^a \neq K - S_0$. We want to prove by contradiction that, if this is the case, arbitrage opportunities exist.

- If $P_0^a > K - S_0$, we can sell the American put and buy the European put at a lower price since $P_0 < K - S_0$.
- If $P_0^a < K - S_0$, we can buy the American put and exercise it immediately, yielding a profit of $(K - S_0) - P_0^a > 0$. This implies $P_0^a = K - S_0$, which is a contradiction.

Also, if $P_0 < K - S_0$, it must be that $P_0^a = K - S_0 > P_0$.

If the holder of the put fails to exercise it at time 0, the option writer is still able to lock in a profit. Thus, if $P_0 < K - S_0$, either P_0^a is exercised immediately or arbitrage opportunities arise.

If $r = 0$, the condition $K - S_0 \leq \frac{S^u - S_0}{S^u - S^d}(K - S^d)$ is always valid. $\square$

Remark 146. *Notice that the relation $C_0 = C_0^a$ is more general. Consider S_T to be a $\mathbb{P}^*$-integrable random variable. Then if $r \geq 0$, we have by* **Jensen's inequality***:*

$$\begin{aligned} \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}(S_T - K)^+\right] &\geq \left(\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}S_T - (1+r)^{-1}K\right]\right)^+ \\ &= \left(S_0 - (1+r)^{-1}K\right)^+ \geq S_0 - K \end{aligned}$$

and then $C_0 = C_0^a$ even in this setup.

For the put option, we have:

$$\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}(K-S_T)^+\right] \geq \left(\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}K-(1+r)^{-1}S_T\right]\right)^+$$
$$= \left[(1+r)^{-1}K-S_0\right]^+$$

Note that $\left[(1+r)^{-1}K-S_0\right]^+ > K-S_0$ *if* $-1<r<0$.

If $r=0$*, then the lower bound is* $K-S_0$.

If $r>0$*, no obvious relationship between* P_0 *and* S_0-K *exists.*

Remark 147. *Let us now focus on the* **"rational" exercise rule** *(which extends to the case of a discrete multi-period setup or to a continuous time setup).*

At any t before expiration, it is necessary to find the maximal possible expected payoff over all admissible exercise rules (under the **risk-neutral measure***!) and compare it to the payoff received by exercising immediately.*

Assumption: *exercise is executed at the dates for which subjective probabilities = risk-neutral ones.*

Proposition 148

The arbitrage prices of an American call and an American put in an arbitrage-free market with $t=\{0,1\}$ are:

$$C_0^a = \max_{\tau\in\mathcal{T}}\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-\tau}(S_\tau-K)^+\right] = \max\left((S_0-K)^+, \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}(S_1-K)^+\right]\right)$$
$$P_0^a = \max_{\tau\in\mathcal{T}}\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-\tau}(K-S_\tau)^+\right] = \max\left((K-S_0)^+, \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}(K-S_1)^+\right]\right)$$

where $\mathcal{T}$ is the set of exercise times, $\mathcal{T}=\{0,1\}$.

In general, if $X^a=(X_0,X_T)$ is an American contingent claim, its arbitrage price $\pi(X^a)$ is:

$$\pi(X^a) = \max_{\tau\in\mathcal{T}}\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-\tau}X_\tau\right]$$

where $\pi(X_T^a)=X_T$ (cf. problem of Ch. 3, 1st part of the course!).

16 Binomial Pricing

Remark 149. *Let us consider a discrete-time setup to see the* **backward induction** *method at play and learn about a methodology to price contingent claims,* **Binomial Pricing** *(Cox, Ross, Rubinstein 1979).*

- **Dates:** $0,1,\dots,T^*$
- **Assets:**
  - $S\to$ *stock:* $\xi_{t+1}=\frac{S_{t+1}}{S_t}\in\{u,d\}\ \forall t,\ 0<d<u.$
    $S_0>0$; $\mathbb{P}(S_{t+1}=uS_t|\mathcal{F}_t)>0$; $\mathbb{P}(S_{t+1}=dS_t|\mathcal{F}_t)>0$.
  - $B\to$ *bank account:* $B_0=1, B_t=(1+r)^t$.

Assume $\mathbb{P}(\xi_t=u)=p=1-\mathbb{P}(\xi_t=d)$, $\forall t$. *For no arbitrage, we require* $0<d<1+r<u$.

The stock price at time t is $S_t = S_0 \prod_{j=1}^{t} \xi_j$. Let $\mu_j = \log \xi_j$ be the **log-returns**, *which are i.i.d.*

Remark 150. *We now focus on contingent claim pricing in this model. Let us consider a simple claim first, with maturity $T \leq T^*$, $X = g(S_T)$ (path independent).*

To determine the arbitrage price of X at any $t \in \mathcal{T}$, we show that it is possible, at any t, to consider a portfolio $\phi_t = (\alpha_t, \beta_t)$ of positions in S and B that replicates the payoff of X at time T.

The portfolio ϕ_t is a **dynamic, replicating, self-financing portfolio (strategy)**.

Backward induction can be readily used, since X_T is known. We start by considering $[T-1, T]$ and constructing the replicating portfolio at $T-1$. $\phi_{T-1} = (\alpha_{T-1}, \beta_{T-1})$ must be s.t.

$$V_T(\phi) = \alpha_{T-1} S_T + \beta_{T-1}(1+r)$$

replicates the payoff of $X_T = g(S_T) = \begin{cases} g(S_{T-1} \cdot u) & \text{if } \xi_T = u \\ g(S_{T-1} \cdot d) & \text{if } \xi_T = d \end{cases}$

Hence:

$$\begin{cases} \alpha_{T-1} u S_{T-1} + \beta_{T-1}(1+r) = g(S_{T-1} \cdot u) \\ \alpha_{T-1} d S_{T-1} + \beta_{T-1}(1+r) = g(S_{T-1} \cdot d) \end{cases}$$

Homework 151

Show that solving the system above yields:

$$\alpha_{T-1} = \frac{g(S_{T-1}u) - g(S_{T-1}d)}{uS_{T-1} - dS_{T-1}}$$

$$\beta_{T-1} = (1+r)^{-1} \frac{g(S_{T-1}d)uS_{T-1} - g(S_{T-1}u)dS_{T-1}}{uS_{T-1} - dS_{T-1}}$$

Note that $\alpha_{T-1}, \beta_{T-1}$ are functions of S_{T-1}.

Also, show that the value of the portfolio at $T-1$ is:

$$V_{T-1}(\phi) = \alpha_{T-1} S_{T-1} + \beta_{T-1} = (1+r)^{-1} \left[p^* g(S_{T-1}u) + (1-p^*) g(S_{T-1}d) \right]$$

where $p^* = \frac{(1+r)-d}{u-d}$ is the **risk-neutral probability**. Note that $p^* \in (0,1)$ requires $d < 1+r < u$.

Remark 152. *Assuming absence of arbitrage (and following the reasoning of the 1-period model), $V_{T-1}(\phi)$ is the value of the contingent claim X at time $T-1$:*

$$X_{T-1} = V_{T-1}(\phi)$$

This is a function of S_{T-1}!

Remark 153. *We can now continue backwards, consider $[T-2, T-1]$ and look for a portfolio $\phi_{T-2} = (\alpha_{T-2}, \beta_{T-2})$ that replicates X_{T-1}, i.e. s.t.*

$$\alpha_{T-2} S_{T-1} + \beta_{T-2}(1+r) = X_{T-1}$$

(notice that $\alpha_{T-2} S_{T-1} + \beta_{T-2}(1+r) = \alpha_{T-1} S_{T-1} + \beta_{T-1}$)

*(***self-financing***)*

Then:

$$\begin{cases} \alpha_{T-2} u S_{T-2} + \beta_{T-2}(1+r) = X^u_{T-1} \\ \alpha_{T-2} d S_{T-2} + \beta_{T-2}(1+r) = X^d_{T-1} \end{cases}$$

where

$$X^u_{T-1} = (1+r)^{-1} \left[p^* g(u \cdot S_{T-2} \cdot u) + (1-p^*) g(u \cdot S_{T-2} \cdot d) \right]$$
$$X^d_{T-1} = (1+r)^{-1} \left[p^* g(d \cdot S_{T-2} \cdot u) + (1-p^*) g(d \cdot S_{T-2} \cdot d) \right]$$

and, finding $\alpha_{T-2}, \beta_{T-2}$,

$$V_{T-2}(\phi) = \alpha_{T-2} S_{T-2} + \beta_{T-2} = (1+r)^{-1} \left(p^* X^u_{T-1} + (1-p^*) X^d_{T-1} \right)$$

which, using the usual NA arguments, gives

$$X_{T-2} = V_{T-2}(\phi).$$

Remark 154. Path-dependent claims

The feature of a path-dependent European claim is that its final payoff is $g(S_0, \ldots, S_T)$. Consider a sample path $(s_0, s_1, \ldots, s_{T-1})$. Consider the interval $[T-1, T]$. Then to find the replicating portfolio:

$$\begin{cases} \alpha_{T-1}(s_0, s_1, \ldots, s_{T-1}) u S_{T-1} + \beta_{T-1}(s_0, \ldots, s_{T-1})(1+r) = g^u \\ \alpha_{T-1}(s_0, s_1, \ldots, s_{T-1}) d S_{T-1} + \beta_{T-1}(s_0, \ldots, s_{T-1})(1+r) = g^d \end{cases}$$

Note that s_0 and S_{T-1} are connected by a number of paths.

Thus, the price at $T-1$ is $\pi_{T-1}(X) = V_{T-1}(\phi) = \alpha_{T-1}(s_0, s_1, \ldots, s_{T-1}) S_{T-1} + \beta_{T-1}(s_0, \ldots, s_{T-1})$, which will be different at any node for each sample path connecting s_0 and S_{T-1}.

As long as $\pi_{T-1}(X)$ can be found at S_{T-1} for every possible path, we can apply the same procedure to obtain $\pi_{T-2}(X)$ by looking at $[T-2, T-1], \ldots,$ yielding a unique solution!

Remark 155. *The binomial (CRR) model is* **COMPLETE** *because any European (path-dependent or independent) claim can be replicated; $V(\phi)$ and ϕ are adapted to the filtration generated by S, and $\pi(X) = V(\phi)$.*

The CRR model gives a closed-form formula for the price of a European call or put option (see MR p. 43–46 for the formula and proof).

17 No Arbitrage in the CRR model

Proposition 156

A martingale measure $\mathbb{P}^*$ for S^* exists iff $d < 1 + r < u$. In this case, $\mathbb{P}^*$ is the unique element of the class of probability measures such that $p^* = \frac{1+r-d}{u-d}$ and $S^*_t = \frac{S_t}{B_t} = S_t(1+r)^{-t}$.

Proof. We want to show $\mathbb{E}_{\mathbb{P}^*}(S^*_{t+1} | \mathcal{F}^S_t) = S^*_t$.

Since $S_{t+1} = \xi_{t+1} S_t$ for all $t < T^*$, we have:

$$\begin{aligned}\mathbb{E}_{\mathbb{P}^*}(S^*_{t+1}|\mathcal{F}^S_t) &= \mathbb{E}_{\mathbb{P}^*}\left[S_t \xi_{t+1}(1+r)^{-(t+1)}\middle|\mathcal{F}^S_t\right] \\ &= S_t(1+r)^{-(t+1)}\mathbb{E}_{\mathbb{P}^*}\left[\xi_{t+1}\middle|\mathcal{F}^S_t\right]\end{aligned}$$

The condition $\mathbb{E}_{\mathbb{P}^*}(S^*_{t+1}|\mathcal{F}^S_t) = S^*_t$ implies:

$$(1+r)^{-(t+1)} S_t \mathbb{E}_{\mathbb{P}^*}\left[\xi_{t+1}\middle|\mathcal{F}^S_t\right] = (1+r)^{-t} S_t$$

Since $S_t > 0$, the equality holds iff:

$$\mathbb{E}_{\mathbb{P}^*}[\xi_{t+1}|\mathcal{F}^S_t] = 1 + r \implies \mathbb{E}_{\mathbb{P}^*}[\xi_{t+1}|\mathcal{F}^S_t] = \mathbb{E}_{\mathbb{P}^*}(\xi_{t+1})$$

since $(1 + r)$ is constant. This means ξ_{t+1} is independent of $\mathcal{F}^S_t$ under $\mathbb{P}^*$ and thus independent of $\xi_1, \dots, \xi_t$ for all t.

Hence:

$$\mathbb{E}_{\mathbb{P}^*}[\xi_t] = u\mathbb{P}^*\{\xi_t = u\} + d\mathbb{P}^*\{\xi_t = d\} = 1 + r$$

Using $\mathbb{P}^*\{\xi_t = d\} = 1 - \mathbb{P}^*\{\xi_t = u\}$, we get:

$$up^* + d(1 - p^*) = 1 + r \implies p^* = \frac{1 + r - d}{u - d} \quad \forall t$$

□

Remark 157. *From this result, it is clear that $\mathbb{P}^*$ is a probability measure only if $d \le 1 + r \le u$ and is equivalent to $\mathbb{P}$ only if $d < 1 + r < u$.*

Homework 158

Why is $\mathbb{P}^*$ equivalent to $\mathbb{P}$ only if $d < 1 + r < u$?

Remark 159. *Hence, we have established that we can use the binomial model as an arbitrage pricing model!*

p^* defines the martingale measure, with our choice of the **NUMERAIRE** *B_t.*

Indeed:

Proposition 160

Consider a European call option. Then the arbitrage price formula obtained by evaluating by backward induction the replicating portfolios coincides $\forall m = 0, 1, \dots, T$ with

$$C^*_{T-m} = \mathbb{E}_{\mathbb{P}^*}\left((1 + r)^{-m}(S_T - K)^+ \mid \mathcal{F}^S_{T-m}\right)$$

Proof. See Musiela and Rutkowski p. 50. □

Remark 161. $\to$ *Notice that the B-S pricing formula can be obtained as a limit version of the CRR formula.*

18 American Options in the CRR Model

Remark 162. *Recall that, given our "rational" reasoning about the exercise of the option; at any valuation time the holder prices the option as the max of the exercise value and the max (R-N) expected payoff that can be earned by exercising at any future possible exercise date.*

Reasoning by backward induction again:

At T, we have $\pi_T(X_T) = X_T$.

Consider $[T-1, T]$. Then, at $T-1$ the holder of the option can exercise immediately, obtaining X_{T-1}, or keep the option, valued $\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}X_T \mid \mathcal{F}_{T-1}\right] \implies$

$$\pi_{T-1} = \max\left(X_{T-1}, \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}X_T \mid \mathcal{F}_{T-1}\right]\right).$$

By induction, for every $t = 1, \ldots, T$

$$\pi_{t-1} = \max\left(X_{t-1}, \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-1}\pi_t \mid \mathcal{F}_{t-1}\right]\right).$$

(This is the **Snell envelope** *$S_n(\omega)$ from ch. 3 - part 1!)*

And, as already said, $\pi_T = X_T$.

Remark 163. *We know from the 1st part of the course that $(\tilde{\pi}_t)_{0\le t\le T}$, where $\tilde{\pi}_t = (1+r)^{-t}\pi_t$ is the smallest supermartingale which dominates $(X_t)_{0\le t\le T}$ and that $(\pi_t)_{0\le t\le T}$ is the* **Snell's envelope** *of $(X_t)_{0\le t\le T}$.*

We know also that τ_m^T s.t. $\pi_m = \mathbb{E}\left[X_{\tau_m^T} \mid \mathcal{F}_m\right]$ is the optimal stopping time.

$\to$ Results extend to the case of $T \to +\infty$.

19 Call Option Pricing

Remark 164. *As already hinted at, the case of a call option is peculiar. We first notice that denoting with C_t the price of a call, the following inequality holds:*

$$C_t \ge (S_t - K)^+ \quad \forall t \le T$$

Indeed, by **Jensen's inequality***:*

$$\begin{aligned}\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-(T-t)}(S_T-K)^+ \mid \mathcal{F}_t^S\right] &\ge \left(\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-(T-t)}S_T - (1+r)^{-(T-t)}K \mid \mathcal{F}_t^S\right]\right)^+ \\ &\ge (S_t - K)^+ \quad \forall t \le T\end{aligned}$$

where we used the fact that $\mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-(T-t)}S_T \mid \mathcal{F}_t^S\right] = S_t$ and $(1+r)^{-(T-t)}K \le K$ (with strict inequality if $r > 0$ and $t < T$).

Homework 165

Prove the inequality $C_t \ge (S_t - K)^+$ alternatively by no arbitrage arguments.

Remark 166. *Then, $C_t^a = C_t$ can be deduced by no arbitrage. Take any $t < T$ at which the holder wants to exercise. Exercising at t yields $S_t - K$, which is equivalent to $(1+r)^{T-t}(S_t - K)$ at time T.*

Alternatively, short the stock and hold the option, getting a payoff at T of:

$$S_t(1+r)^{T-t} - S_T + (S_T - K)^+ = \begin{cases} S_t(1+r)^{T-t} - K & \text{if } S_T > K \\ S_t(1+r)^{T-t} - S_T & \text{if } S_T < K \end{cases}$$

The second portfolio is always better!!

$\implies$ *exercising immediately is never optimal* $\implies$ *it is like having a European call!!!*

The result extends to infinite horizon and continuous time, but not to dividend-paying stocks!

> **Homework 167**
> Show more directly that $\tilde{C}_t^a$ is a supermartingale and the European call is a lower bound.

20 American Put Option

Remark 168. *The arbitrage price of an American put option is*

$$P_t^a = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*}\left[(1+r)^{-(\tau-t)}(K - S_\tau)^+ \mid \mathcal{F}_t\right]$$

$$\tau_t^* = \min\{u \in \{t, \dots, T\} \mid (K - S_u)^+ \geq P_u^a\}$$

where τ_t^* is the **"rational" exercise time** *that solves the problem for the risk-neutral agent.*

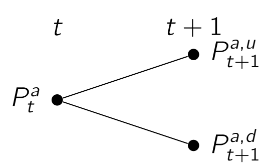
- *This is equivalent to (the proof by NA is similar to the one in one period, see MR Thm 5.1.1, cont. time):*

$$\tau_t^* = \min\{u \in \{t, \dots, T\} \mid K - S_u \geq V_u^p\}$$

where $V_u^p = \max\{K - S_u, \mathbb{E}_{\mathbb{P}^*}[(1+r)^{-1}V_{u+1}^p \mid \mathcal{F}_u]\}$, which is the approach used in the CRR model. Indeed, recursively, one applies $\forall t \leq T-1$:

$$P_t^a = \max\{K - S_t, (1+r)^{-1}[p^* P_{t+1}^{a,u} + (1-p^*)P_{t+1}^{a,d}]\}$$

and $P_T^a = (K - S_T)^+$.



$\rightarrow$ *The result extends to any American contingent claim!*

21 Continuous Time

Remark 169. **Infinite Horizon Put** *(Peskir & Shiryaev, Ch 2/11, Bjork, Ch 21)*

Generalising the results in discrete time, the arbitrage-free price of an American put is

$$V_s = \sup_{\tau} \mathbb{E}_s \left[e^{-r\tau}(K - S_\tau)^+ \right] = V_*(s)$$

where $g(s) = (K - s)^+$ is the payoff function, and we consider a **Geometric Brownian Motion** *(GBM) for the stock price:*

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 > 0 \text{ under } \mathbb{P}^*$$

where $r > 0$, $(W_t)_{t \geq 0}$ is a standard BM with infinitesimal generator

$$\mathcal{L}_S = rs\frac{\partial}{\partial s} + \frac{\sigma^2}{2}s^2\frac{\partial^2}{\partial s^2}$$

The unique solution is $S_t = S_0 e^{(r - \frac{\sigma^2}{2})t + \sigma W_t}$.

Recall that the optimal stopping problem above is connected to the solution of a **variational inequality** *and to a* **free-boundary problem***:*

$$\max\{\mathcal{L}_S V(s) - rV(s), g(s) - V(s)\} = 0, \quad s \in \mathbb{R}^+$$

Notice that the closer S_t gets to zero, the greater the current exercise value and the less likely the gain function is to increase upon continuation. It is natural thus to guess that the **continuation region** *$\mathcal{C} = (b, +\infty)$, $b \in (0, K)$, and thus if $S < b$ it will be optimal to exercise the option, i.e. the stopping time $\tau_b = \inf\{t \geq 0 : X_t \leq b\}$ is optimal.*

Remark 170. *Thus, the free-boundary problem reads:*

(i) $\mathcal{L}_S V = rV$ *for $s > b$ (PDE in the continuation region)*

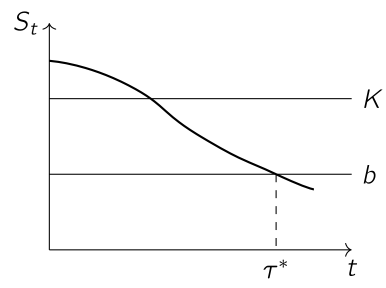
(ii) $V(s) = (K - s)^+$ *for $s = b$ (exercise value at a stopping time, continuity)*

(iii) $V(s) > (K - s)^+$ *for $s > b$ (continuation)*

(iv) $V(s) = (K - s)^+$ *for $0 < s < b$ (exercise)*

(v) $V'(s) = -1$ *for $s = b$ (***smooth-fit condition***) $\implies$ V is C^1 at b.*

Let's solve it. This will be our solution guess, which we will then verify.



Remark 171. *(i) reads*

$$rsV' + \frac{\sigma^2}{2}s^2V'' - rV = 0 \tag{*}$$

We guess a solution of the form $V(s) = s^p$. Since $V' = ps^{p-1}$ and $V'' = p(p-1)s^{p-2}$, we substitute in (*) and obtain

$$rsps^{p-1} + \frac{\sigma^2}{2}s^2p(p-1)s^{p-2} - rs^p = 0$$

$$\implies \frac{\sigma^2}{2}p^2 + p\left(r - \frac{\sigma^2}{2}\right) - r = 0$$

$$p_{1,2} = \frac{-(r-\frac{\sigma^2}{2}) \pm \sqrt{(r-\frac{\sigma^2}{2})^2 + 2\sigma^2 r}}{\sigma^2} = \frac{-(r-\frac{\sigma^2}{2}) \pm \sqrt{(r+\frac{\sigma^2}{2})^2}}{\sigma^2}$$

where we used $(r-\frac{\sigma^2}{2})^2 + 2\sigma^2 r = r^2 - r\sigma^2 + \frac{\sigma^4}{4} + 2\sigma^2 r = r^2 + r\sigma^2 + \frac{\sigma^4}{4} = (r+\frac{\sigma^2}{2})^2$. *The roots are:*

$$p_1 = 1, \quad p_2 = -\frac{2r}{\sigma^2}$$

The general solution of (*) then reads $V(s) = C_1 s + C_2 s^{-\frac{2r}{\sigma^2}}$.

Remark 172. *Now,* $\mathbb{P}_s(S_t = b) = 0$ *for all* s, t *and we have that*

$$e^{-rt}(K - S_t)^+ \leq e^{-rt}V(S_t) \leq V(s) + M_t, \quad \forall s$$

where the first inequality follows from properties (i), (iii), and (iv) established previously, and the second follows from $(*)$. *Here* $(M_t)_{t\geq 0}$ *is a continuous local martingale defined as*

$$M_t = \int_0^t e^{-rs}\sigma S_s V'(S_s)dB_s.$$

> **Homework 173**
> Verify that $(M_t)_{t\geq 0}$ is indeed a martingale.

Let $(\tau_n)_{n\geq 1}$ *be a sequence of bounded stopping times for* M. *Then for any stopping time* τ *of* S *we have that*

$$e^{-r(\tau\wedge\tau_n)}(K - S_{\tau\wedge\tau_n})^+ \leq V(s) + M_{\tau\wedge\tau_n}, \quad n \geq 1.$$

Taking $\mathbb{P}_s$*-expectation and noting that* $\mathbb{E}_s[M_{\tau\wedge\tau_n}] = 0$ *by the* **optional sampling theorem**, *letting* $n \to +\infty$, *we have by* **Fatou's lemma**:

$$\mathbb{E}_s\left(e^{-r\tau}(K - S_\tau)^+\right) \leq V(s).$$

Taking the supremum over all stopping times τ, *we obtain*

$$\sup_\tau \mathbb{E}_s\left(e^{-r\tau}(K - S_\tau)^+\right) = V_*(s) \leq V(s), \quad \forall s.$$

Remark 174. *2) To prove* $\geq$, *taking* $(*)$ *and (i) and using again the optional sampling theorem we have that,* $\forall \tau$ *and for* τ_b

$$\mathbb{E}_s\left(e^{-r(\tau_b\wedge\tau_n)}V(S_{\tau_b\wedge\tau_n})\right) = V(s) \quad \forall n \geq 1$$

Letting $n \to \infty$, *since* $e^{-r\tau_b}V(S_{\tau_b}) = e^{-r\tau_b}(K - S_{\tau_b})^+$, *by dominated convergence we have*

$$\mathbb{E}_s\left(e^{-r\tau_b}(K - S_{\tau_b})^+\right) = V(s).$$

$\implies \tau_b$ *is optimal and* $V_*(s) = V(s) \quad \forall s > 0$. $\square$

22 Finite Horizon

Remark 175. **FINITE HORIZON**

The finite horizon case is not so analytically tractable as the infinite horizon one.

The problem becomes

$$V(t,s) := \sup_{0 \leq \tau \leq T-t} \mathbb{E}_{t,s}\left[e^{-r\tau}(K - S_{t+\tau})^+\right]$$

where τ is a stopping time.

$$dS_{t+u} = rS_{t+u}du + \sigma S_{t+u}dB_u, \quad S_t = s > 0$$

Here again (see 1st part of the course) we have a variational inequality:

$$\max\left\{g(s) - V(t,s), \frac{\partial V(t,s)}{\partial t} + \mathcal{L}_X V(t,s) - rV(t,s)\right\} = 0$$

for $(t,s) \in [0,T) \times \mathbb{R}^d$, with $V(T,s) = g(s)$.

The problem is, in this case, bi-dimensional.

Remark 176. *In this case, the* **continuation region** *is*

$$\mathcal{C} = \{(t,s) \in [0,T) \times (0,\infty) : V(t,s) > g(s)\}$$

while the **stopping set** *is*

$$\bar{D} = \{(t,s) \in [0,T] \times (0,\infty) : V(t,s) = g(s)\}$$

and $\tau_D^* = \inf\{0 \leq u \leq T-t : X_{t+u} \in \bar{D}\}$ is the optimal stopping time.

It is possible to prove that:

a) $\exists b : [0,T] \to \mathbb{R}$ *s.t.*

$$\mathcal{C} = \{(t,s) \in [0,T] \times (0,\infty) : s > b(t)\},$$

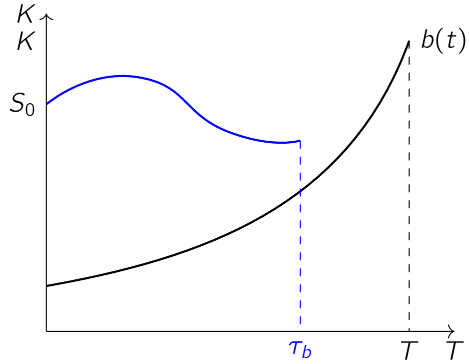
$\bar{D}$ is the closure of $D = \{(t,s) \in [0,T] \times (0,\infty) : s \leq b(t)\}$.

b) *$V(t,s)$ is decreasing on $[0,T]$ for each $s \in (0,\infty)$ and $t \to b(t)$ is increasing on $[0,T]$.*

c) *$(t,s) \to V(t,s)$ is continuous.*

d) *$s \to V(t,s)$ is C^1 at $b(t)$.*

e) *b is continuous on $[0,T]$ and $b(T) = K$.*



Remark 177.

Hence, it follows that we have the following free-boundary problem for the unknown value function V and boundary b:

$$
\begin{aligned}
&V_t + \mathcal{L}_S V = rV && \text{in } \mathcal{C} \\
&V(t,s) = (K-s)^+ && s = b(t) \\
&V_s(t,s) = -1 && s = b(t) \quad \to \text{smooth fit, guarantees } V \text{ is } C^{1,2} \text{ in } \mathcal{C},\ C^1 \text{ in } D \\
&V(t,s) > (K-s)^+ && \text{in } \mathcal{C} \\
&V(t,s) = (K-s)^+ && \text{in } D
\end{aligned}
$$

- $\tau^* = \inf\{t \geq 0 : S_t = b(t)\}$ *is the optimal stopping time.*

Remark 178. *No analytical formula is available. Alternatives:*

1. *Solve the free-boundary problem numerically.*
2. *Solve the variational inequalities.*
3. *Approximate via a binomial model and compute the binomial price.*

23 Optimal Sale of an Asset

Remark 179. *Let us consider an investor who holds an asset (e.g., a firm) with value process X. He/she can sell at any time paying a fixed cost K. The problem is to find the time at which it is optimal to sell, solving*

$$v(n) = \sup_{\tau \in \mathcal{T}} \mathbb{E}\left[e^{-\rho\tau}(X_\tau^n - K)\right]$$

where $\rho > 0$ is the **DISCOUNT FACTOR** *(under the risk-neutral measure).*

This time, $g(n) = (n - K)$, which is C^2 on $(0, \infty)$. We assume the asset value follows a Geometric Brownian Motion:

$$dX_t = \mu X_t dt + \sigma X_t dW_t$$

The **stopping region** *is in the set where $\rho g(n) - \mathcal{L}g(n) \geq 0$. Let us define*

$$\mathcal{D} = \{n \in (0, \infty) : (\rho - \mu)n \geq \rho K\}$$

We have three cases:

(i) $\rho < \mu$. In this case, $v(n) = \infty$ for all $n > 0$. Indeed, for any fixed time T, we have

$$v(n) \geq \mathbb{E}\left[e^{-\rho T}(X_T^n - K)\right] = ne^{(\mu-\rho)T}\mathbb{E}\left[e^{\sigma W_T - \frac{\sigma^2}{2}T}\right] - Ke^{-\rho T}$$

Since X_t is a GBM, the expectation term is 1, yielding

$$= ne^{(\mu-\rho)T} - Ke^{-\rho T}$$

As $T \to +\infty$, this expression goes to infinity.

(ii) $\mu = \rho$. In this case, $v(n) = n$. Indeed, for any stopping time τ and for all $m \in \mathbb{N}$, by the optional sampling theorem we have:

$$\mathbb{E}\left[e^{-\rho(\tau\wedge m)}(X_{\tau\wedge m}^n - K)\right] = n - K\mathbb{E}\left[e^{-\rho(\tau\wedge m)}\right] \leq n$$

Sending $m \to \infty$ and using Fatou's lemma, for all $\tau \in \mathcal{T}$ we have

$$\mathbb{E}\left[e^{-\rho\tau}(X_\tau^n - K)\right] \leq n \implies v(n) \leq n$$

By taking $\tau = m$ and sending $m \to +\infty$, we find $v(n) \geq n$, hence $v(n) = n$.

(iii) $\rho > \mu$. This is the interesting case. We have $\mathcal{D}$ defined as $n \in [a, \infty)$, where $a = \frac{\rho K}{\rho - \mu} > 0$.

Notice that if we never stop the process, the value we get is $0 < g(n) = n - K$ for all $n > K$. We can prove that the **stopping region** *is of the form $\mathcal{S} = [n^*, \infty)$ for some $n^* \in (0, \infty)$ with $n^* \geq a$ (omitted, see Pham p. 104).*

We have the following variational inequality: $\max\{\mathcal{L}v - \rho V, v(n) - g(n)\} = 0$, which leads to:

1. *$\mathcal{L}v - \rho V = 0$ for $n < n^*$*

2. *$V(n) = n - K$ for $n \geq n^*$*

3. *$V'(n^*) = 1$ (the* **smooth-fit** *condition)*

Let us start from the **continuation region***, where v satisfies the ODE:*

$$\mu n v'(n) + \frac{1}{2}\sigma^2 n^2 v''(n) - \rho v = 0, \quad n < n^*$$

This is a second-order ODE whose general solution is $v(n) = An^m + Bn^n$*, where m and n are the roots of the characteristic equation:*

$$\frac{1}{2}\sigma^2 x^2 + \gamma x - \rho = 0, \quad \text{with } \gamma = \mu - \frac{1}{2}\sigma^2.$$

The roots are given by $m, n = \frac{-\gamma \pm \sqrt{\gamma^2 + 2\rho\sigma^2}}{\sigma^2}$*. Specifically:*

- $m = \frac{-\gamma - \sqrt{\gamma^2 + 2\rho\sigma^2}}{\sigma^2} < 0$
- $n = \frac{-\gamma + \sqrt{\gamma^2 + 2\rho\sigma^2}}{\sigma^2} > 1$

We also know that $v(0^+) = \max(-K, 0) = 0$*, which implies* $A = 0$*. Using the* **smooth-fit** *condition* $V'(n^*) = 1$*, we have:*

$$Bn(n^*)^{n-1} = 1 \implies B = \frac{1}{n} \cdot \frac{1}{(n^*)^{n-1}}.$$

While using the boundary condition $V(n^*) = n^* - K$*, we get:*

$$\begin{aligned} B(n^*)^n = n^* - K &\implies K = n^* - B(n^*)^n \\ &\implies K = n^* - \frac{1}{n}\frac{(n^*)^n}{(n^*)^{n-1}} = n^* - \frac{1}{n}n^* \\ &\implies n^*\left(1 - \frac{1}{n}\right) = K \implies n^* = \frac{K}{1 - \frac{1}{n}}. \end{aligned}$$

Remark 180. *This implies that the value function* $v(n)$ *is*

$$v(n) = \begin{cases} \frac{1}{n}(n^*)^{1-n} n^n & n < n^* \\ n - K & n \geq n^* \end{cases}$$

and the optimal stopping time is $\tau^* = \inf\{t \geq 0 : X_t \geq n^*\}$*.*

24 Quick Detection of a Wiener Process

Remark 181. *Assume that we observe the trajectory of a Wiener process $X = (X_t)_{t\geq 0}$ whose drift may change from 0 to $\mu \neq 0$ at a random time θ (PS Ch. 22).*

Let $\mathbb{P}(\theta = 0) = \pi$ and $\theta|_{\theta>0} \sim Exp(\lambda)$ with $\lambda > 0$. We consider the space $(\Omega, \mathcal{F}, \mathbb{P}_\pi, \pi \in [0,1])$ where

$$\mathbb{P}_\pi = \pi \mathbb{P}^0 + (1-\pi)\int_0^\infty \lambda e^{-\lambda s}\mathbb{P}^s ds$$

The process X_t is such that

$$dX_t = \mu \mathbb{1}_{\{t\geq\theta\}}dt + \sigma dW_t, \quad X_0 = 0, \sigma > 0$$

which implies

$$X_t = \begin{cases} \sigma W_t & \text{if } \theta > t \\ \mu(t-\theta) + \sigma W_t & \text{if } t \geq \theta \end{cases}$$

Here θ is unobservable. Our task is to find a stopping time τ^* of X (relative to the natural filtration generated by X_t) that is as close as possible to θ.

Remark 182. **Quickest detection problem**. *The value function is defined as*

$$V(\pi) = \inf_\tau \left(\underbrace{\mathbb{P}_\pi(\tau < \theta)}_{\text{probability of a false alarm}} + c \underbrace{\mathbb{E}_\pi[\tau - \theta]^+}_{\text{average delay in detecting the disorder}} \right)$$

This can be reduced to

$$V(\pi) = \inf_\tau \mathbb{E}_\pi \left[1 - \pi_\tau + c\int_0^\tau \pi_t dt \right],$$

where $\pi_t = \mathbb{P}_\pi(\theta \leq t \mid \mathcal{F}_t^X)$ for $t \geq 0$ is the **a posteriori probability**, *with $\mathbb{P}_\pi(\pi_0 = \pi) = 1$.*

By Bayes formula,

$$\pi_t = \pi \frac{d\mathbb{P}^0}{d\mathbb{P}_\pi}(t,X) + (1-\pi)\int_0^t \frac{d\mathbb{P}^s}{d\mathbb{P}_\pi}(t,X)\lambda e^{-\lambda s}ds$$

where $\frac{d\mathbb{P}^s}{d\mathbb{P}_\pi}(t,X)$ is the **Radon-Nikodym density** *of $\mathbb{P}^s \mid \mathcal{F}_t^X$ with respect to $\mathbb{P}_\pi \mid \mathcal{F}_t^X$. Furthermore,*

$$1 - \pi_t = (1-\pi)e^{-\lambda t}\frac{d\mathbb{P}^t}{d\mathbb{P}_\pi}(t,X) = (1-\pi)e^{-\lambda t}\frac{d\mathbb{P}^\infty}{d\mathbb{P}_\pi}(t,X)$$

where $\mathbb{P}^\infty$ corresponds to the law of $(\sigma W_t)_{t\geq 0}$.

Remark 183. *The process for the likelihood ratio $\rho_t = \frac{\pi_t}{1-\pi_t}$ can be derived as*

$$\rho_t = e^{\lambda t} z_t \left(\rho_0 + \lambda \int_0^t \frac{e^{-\lambda s}}{z_s} ds \right) = e^{Y_t}\left(\frac{\pi}{1-\pi} + \lambda \int_0^t e^{-Y_s} ds \right)$$

where $z_t = \frac{d\mathbb{P}^0}{d\mathbb{P}^\infty}(t,X) \equiv \frac{d(\mathbb{P}^0|\mathcal{F}_t^X)}{d(\mathbb{P}^\infty|\mathcal{F}_t^X)} = \exp\left(\frac{\mu}{\sigma^2}\left(X_t - \frac{\mu}{2}t\right)\right)$ and $Y_t = \lambda t + \frac{\mu}{\sigma^2}\left(X_t - \frac{\mu}{2}t\right)$.

Applying Ito's formula to $z_t = f(X_t, t)$, we get

$$\begin{aligned} dz_t &= \frac{\mu}{\sigma^2} z_t dX_t - \left[\frac{\mu^2}{2\sigma^2} z_t dt \right] + \frac{1}{2}\sigma^2 \frac{\mu^2}{\sigma^4} z_t dt \\ &= \frac{\mu}{\sigma^2} z_t dX_t \end{aligned}$$

so that the dynamics of ρ_t becomes

$$d\rho_t = \lambda(1+\rho_t)dt + \frac{\mu}{\sigma^2}\rho_t dX_t$$

Remark 184. *The problem is to compute*

$$V(\pi) = \inf_{(\tau,d)} \mathbb{E}_\pi \left[\tau + a\mathbb{1}_{\{d=0,\theta=1\}} + b\mathbb{1}_{\{d=1,\theta=0\}}\right]$$

25 Sequential Testing

Remark 185. Problem: *we observe a trajectory of the* **Wiener process** $X = (X_t)_{t \geq 0}$ *with drift* $\theta\mu$*, where* θ *is a random variable such that*

$$\theta = \begin{cases} 1 & \text{with prob } \pi \\ 0 & \text{with prob } 1 - \pi \end{cases}$$

in the probability space $(\Omega, \mathcal{F}, \mathbb{P}_\pi, \pi \in [0, 1])$.

The probability measure is defined as

$$\mathbb{P}_\pi = \pi \mathbb{P}_1 + (1 - \pi)\mathbb{P}_0$$

so that $\mathbb{P}_\pi(\theta = 1) = \pi$. *The process* X_t *is given by*

$$X_t = \theta \mu t + \sigma W_t$$

where $(W_t)_{t \geq 0}$ *is a* **standard BM**, *with* $\mu \neq 0$ *and* $\sigma > 0$.

The task is to test

$$H_1 : \theta = 1, \quad H_0 : \theta = 0$$

sequentially with minimal loss. We then define a **stopping time** τ *of* X *and an* $\mathcal{F}_\tau^X$*-measurable random variable* d *taking values 0 or 1.*

26 Discrete Stochastic Control

26.1 Optimal Control in Discrete Time

Discrete time is generally used in macroeconomics.

Problem (Typical problem)

Find the value function $V_T(n)$:

$$V_T(n) = \sup_{U \in \hat{\mathcal{A}}} J_n^T(U) = \sup_{U \in \hat{\mathcal{A}}} \mathbb{E}\left[\sum_{m=1}^{T} g(X_m, U_{m-1}) e^{-rm}\right] \tag{7}$$

where:

- X_m is the **state**;
- U_{m-1} are the **controls**;
- e^{-rm} is the **discount factor**.

26.1.1 Method to tackle these problems

DPP (Dynamic Programming Principle): Assuming $|V_T(n)| < \infty$, we have:

$$V_T(n) = \sup_{\mu \in \mathcal{U}} \left\{\mathbb{E}\left[e^{-r} g(X_1, u) + e^{-r} V_{T-1}(X_1)\right]\right\} \tag{8}$$

26.2 A Portfolio Consumption/Investment Problem

We now consider a **portfolio consumption/investment problem**. We will first tackle it in discrete time, then in continuous time.

Consider a horizon $[0, T]$ (possibly $T \to +\infty$) and an agent who has to select an **investment strategy** in the market. Suppose there is 1 riskless asset and K risky ones.

Let $\alpha_t = (\alpha_t^0, \alpha_t^1, \ldots, \alpha_t^K) = (\alpha_t^i)_{i \geq 0}$ be a **predictable process** defining the strategy, such that $\alpha_t \in \mathcal{F}_{t-1}$. Here, α_t^i represents the **number of units invested in asset *i***.

The **portfolio value (wealth)** at time t, usually denoted by X_t (or V_t in discrete-time), is given by:

$$X_t = \alpha_t^0 B_t + \sum_{i=1}^{K} \alpha_t^i S_t^i \tag{9}$$

where B_t is the riskless asset and S_t^i are the risky assets.

If the strategy α is **self-financing**, then:

$$X_{t+\Delta t} - X_t = \alpha_t^0 \Delta B_t + \sum_{i=1}^{K} \alpha_t^i \Delta S_t^i \tag{10}$$

where $\Delta B_t = B_{t+\Delta t} - B_t$ and $\Delta S_t^i = S_{t+\Delta t}^i - S_t^i$.

26.2.1 The Agent's Decision

The agent decides α_t at any t by maximizing the expected utility from:

1. **Terminal wealth** (solved in discrete time);

2. **Consumption** (+ terminal wealth) (tackled in continuous time).

26.2.2 Utility Function

The **utility function** $u(X)$ is a function of a monetary amount X representing a "measure" of the well-being of an individual with wealth X. It satisfies the following properties:

- $u(X)$ is strictly increasing (**non-satiation**, increasing marginal utility);

- $u(X)$ is strictly concave and differentiable (**decreasing marginal utility**);

- $\lim_{X\to\infty} u'(X) = 0$;

- $\lim_{X\to 0^+} u'(X) = +\infty$.

26.3 Consumption

The individual may use its wealth to "generate" utility (buy goods, services, ...).

- The consumption process C will be defined as a non-negative, adapted process $C = (C_t)_{t\geq 0}$.

Hence (C, α) will be a **CONSUMPTION/INVESTMENT STRATEGY PAIR**.

If we allow for consumption, the wealth at time t is allocated as:

$$X_t = C_t + \alpha^0_{t+1} B_t + \sum_{i=1}^{K} \alpha^i_{t+1} S^i_t \tag{11}$$

The **self-financing condition** with consumption is:

$$X_{t+1} - X_t = -C_t + \alpha^0_{t+1} \Delta B_t + \sum_{i=1}^{K} \alpha^i_{t+1} \Delta S^i_t \tag{12}$$

So, the problem of the agent is:

1. **Max Terminal Utility**:
$$\max_{\alpha\in\hat{\mathcal{A}}} \mathbb{E}\left[\beta^T u(X_T;\alpha)\right], \quad X_0 = x_0 \tag{13}$$
where α is predictable and self-financing, and β^T is the discount factor.

2. **Max Consumption and Terminal Utility**:
$$\max_{(C,\alpha)\in\hat{\mathcal{A}}} \mathbb{E}\left[\sum_{t=0}^{T} \beta^t u_c(C_t) + \beta^T u_x(X_T)\right] \tag{14}$$

where u_c and u_x are the utility functions for consumption and terminal wealth respectively (in principle, they can be different).

26.4 The CRR Binomial Model

Let us put ourselves in the **CRR binomial model**, with one risky asset. Then:

$$S_{t+1} = \xi_{t+1} S_t, \quad \text{where } \xi_t = \begin{cases} u & \text{with prob. } p \\ d & \text{with prob. } 1-p \end{cases} \tag{15}$$

We assume $u > (1+r) > d$ to ensure the existence of a unique risk-neutral martingale measure $\mathbb{P}^*$ with:

$$p^* = \frac{1+r-d}{u-d} \tag{16}$$

26.4.1 Focus on Problem 1: Max Terminal Utility

Let us rewrite the strategy α in terms of the **share of wealth** π_t invested in the risky asset:

$$\pi_t = \frac{\alpha^1_{t+1} S_t}{X_t}, \quad \pi^0_t = 1 - \pi_t = \frac{\alpha^0_{t+1} B_t}{X_t} \tag{17}$$

The wealth evolution becomes:

$$\begin{aligned} X_{t+1} &= X_t + \alpha^0_{t+1} \Delta B_t + \alpha^1_{t+1} \Delta S_t \\ &= X_t + \alpha^0_{t+1} r B_t + \alpha^1_{t+1} S_t (\xi_{t+1} - 1) \\ &= X_t + X_t \left[\pi^0_t r + \pi_t (\xi_{t+1} - 1)\right] \\ &= X_t \left[1 + (1-\pi_t) r + \pi_t (\xi_{t+1} - 1)\right] \end{aligned}$$

The term in the brackets is the **gain process** $G(X_t, \pi_t, \xi_{t+1})$.

26.4.2 Applying the DPP

Applying the Dynamic Programming Principle, the value function $V_t(x)$ satisfies:

$$\begin{cases} V_T(x) = u(x) & \text{for } t = T \\ V_t(x) = \sup_\pi \mathbb{E}\left[V_{t+1}\left(x(1 + (1-\pi) r + \pi(\xi_{t+1} - 1))\right)\right] & \text{for } t < T \end{cases} \tag{18}$$

The recursive step can be simplified to:

$$V_t(x) = \sup_\pi \mathbb{E}\left[V_{t+1}\left(x(1+r) + x\pi(\xi_{t+1} - 1 - r)\right)\right] \tag{19}$$

Note that we require $X_t \geq 0$, which is satisfied as long as $\pi \geq 0$ and $\xi_{t+1} > 0$.

26.4.3 Example: Log Utility

Consider the simplest case: $u(x) = \log x$ and $r = 0$. Notice that in the CRR model:

$$S_t = S_0 u^m d^{t-m} \tag{20}$$

where m is the realization of the Bernoulli random variable $N_t \sim B(t, p^*)$, which models the number of upward movements in the stock price up to time t.

In our case, we have:

$$\begin{aligned} V_T(n) &= \log n; \\ V_t(n) &= \sup_{\pi} \mathbb{E}\left[V_{t+1}\left(n(1+\pi(\xi_{t+1}-1))\right) \mid X_t = n\right] \quad \forall t \leq T \end{aligned}$$

Proceeding backwards:

For $T-1$:

$$\begin{aligned} V_{T-1}(n) &= \sup_{\pi_{T-1}} \mathbb{E}\left[\log\left[n(1+\pi_{T-1}(\xi_T-1))\right] \mid X_t = n\right] \\ &= \sup_{\pi_{T-1}} \left[p \log\left[n(1+\pi_{T-1}(u-1))\right] + (1-p)\log\left[n(1+\pi_{T-1}(d-1))\right]\right] \\ &= \log n + \sup_{\pi_{T-1}} \left[p\log\left[1+\pi_{T-1}(u-1)\right] + (1-p)\log\left[1+\pi_{T-1}(d-1)\right]\right] \end{aligned}$$

Notice that the supremum term does not depend on n!

F.O.C.:

$$p \cdot \frac{u-1}{1+\pi_{T-1}(u-1)} + (1-p)\frac{d-1}{1+\pi_{T-1}(d-1)} = 0$$

$$p(u-1)[1+\pi_{T-1}(d-1)] + (1-p)[1+\pi_{T-1}(u-1)](d-1) = 0$$

$$\pi_{T-1}(u-1)(d-1)[p+1-p] = p(1-u) + (1-p)(1-d)$$

$$\begin{aligned} \Rightarrow \pi_{T-1} &= \frac{1-pu-d+pd}{(u-1)(d-1)} = \frac{p(d-u)+1-d}{(u-1)(d-1)} \\ &= -\frac{p(u-d)+d-1}{(u-1)(d-1)} = \frac{p(u-d)+d-1}{(u-1)(1-d)} \end{aligned}$$

In the last expression, the denominator $(u-1)(1-d) > 0$.

For the condition of **no short-selling**:

$$pu - pd + d - 1 \geq 0 \implies p \geq \frac{1-d}{u-d} = p^*!$$

We can then iterate backwards! As an alternative, we can guess a functional form for V_t, for instance:

$$V_t(n) = \log n + a_t$$

and proceed backwards using the Bellman equation, using the terminal condition $a_T = 0$ since $V_T(n) = \log n$. **(HW)**

27 Continuous Time Stochastic Control

27.1 Merton's optimal consumption/investment problem

Let us now consider the problem of an agent that maximizes consumption utility over his/her lifetime / a fixed horizon. (Brief recap):

Let us consider that his/her wealth X_t evolves according to

$$dX_s = \mu(s, X_s, u_s)ds + \sigma(s, X_s, u_s)dW_s$$

with initial condition $X_t = n$. In this formulation:

- $u_s = \alpha(s, X_s) \in \mathcal{U}$ is the control process;
- W_s is a **Brownian Motion** (BM).

We assume the SDE has a unique strong solution (i.e., the control u is **admissible**, denoted by $u \in \mathcal{U}$).

Let us consider the problem:

$$V(t, n) = \sup_{u \in \mathcal{U}} J_{t,n}(u) = \mathbb{E}\left[\int_t^T e^{-\rho(s-t)} f(s, X_s^u, u_s)ds + e^{-\rho(T-t)} g(X_T^u)\right]$$

Then, under some assumptions (regularity, existence of an optimal u, ...) the solution to the problem can be found as a solution to the **HJB Equation**:

$$\begin{cases} \partial_t v(t, n) + \sup_{u \in \mathcal{U}} \left\{ \mathrm{tr}(\sigma\sigma^T(n, u) D^2 v(t, n)) + \langle \mu(n, u), \nabla v(t, n) \rangle + f(t, n, u) \right\} = \rho v(t, n) \\ v(T, n) = g(n), \quad n \in \mathbb{R}^d \end{cases} \tag{21}$$

where $(t, n) \in [0, T] \times \mathbb{R}^d$. The HJB equation is a PDE in the unknown function $v(t, n)$.

27.1.1 Steps to solve the problem:

1. Consider the static optimization problem for arbitrary (t, n) and solve:

   $$\sup_{u \in \mathcal{U}} \left\{ \mathrm{tr}(\sigma\sigma^T(n, u) D^2 v(t, n)) + \langle \mu(n, u), \nabla v(t, n) \rangle + f(t, n, u) \right\}$$

   treating t and n as fixed parameters and $f, \mu, \sigma\sigma^T, v$ as given functions or parameters.

2. The solution $\hat{u}$ will depend on t, n and v:

   $$\hat{u} = u(t, n; v)$$

3. We now need to find v. We thus substitute our candidate optimal control $\hat{u}$ into the HJB and solve it.

The $\hat{v}$ we find is the candidate value function and using a verification theorem we can say it is the optimal value function, and $\hat{u} = u(t, n; \hat{v})$ is the optimal control.

27.2 Market Model with Two Securities

We consider now a market in which 2 securities are traded:

- **Risk-less** (bank account):
$$dB_t = rB_t dt$$

- **Risky** (stock):
$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 = s$$

The wealth X_t will follow the dynamics:

$$dX_t = X_t \left[\pi_t^0 r + \pi_t^1 \mu\right] dt - C_t dt + \pi_t^1 \sigma X_t dW_t \tag{$*$}$$

where (π_t^0, π_t^1) is the **investment strategy** and C_t is **consumption**. We require:

$$C_t \geq 0; \quad \pi_t^0 + \pi_t^1 = 1 \quad \forall t \geq 0$$

The agent's problem will be:

$$\max_{\pi^0, \pi^1, C} \mathbb{E}\left[\int_0^T e^{-\rho t} f(t, C_t) dt + e^{-\rho T} g(X_T)\right] \tag{22}$$

subject to:

- Wealth dynamics $(*)$ with $X_0 = n_0$;
- $C_t \geq 0$;
- $\pi_t^0 + \pi_t^1 = 1 \quad \forall t \geq 0$.

Here, ρ is the "subjective" discount factor.

Optimizing Terminal Utility from Wealth

Let us first consider again the problem of maximizing terminal utility in this context.

Hence, $C_t = 0 \quad \forall t$, and the performance criterion is $J_{t,n} = \mathbb{E}\left[u(X_T^{t,n})\right]$ for $(t, n) \in [0, T] \times \mathbb{R}$.

For simplicity, let us assume no discounting.

First, we have to prove that the value function v is increasing and concave in n.

Fix $0 < n \leq y$, and let π_t be an admissible control. The process $Z_s = X_s^{t,y} - X_s^{t,n}$ satisfies the following SDE:

$$dZ_s = Z_s \left[(\pi_s \mu + (1 - \pi_s) r) ds + \pi_s \sigma dW_s\right];$$

Since $Z_t = y - n \geq 0$, it follows that $Z_s \geq 0$ for all $s \geq t$, which implies:

$$X_s^{t,y} \geq X_s^{t,n} \quad \forall s \geq t$$

Since the utility function u is increasing, we have $u(X_T^{t,n}) \leq u(X_T^{t,y})$, and consequently:

$$\mathbb{E}\left[u(X_T^{t,n})\right] \leq \mathbb{E}\left[u(X_T^{t,y})\right] \leq v(t,y) \quad \forall \pi \in \mathcal{U}$$

Taking the supremum over $\pi \in \mathcal{U}$ on the left-hand side shows that $v(t,n) \leq v(t,y)$, so v is increasing in n.

Now, consider $n_1, n_2 > 0$, two control processes $\pi^1, \pi^2 \in \mathcal{U}$, and a constant $\lambda \in [0,1]$.

Consider $n_\lambda = \lambda n_1 + (1-\lambda)n_2$. Let X^{t,n_i} be the wealth process starting from n_i at time t and controlled by π^i for $i = 1, 2$.

Then,

$$\pi_s^\lambda = \frac{\lambda X_s^{t,n_1}\pi_s^1 + (1-\lambda)X_s^{t,n_2}\pi_s^2}{\lambda X_s^{t,n_1} + (1-\lambda)X_s^{t,n_2}} \in \mathcal{U} \text{ by convexity}$$

Consider now $X^\lambda = \lambda X^{t,n_1} + (1-\lambda)X^{t,n_2}$, which evolves according to

$$dX_s^\lambda = X_s^\lambda\left[\pi_s^\lambda \mu + (1-\pi_s^\lambda)r\right]ds + X_s^\lambda \pi_s^\lambda \sigma dW_s$$

$$X_t^\lambda = n_\lambda$$

In other words, X^λ is a wealth process starting from n_λ at t and controlled by π^λ.

Since u is concave:

$$u\left(\lambda X_T^{t,n_1} + (1-\lambda)X_T^{t,n_2}\right) \geq \lambda u\left(X_T^{t,n_1}\right) + (1-\lambda)u\left(X_T^{t,n_2}\right)$$

$$\Rightarrow J_{t,n_\lambda} \geq \lambda \mathbb{E}\left[u\left(X_T^{t,n_1}\right)\right] + (1-\lambda)\mathbb{E}\left[u\left(X_T^{t,n_2}\right)\right]$$

$\forall \pi^1, \pi^2 \Rightarrow J_{t,n_\lambda} \geq \lambda J_{t,n_1} + (1-\lambda)J_{t,n_2} \Rightarrow$ the value function is concave in n.

Solving the HJB Equation

We are now ready then to solve the problem. The HJB reads:

$$\frac{\partial V}{\partial t} + \sup_{\pi \in \mathcal{U}}\left[\mathcal{L}^\pi V(t,n)\right] = 0, \quad V(T,n) = u(n)$$

where the infinitesimal generator is:

$$\mathcal{L}^\pi V(t,n) = n(\pi\mu + (1-\pi)r)\frac{\partial V}{\partial n} + \frac{1}{2}n^2\pi^2\sigma^2\frac{\partial^2 V}{\partial n^2}$$

Let us consider the **CRRA utility function** (Constant Relative Risk Aversion: $-n\frac{u''(n)}{u'(n)}$):

$$u(n) = \frac{n^\gamma}{\gamma}, \quad n \geq 0, \quad \gamma \neq 0, \gamma < 1$$

Then, it is possible to find an explicit solution to the problem. Let's guess a form for the candidate solution:

$$V(t,n) = \phi(t) \cdot u(n)$$

with derivatives:

$$V_t(t,n) = \phi'(t) \cdot u(n)$$
$$V_n(t,n) = \phi(t) u_n(n)$$
$$V_{nn}(t,n) = \phi(t) u_{nn}(n)$$

where $u_n(n) = n^{\gamma-1}$ and $u_{nn}(n) = (\gamma-1)n^{\gamma-2}$.

Substituting into the HJB we have:

$$\phi'(t) \cdot \frac{n^\gamma}{\gamma} + \sup_{\pi \in \mathcal{U}} \left[n(\pi\mu + (1-\pi)r)\phi(t)n^{\gamma-1} + \frac{1}{2} n^2 \pi^2 \sigma^2 (\gamma-1) n^{\gamma-2} \phi(t) \right] = 0$$

$$\Rightarrow \phi'(t) + \phi(t) \underbrace{\gamma \sup_{\pi \in \mathcal{U}} \left[\pi\mu + (1-\pi)r - \frac{1}{2}\pi^2\sigma^2(1-\gamma) \right]}_{P} = 0$$

Hence, this is a first order linear ODE, and thus (with boundary condition $\phi(T) = 1$):

$$\phi(t) = e^{P(T-t)}$$

Thus, the value function is $V(t,n) = e^{P(T-t)} u(n)$.

$V(t,n)$ is strictly increasing and concave in n. The term $\pi\mu + (1-\pi)r - \frac{1}{2}\pi^2\sigma^2(1-\gamma)$ is strictly concave in π and has a maximum at $\hat{\pi}$:

$$\text{F.O.C. for } \hat{\pi} : \mu - r - \pi\sigma^2(1-\gamma) = 0 \Rightarrow \hat{\pi} = \frac{\mu - r}{\sigma^2(1-\gamma)}$$

and the optimal wealth process evolves as:

$$dX_t = X_t(\hat{\pi}\mu + (1-\hat{\pi})r)dt + \sigma X_t \hat{\pi} dW_t$$

Is the controlled wealth process which admits a unique solution.

OPTIMAL CONSUMPTION

Let's consider the case in which the agent is interested in maximizing his/her (discounted) consumption utility. The objective function is of the form:

$$\mathbb{E}\left[\int_0^T f(t, c_t) dt \right]$$

with wealth dynamics:

$$dX_t = X_t \left[\pi_t^0 r + \pi_t^1 \mu \right] dt - c_t dt + \pi_t^1 \sigma X_t dW_t$$

subject to:

$$\left. \begin{aligned} c_t &\geq 0 \\ \pi_t^0 + \pi_t^1 &= 1 \end{aligned} \right\} \quad \forall t \geq 0. \text{ This defines the set of admissible controls } \mathcal{U}.$$

We set $\pi_t^1 = \pi_t$ and $\pi_t^0 = 1 - \pi_t$.

Problem: If f is increasing and unbounded, the agent might consume infinitely, causing wealth to become negative, as the current constraints do not prohibit it.

Solution: Consider a fixed domain $D \subseteq [0,T] \times \mathbb{R}$. If the state process X hits the boundary ∂D, the activity is stopped.

Hence, define the stopping time $\tau = \inf\{t \geq 0 : (t, X_t) \in \partial D\} \wedge T$ and consider the domain:

$$D = [0, T] \times \{n : n > 0\} \quad \text{(we require wealth to be positive, otherwise we stop!)}$$

The performance criterion becomes:

$$J_{t,n} = \mathbb{E}\left[\int_t^\tau f(s, c_s)ds\right]$$

Rewriting the wealth dynamics dX_t:

$$dX_t = X_t[(1-\pi_t)r + \pi_t\mu]dt - c_t dt + \pi_t\sigma X_t dW_t$$
$$dX_t = X_t\pi_t(\mu - r)dt + [X_t r - c_t]dt + \pi_t\sigma X_t dW_t$$

The corresponding **HJB Equation** is:

$$\partial_t v(t,n) + \sup_{c,\pi}\left\{f(t,c) + [\pi n(\mu - r) + nr - c]\partial_n v(t,n) + \frac{1}{2}\sigma^2\pi^2 n^2 \partial_{nn} v(t,n)\right\} = 0,$$

with boundary conditions:

$$v(T, n) = 0$$
$$v(t, 0) = 0$$

Let us now consider a **power utility** function with **subjective discounting**:

$$f(t, c_t) = e^{-\rho t}c^\gamma, \quad \gamma < 1$$

This is a **CRRA** utility with relative risk aversion:

$$R(c) = -\frac{cu''(c)}{u'(c)} = \frac{-c\gamma(\gamma-1)c^{\gamma-2}}{\gamma c^{\gamma-1}} = 1 - \gamma$$

The static optimization problem inside the HJB is:

$$\sup_{\pi,c}\left\{e^{-\rho t}c^\gamma + [\pi n(\mu - r) + nr - c]\partial_n v(t,n) + \frac{1}{2}\sigma^2 n^2\pi^2\partial_{nn}v(t,n)\right\}$$

The **First Order Conditions (F.O.C.s)** are:

(a) For c: $e^{-\rho t}\gamma c^{\gamma-1} = \partial_n v(t,n) \implies \gamma c^{\gamma-1} = e^{\rho t}\partial_n v(t,n)$

(b) For π: $n(\mu - r)\partial_n v(t,n) + \pi\sigma^2 n^2 \partial_{nn}v(t,n) = 0 \implies \hat{\pi} = -\frac{\partial_n v(t,n)}{n\partial_{nn}v(t,n)} \cdot \frac{\mu - r}{\sigma^2}$

We now have to find the function v.

We guess it, given the shape of the utility function, as:

$$v(t,n) = e^{-\rho t}h(t)n^\gamma$$

Since $v(T, n) = 0$, we have $h(T) = 0$. Here $h'(t)$ is the time-derivative of $h(t)$. The partial derivatives are:

$$\begin{aligned}
\partial_t v(t,n) &= e^{-\rho t} h'(t) n^\gamma - \rho e^{-\rho t} h(t) n^\gamma \\
\partial_n v(t,n) &= \gamma e^{-\rho t} h(t) n^{\gamma-1} \\
\partial_{nn} v(t,n) &= \gamma(\gamma-1) e^{-\rho t} h(t) n^{\gamma-2}
\end{aligned}$$

The optimal controls become then:

$$\begin{aligned}
\hat{\pi}(t,n) &= \frac{\mu - r}{\sigma^2}\left(-\frac{\gamma e^{-\rho t} h(t) n^{\gamma-1}}{n\gamma(\gamma-1)e^{-\rho t} h(t) n^{\gamma-2}}\right) = \boldsymbol{\frac{\mu - r}{\sigma^2(1-\gamma)}} \quad \text{CONSTANT!} \\
\hat{c}(t,n) &= \left[\frac{e^{\rho t}}{\gamma} \cdot \gamma e^{-\rho t} h(t) n^{\gamma-1}\right]^{\frac{1}{\gamma-1}} = \boldsymbol{n h(t)^{\frac{1}{\gamma-1}}} \quad \text{linear in wealth!}
\end{aligned}$$

We now substitute in the HJB and get:

$$\begin{aligned}
& e^{-\rho t} h'(t) n^\gamma - \rho e^{-\rho t} h(t) n^\gamma + e^{-\rho t} n^\gamma h(t)^{\frac{\gamma}{\gamma-1}} + \\
& \quad + \left[\frac{\mu - r}{\sigma^2(1-\gamma)} n(\mu - r) + rn - n h(t)^{\frac{1}{\gamma-1}}\right] \gamma e^{-\rho t} h(t) n^{\gamma-1} + \\
& \quad + \frac{1}{2} n^2 \left[\frac{\mu - r}{\sigma^2(1-\gamma)}\right]^2 \sigma^2 \gamma(\gamma-1) e^{-\rho t} h(t) n^{\gamma-2} = 0
\end{aligned}$$

Simplifying, we obtain:

$$n^\gamma \left\{ h'(t) + h(t) \underbrace{\left[\frac{\gamma(\mu - r)^2}{\sigma^2(1-\gamma)} + r\gamma - \frac{1}{2}\frac{(\mu - r)^2\gamma}{\sigma^2(1-\gamma)} - \rho\right]}_{=A} + h(t)^{\frac{\gamma}{\gamma-1}} \underbrace{(1-\gamma)}_{=B} \right\} = 0$$

$\Rightarrow$ This is an ODE which must be solved by $h(t)$:

$$h'(t) + A h(t) + B h(t)^{\frac{\gamma}{\gamma-1}} = 0, \quad h(T) = 0$$

This is a **Bernoulli equation** which can be solved.

Homework

Solve it explicitly. Hint: consider $y(t) = h(t)^{1-\alpha}$ with $\alpha = \frac{\gamma}{\gamma-1}$

$$\begin{aligned}
y'(t) &= (1-\alpha) h(t)^{-\alpha} h'(t) \\
h'(t) &= -A h(t) - B h(t)^\alpha \\
\Rightarrow y'(t) &= [-A h(t) - B h(t)^\alpha](1-\alpha) h(t)^{-\alpha} \\
&= -A(1-\alpha) h(t)^{1-\alpha} - (1-\alpha) B \\
y'(t) &= -A(1-\alpha) y(t) - (1-\alpha) B \\
\Rightarrow y'(t) &+ A(1-\alpha) y(t) + (1-\alpha) B = 0
\end{aligned}$$

This is **linear** in y', with $y(T) = 0$.

Once solved, it gives an $h(t)$ for which v solves the HJB equation.

The case of a random time horizon

When planning investments, it is reasonable to assume that the agent has an uncertain investment horizon (lifetime). This was originally already present in Merton's 1969 paper.

$$V(n) = \sup_{(\pi,c)\in\diamond\times\mathcal{C}} \mathbb{E}\left[\int_0^\tau e^{-\rho t} u(c_t X_t^n) dt\right]$$

where $(\pi, c) \in \diamond \times \mathcal{C}$ are **admissible** controls, with $\mathcal{C} \subseteq \mathbb{R}^+$ and the condition $\int_0^\infty |\pi_t|^2 dt + \int_0^\infty c_t dt < \infty$ a.s.

We assume that τ is a random time independent of the security processes. Its cdf will be:

$$F(t) = \mathbb{P}(\tau \leq t) = \mathbb{P}(\tau \leq t \mid \mathcal{F}_\infty)$$

where $\mathcal{F}_\infty$ is the filtration generated by W, the Brownian motion of the stock.

Hence, $V(n)$ can be written as:

$$\begin{aligned} V(n) &= \sup_{(\pi,c)\in\diamond\times\mathcal{C}} \mathbb{E}\left[\int_0^\infty e^{-\rho t} u(c_t X_t^n) \mathbb{1}_{\{t<\tau\}} dt\right] \\ &= \sup_{(\pi,c)\in\diamond\times\mathcal{C}} \mathbb{E}\left[\int_0^\infty e^{-\rho t} u(c_t X_t^n)(1 - F(t)) dt\right] \end{aligned}$$

We assume that τ is **exponential**:

$$1 - F(t) = e^{-\lambda t} \implies V(n) = \sup_{(\pi,c)\in\diamond\times\mathcal{C}} \mathbb{E}\left[\int_0^\infty e^{-(\rho+\lambda)t} u(c_t X_t^n) dt\right]$$

Let $\hat{\rho} = \rho + \lambda$. We consider the power utility $u(c) = \frac{c^\gamma}{\gamma}$ with $\gamma \neq 1$ (typically $\gamma < 1, \gamma \neq 0$).

The **HJB Equation** attached to this problem is:

$$\sup_{\pi,c}\left\{\left[n((\mu - r)\pi + r - c)\right]\partial_n v + \frac{1}{2} n^2 \pi^2 \sigma^2 \partial_{nn} v + u(cn)\right\} = \hat{\rho} v$$

where $u(cn) = \frac{(cn)^\gamma}{\gamma}$. From the **First Order Conditions (F.O.C.s)**:

($\star$) For c: $-\partial_n v + c^{\gamma-1} n^\gamma = 0 \implies \hat{c} = \left(\frac{\partial_n v}{n^\gamma}\right)^{\frac{1}{\gamma-1}}$

($\star\star$) For π: $n(\mu - r)\partial_n v + \pi n^2 \sigma^2 \partial_{nn} v = 0 \implies \hat{\pi} = -\frac{\partial_n v}{n \partial_{nn} v} \cdot \frac{\mu - r}{\sigma^2}$

Note that in the HJB above, we can separate the optimizations. The term $\sup_c\{u(cn) - cn\partial_n v\}$ is the **Legendre transform** of $u(cn)$.

We look for a solution with the candidate form:

$$v(n) = Ku(n) = K\frac{n^\gamma}{\gamma} \implies \partial_n v = Kn^{\gamma-1}, \quad \partial_{nn} v = K(\gamma - 1)n^{\gamma-2}$$

Substituting into the HJB, and letting $p = \gamma \sup_\pi \left\{(\mu - r)\pi + r - \frac{1}{2}\pi^2\sigma^2(1 - \gamma)\right\}$ be the optimized growth rate from the previous section:

$$\hat{\rho} K \frac{n^\gamma}{\gamma} = pK\frac{n^\gamma}{\gamma} + \sup_c\left\{\frac{c^\gamma n^\gamma}{\gamma} - cnKn^{\gamma-1}\right\}$$

The second supremum is attained at $\hat{c} = K^{\frac{1}{\gamma-1}}$, yielding $\frac{1-\gamma}{\gamma} n^\gamma K^{\frac{\gamma}{\gamma-1}}$. Dividing by $\frac{n^\gamma}{\gamma}$:

$$\hat{\rho} K = pK + (1-\gamma) K^{\frac{\gamma}{\gamma-1}} \implies \frac{K}{\gamma}(\hat{\rho} - p) - \left[\frac{K^{\frac{\gamma}{\gamma-1}}}{\gamma} - K^{1+\frac{1}{\gamma-1}} \right] = 0$$

This simplifies to an equation in the unknown K:

$$K^{\frac{1}{\gamma-1}}(1-\gamma) = \hat{\rho} - p \implies K = \left(\frac{1-\gamma}{\hat{\rho} - p} \right)^{1-\gamma}$$

where we require $\hat{\rho} - p = \rho + \lambda - p > 0$. We have thus found an explicit solution to the HJB.

Remarks:

1. The same problem is faced by a life insurer or a pension fund that invests on behalf of a pensioner who withdraws (consumes) optimally. However, there might be misalignment between the fund manager's and the pensioner's objectives.

2. The problem can be modified to feature an inflow $a(t)dt$ (from labour income, etc.). Notice that this won't modify the optimal controls' ($\star$ and $\star\star$) general expressions, but obviously modifies the value function v.

THE CONSUMPTION / INVESTMENT / INSURANCE PROBLEM

Suppose now that the individual pays an insurance premium to obtain an insured sum b (determined at t) which happens as a jump of a Poisson process with intensity λ_t upon his/her death. We generalize the above setup in a few directions. While alive, wealth evolves according to:

$$dX_t = (r + \pi_t(\mu - r))X_t dt + \pi_t \sigma X_t dW_t + a_t dt - c_t dt - \lambda_t^* b_t dt$$

In this formulation:

- $a_t dt$: **labour / non-financial income**;
- $c_t dt$: **consumption**;
- $\lambda_t^* b_t dt$: **insurance premium** (where λ_t^* is the mortality rate used to price the contract, which may be different from the actual mortality λ_t).

The objective is to maximize both the utility from consumption and from terminal wealth:

$$V(t, X_t) = \sup_{\pi,c,b} \mathbb{E} \left[\int_t^T e^{-\int_t^s (\rho+\lambda_u) du} \left[w_s^\gamma u(c_s) + \lambda_s (w_s^T)^\gamma u(X_s + b_s) \right] ds \right.$$
$$\left. + e^{-\int_t^T (\rho+\lambda_u) du} \Delta W^\gamma(T) u(X_T) \right]$$

where the term $\lambda_s (w_s^T)^\gamma u(X_s + b_s)$ represents the utility **in case of death**, and the final term represents the **terminal utility**.

The associated **HJB Equation** reads:

$$\partial_t v(t,n) + \sup_{\pi,c,b} \Big\{ \partial_n v(t,n) \left[r + \pi(\mu - r)n + a_t - c - \lambda_t^* b \right] \\ + \frac{1}{2}\partial_{nn} v(t,n)\pi^2\sigma^2 n^2 + w_t^\gamma u(c) + \lambda_t (w_t^T)^\gamma u(n+b) \Big\} = (\rho + \lambda_t) v(t,n)$$

with boundary condition $v(T,n) = \Delta W^\gamma(T) u(n)$.

Notice that in the supremum, the terms involving b are separated from those with c and π. Thus, the **F.O.C.s** for π and c are the same as in the case without life insurance (the addition of a_t also does not change these formulas).

The **F.O.C.** for c is:

$$-\partial_n v(t,n) + w_t^\gamma u'(c) = 0 \implies u'(c) = \frac{\partial_n v(t,n)}{w_t^\gamma}$$

This implies that the **marginal utility of consumption** equals the (rescaled) **marginal gain of v**. The **F.O.C.** for b is:

$$-\partial_n v(t,n)\lambda_t^* + \lambda_t (w_t^T)^\gamma u'(n+b) = 0$$

Let us now focus on the **power utility** case: $u(n) = \frac{n^{1-\gamma}}{1-\gamma}$. We consider the candidate value function (*ansatz*):

$$v(t,n) = \frac{1}{1-\gamma} f(t)^\gamma (n + g(t))^{1-\gamma}$$

where $g(t)$ represents the **human capital** (arising due to the income stream a_t).

Steps to solve the HJB:

1. **Evaluate the derivatives of the candidate value function:**

$$\begin{aligned} \partial_t v(t,n) &= \frac{\gamma}{1-\gamma}\left(\frac{n+g(t)}{f(t)}\right)^{1-\gamma} f'(t) + \left(\frac{n+g(t)}{f(t)}\right)^{-\gamma} g'(t) \\ \partial_n v(t,n) &= \left(\frac{n+g(t)}{f(t)}\right)^{-\gamma} \\ \partial_{nn} v(t,n) &= -\gamma\left(\frac{n+g(t)}{f(t)}\right)^{-\gamma-1} \frac{1}{f(t)} \end{aligned}$$

2. **Plug these derivatives into the F.O.C.s:**

   (a) **For c:** $-\left(\frac{n+g(t)}{f(t)}\right)^{-\gamma} + w_t^\gamma c^{-\gamma} = 0 \implies \hat{c} = w_t \frac{n+g(t)}{f(t)}$

   (b) **For π:** $\left(\frac{n+g(t)}{f(t)}\right)^{-\gamma} (\mu - r)n - \gamma\left(\frac{n+g(t)}{f(t)}\right)^{-\gamma-1} \frac{1}{f(t)}\pi\sigma^2 n^2 = 0$

$$\implies \hat{\pi} = \frac{1}{\gamma}\frac{\mu - r}{\sigma^2}\frac{n+g(t)}{n}$$

   (c) **For b:** $-\left(\frac{n+g(t)}{f(t)}\right)^{-\gamma} \lambda_t^* + \lambda_t (w_t^T)^\gamma (n+b)^{-\gamma} = 0$

$$\implies \hat{b} = \left(\frac{n+g(t)}{f(t)}\right)\left(\frac{\lambda_t^*}{\lambda_t}\right)^{-\frac{1}{\gamma}} w_t^T - n$$

2. **Optimal Investment Strategy ($\hat{\pi}$):**

$$\hat{\pi} = \underbrace{\left(\frac{\mu - r}{\sigma^2 \gamma}\right)}_{\text{Merton's standard}} \cdot \frac{n + g(t)}{n}$$

   where $g(t)$ is the **human capital**.

3. **Optimal Insurance Benefit ($\hat{b}$):**

$$\hat{b} = \underbrace{\frac{w_t^T}{f(t)}}_{A} \underbrace{\left(\frac{\lambda_t}{\lambda_t^*}\right)^{\frac{1}{\gamma}}}_{B} (n + g(t)) - n$$

   Notice that $n + \hat{b}$ is the **total wealth before death** multiplied by two factors.

   If $A \times B = 1 \implies \hat{b} = g(t)$, which means the insurance benefit is exactly the **human capital lost**.

The ratio $\frac{\lambda}{\lambda^*}$ represents the ratio between the objective mortality rate and the pricing insurance rate.

- If $\lambda > \lambda^*$, insurance is "cheap" (one should over-insure!).
- The higher the ratio $\frac{\lambda}{\lambda^*}$, the higher the optimal benefit $\hat{b}$.
- w_t^T weights utility at the time of death: $\uparrow w_t^T \implies \uparrow \hat{b}$.
- $f(t)$ is the preference-adjusted discount factor: $\uparrow f(t) \implies \downarrow \hat{b}$.

CORPORATE FINANCE MODELS (Morellec, Brumberg & Rochet, Ch. 1)

We will study 2 optimal capital structure problems:

- optimal debt issuance
- optimal dividends

Consider a company with the following structure:

Assets			Liabilities		
LIQUID RESERVES	$\rightarrow$	X_t	D_t	$\rightarrow$	DEBT
MARKET VALUE OF PRODUCTIVE ASSETS	$\rightarrow$	S_t	E_t	$\rightarrow$	EQUITY

We consider S_t to evolve as:

$$dS_t = S_t[rdt + \sigma dW_t], \quad S_0 = s$$

with b being the payout rate.

How can we price D_t? How can we select optimal how to finance the company?

Consider $X_t = 0$ and that the firm has a **nominal debt value** F.

DEFAULT AT MATURITY (Merton, 1973)

Then:

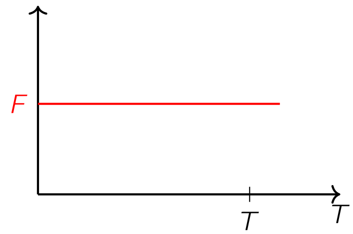
1. The discounted value of assets is a martingale:

$$\mathbb{E}\left[e^{-rT}S_T\right] = S$$

2. The value of equity is:

$$E_0 = \mathbb{E}\left[e^{-rT}\max(0, S_T - F)\right]$$

which corresponds to the **payoff of a European call option**.



If $S_T > F$: firm is solvent, repays F to debtholders, $S_T - F$ is distributed to shareholders

3. The value of debt is:

$$\begin{aligned} D_0 &= \mathbb{E}\left[e^{-rT}\min(F, S_T)\right] \\ &= \mathbb{E}\left[e^{-rT}(S_T - \max(0, S_T - F))\right] \end{aligned}$$

and $E_0 + D_0 = \mathbb{E}\left[e^{-rT}S_T\right] = S$.

Given an assumption (S is **GBM**), E_0 and D_0 can be found using the Black-Scholes option pricing formula.

DEFAULT BEFORE DEBT MATURITY (Black and Cox, 1976)

Suppose now that shareholders can decide to stop paying debt coupons (and determine default) prior to maturity.

→ we specify an exogenous **default BOUNDARY** (at no cost) $\{B_t, t \in [0, T]\}$ and **LIQUIDATION** is triggered whenever $S_t \leq B_t$.

Price of a bond with finite maturity

How do we price a bond described above?

$$\text{Define } \eta = \min\{\tau, T\}, \quad \tau := \inf\{t \geq 0 : S_t = B_t\}$$

$$\text{and} \quad p(\eta) = \begin{cases} B_\eta & \text{if } \eta < T \\ F(S_\tau) & \text{if } \eta = T \end{cases}$$

$$\text{Then} \quad P(t, S) = \mathbb{E}\left[\int_t^\eta e^{-r(l-t)}\underbrace{C(l, S_l)}_{\text{coupon}}\,dl + e^{-r(\eta-t)}p(\eta)\,\middle|\, S_t = S\right]$$

Given an assumption (S is **GBM**), E_0 and D_0 can be found using the Black-Scholes option pricing formula.

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$$p(\eta) = \begin{cases} B_\eta & \text{if } \eta < T \\ F(S_\tau) & \text{if } \eta = T \end{cases}$$

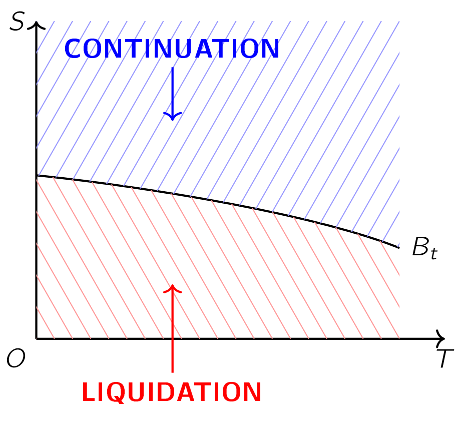
Then

$$P(t, S) = \mathbb{E}\left[\int_t^\eta e^{-r(l-t)} \underbrace{C(l, S_l)}_{\text{coupon}} \, dl + e^{-r(\eta-t)} p(\eta) \,\middle|\, S_t = S \right]$$

which, using the **Feynman-Kac formula***, can be associated to the following free-boundary problem:

$$\begin{cases} \underbrace{rP(t,S)}_{\text{L.H.S. of } (*)} = \underbrace{\overbrace{\partial_t P(t,S) + rS\partial_S P(t,S) + \frac{\sigma^2 S^2}{2}\partial_{SS} P(t,S)}^{\text{EXPECTED CAPITAL GAINS}} + \underbrace{C(t,S)}_{\text{COUPON}}}_{\text{FLOW OF EXPECTED GAINS OF HOLDING DEBT}}, \quad 0 \leq t \leq T;\ S \geq B_t; \\ P(t, B_t) = B_t; \quad P(T, S) = F(S) \end{cases} \tag{$*$}$$

By **no arbitrage**, the R.H.S. of $(*)$ must equal the flow of income obtained by investing $P(t, S)$ at the riskless rate (the L.H.S. of $(*)$).



$P(T, S) = F(S)$

$P(t, S) = B_t$

at the threshold

In the blue region

$P(t, S)$ solves the PDE.

* FEYNMAN-KAC FORMULA (see Øksendal, or Shreve, for instance)

- It establishes a link between **stochastic processes** and **PDEs**.

In particular, for boundary value problems, the function

$$h(n) = \mathbb{E}^n \left[\int_0^{\tau_D} e^{-\int_0^t q(X_s)ds} g(X_t)dt + e^{-\int_0^{\tau_D} q(X_s)ds} \phi(X_{\tau_D}) \right]$$

solves

$$\begin{cases} \mathcal{L}h(n) - q(n)h(n) = -g(n) & \text{on } D \\ \lim_{n \to y} h(n) = \phi(y) & y \in \partial D \end{cases}$$

where ϕ is bounded and g satisfies $\mathbb{E}^n \left[\int_0^{\tau_D} |g(X_s)| ds \right] < \infty$.

Price of a consol bond ($T = +\infty$)

Consider $C(t, S) = C$, $T = +\infty \to$ **CONSOL BOND**, and $B_t = S_B > 0$. Its price $D(S)$ is given by the system:

$$\begin{cases} rD(S) = rS\partial_S D(S) + \frac{\sigma^2}{2} S^2 \partial_{SS} D(S) + C \\ D(S_B) = S_B \end{cases}$$

This is an ODE (the problem is stationary) and thus the price can be found by solving it. The associated homogeneous differential equation has the general solution:

$$f(S) = K_1 S^{\gamma_1} + K_2 S^{\gamma_2}$$

where γ_1, γ_2 solve the characteristic equation:

$$r = r\gamma + \frac{\sigma^2}{2}\gamma(\gamma - 1), \quad \text{with } \gamma_1 < 0 < \gamma_2.$$

Since $\frac{C}{r}$ is a particular solution of the ODE, the general solution is:

$$D(S) = K_1 S^{\gamma_1} + K_2 S^{\gamma_2} + \frac{C}{r}$$

K_1 and K_2 can be found by observing that, when $S \to \infty$, the bond must approach its nominal value:

$$\int_0^\infty C e^{-rt} dt = \frac{C}{r} \implies D(\infty) \to \frac{C}{r}.$$

Since $\gamma_1 < 0 < \gamma_2$, this implies that $K_2 = 0$. K_1 is then determined by the boundary condition $D(S_B) = S_B$:

$$K_1 S_B^{\gamma_1} + \frac{C}{r} = S_B \implies K_1 = S_B^{1-\gamma_1} - \frac{C}{r} S_B^{-\gamma_1}$$

Substituting back, we obtain:

$$\begin{aligned} D(S) &= \left[S_B^{1-\gamma_1} - \frac{C}{r} S_B^{-\gamma_1} \right] \cdot S^{\gamma_1} + \frac{C}{r} \\ &= \left(\frac{S}{S_B} \right)^{\gamma_1} \left[S_B - \frac{C}{r} \right] + \frac{C}{r}. \end{aligned}$$

Homework 1. Similarly, price the equity.

2. How should equityholders choose S_B optimally?

DYNAMIC TRADE-OFF THEORY (How to choose F? or C? or S_B?)

(Leland, 1994)

Why do firms issue debt? Trade-off theory says it is because the interests are **TAX DEDUCTIBLE**.

Suppose the firm's debt is a consol bond with coupon C. Shareholders choose the default threshold S_B:

$$\tau := \inf\{t > 0 : S_t \leq S_B\}$$

The firm pays taxes $\theta \in [0,1]$ on the operating income βS_t (where β is the payout rate), net of coupon payments:

$$T = \theta[\beta S_t - C] \implies (1-\theta)(\beta S_t - C) \text{ is the flow to shareholders.}$$

At date τ, the firm is liquidated and debtholders get $(1-\alpha)S_B$, where $\alpha \in [0,1]$ represents **DEFAULT COSTS**. The asset value evolves as:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad \mu := r - \beta$$

Then:

$$D(S) = \mathbb{E}\left[\int_0^\tau e^{-rt} C dt + e^{-r\tau}(1-\alpha)S_B \,\middle|\, S_0 = S\right]$$

D satisfies the free-boundary problem we characterized above:

$$\begin{cases} rD(S) = \mu S D'(S) + \frac{\sigma^2}{2} S^2 D''(S) + C, & S > S_B \\ D(S_B) = (1-\alpha)S_B, \quad D(S) \to \frac{C}{r} \text{ as } S \to +\infty \end{cases}$$

The value of equity is:

$$E(S) = \mathbb{E}\left[(1-\theta)\int_0^\tau e^{-rt}(\beta S_t - C)dt \,\middle|\, S_0 = S\right]$$

which gives the following problem:

$$\begin{cases} rE(S) = \mu S E'(S) + \frac{\sigma^2}{2} S^2 E''(S) + (1-\theta)(\beta S - C), & S > S_B \\ E(S_B) = 0, \quad E(S) \sim (1-\theta)\left(S - \frac{C}{r}\right) \text{ as } S \to +\infty \end{cases}$$

Homework

Solve these problems explicitly. In particular, show that:

$$D(S) = \frac{C}{r} - \left[\frac{C}{r} - (1-\alpha)S_B\right]\left(\frac{S}{S_B}\right)^{\gamma_1}$$

$$E(S) = (1-\theta)\left[S - \frac{C}{r} + \underbrace{\left(\frac{C}{r} - S_B\right)\left(\frac{S}{S_B}\right)^{\gamma_1}}_{\text{LIMITED-LIABILITY OPTION}}\right]$$

We now consider that C, S_B are chosen optimally:

1. The firm chooses C

2. Debt is issued (at price D_0)

3. Shareholders set S_B (maximizing shareholder value!)

They do not commit ex-ante: they select it ex-post to maximize their value (and creditors anticipate it).

S_B is chosen to maximize $E(S)$ (or, equivalently, the value of the LL option):

$$\max_{S_B}\left[\frac{C}{r}-S_B\right]\cdot S_B^{-\gamma_1}$$

This defines a link between C and S_B!

$$\to -\gamma_1\frac{C}{r}S_B^{-\gamma_1-1}+S_B^{-\gamma_1}(1-\gamma_1)=0 \implies S_B=\frac{\gamma_1}{\gamma_1-1}\frac{C}{r}<\frac{C}{r}$$

Hence, this is what creditors expect!

2) D_0 can be found using debt's valuation formula:

$$D_0=\frac{C}{r}-\left[\frac{C}{r}-(1-\alpha)S_B\right]\left(\frac{S}{S_B}\right)^{\gamma_1}$$

3) Suppose that the total investment needed at time 0 to start the firm is I.

Then, shareholder's value at $t=0$, i.e., the difference between equity value and the amount that shareholders have invested in the firm is:

$$\begin{aligned}
SV_0 &= E_0-(I-D_0)\quad \text{(if } I=0\text{, this is } D_0+E_0\text{!)}\\
&=(1-\theta)\left[S-\frac{C}{r}+\left(\frac{C}{r}-S_B\right)\left(\frac{S}{S_B}\right)^{\gamma_1}\right]-I+\frac{C}{r}-\left[\frac{C}{r}-(1-\alpha)S_B\right]\left(\frac{S}{S_B}\right)^{\gamma_1}\\
&=\underbrace{(1-\theta)S-I}_{\text{NPV of earnings after-tax}}\;\underbrace{+\theta\frac{C}{r}\left[1-\left(\frac{S}{S_B}\right)^{\gamma_1}\right]}_{\text{expected tax rebate}}\underbrace{-(\alpha-\theta)S_B\left(\frac{S}{S_B}\right)^{\gamma_1}}_{\text{expected net liq costs}}
\end{aligned}$$

SV_0 must be maximized given the optimal choice of S_B, S_B^*, i.e., with $S_B=S_B^*=\frac{\gamma_1}{\gamma_1-1}\frac{C}{r}$.

Indeed,

$$SV_0(S_B^*)=(1-\theta)S-I+\frac{\gamma_1-1}{\gamma_1}S_B^*\left[\theta+\frac{\theta-\alpha\gamma_1}{\gamma_1-1}\left(\frac{S}{S_B^*}\right)^{\gamma_1}\right]$$

Shareholder value is maximized with

$$SV_0'(S_B^*)=0\to\frac{\gamma_1-1}{\gamma_1}\theta+\frac{\theta-\alpha\gamma_1}{\gamma_1-1}S^{\gamma_1}(1-\gamma_1)(S_B^*)^{-\gamma_1}=0$$

$$(S_B^*)^{-\gamma_1}=\frac{1-\gamma_1}{\gamma_1}\theta\cdot\frac{1}{1-\gamma_1}S^{-\gamma_1}\frac{\gamma_1-1}{\theta-\alpha\gamma_1}$$

$$S_B^*=S\left[\frac{\theta}{\gamma_1}\frac{\gamma_1-1}{\theta-\alpha\gamma_1}\right]^{-\frac{1}{\gamma_1}}$$

OPTIMAL DIVIDENDS (Jeanblanc & Shiryaev 1995)

Consider X to be the liquid **reserves** $X = (X_t)_{t\geq 0}$ of a company with no investment. Reserves are not remunerated and there is no debt.

$$dX_t = \underbrace{\mu dt + \sigma dW_t}_{\substack{\text{CUMULATIVE} \\ \text{NET EARNINGS}}} - dz_t, \quad X_0 = n \geq 0, \quad \mu, \sigma > 0$$

where W_t is a standard Wiener process. The term $\mu dt + \sigma dW_t$ is an **ABM** (Arithmetic BM). The process z_t is a non-decreasing, adapted **DIVIDEND PAYMENT STRATEGY**. We have $dX_t = dz_t = 0$ when $t \geq \tau$, where τ is the time of **bankruptcy**.

The company (managers) maximize shareholder's value and are risk-neutral (alternatively, they could maximize utility from dividends):

$$V(n) = \sup_z \mathbb{E}_n \left[\int_0^\infty e^{-\rho t} dz_t \right]$$

Note that:

$$\int_0^\infty e^{-\rho t} dz_t = z_0 + \int_{(0,\infty)} e^{-\rho t} dz_t$$

The objective is to find the supremum over all admissible dividend processes $z = (z_t)_{t\geq 0}$.

Remark 186. *We consider Z such that $dZ_t = u(X_t)dt, Z_0 = z_0(n)$.*

Suppose u and z_0 are measurable functions such that $0 \leq u(n) \leq K < \infty$ (this is the **bounded case***) and $0 \leq z_0(n) \leq n$. The dynamics of the process X_t are given by:*

$$dX_t = (\mu - u(X_t))dt + \sigma dW_t,$$

which has a unique strong solution.

Also, we consider τ as the **bankruptcy time***, i.e.,*

$$\tau = \inf\{t : X_t = 0\}, \quad s.t. \ X_t = 0 \ \forall t \geq \tau.$$

Let us assume $z_0 = 0$. Then the value function is defined as:

$$V(n) = \sup_{0\leq u(n)\leq K} \mathbb{E}_n \int_0^\tau e^{-\rho t} u(X_t) dt.$$

As usual, we want to find $\tilde{V}(n)$ using **Bellman's equation** *(and then verify that it is optimal). We thus look for an admissible u that solves:*

$$\sup_{0\leq u\leq K} \left[\mathcal{L}_u \tilde{V}(n) + u \right] = \rho \tilde{V}(n)$$

where the generator is $\mathcal{L}_u \tilde{V}(n) = (\mu - u)\partial_n \tilde{V} + \frac{1}{2}\sigma^2 \partial_{nn} \tilde{V}$. This leads to:

$$\mu \partial_n \tilde{V} + \frac{1}{2}\sigma^2 \partial_{nn} \tilde{V} - \rho \tilde{V} + \sup_{0\leq u\leq K} \left[u(1 - \partial_n \tilde{V}) \right] = 0.$$

Hence, given $\tilde{V}$, the optimal control $\tilde{u}(n)$ is given by:

$$\tilde{u}(n) = \begin{cases} K & \text{if } 1-\partial_n\tilde{V} \geq 0 \\ 0 & \text{if } 1-\partial_n\tilde{V} < 0 \end{cases}$$

For n such that $\tilde{u}(n)=0$, it must hold:

$$\mu\partial_n\tilde{V}(n) + \frac{1}{2}\sigma^2\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) = 0,$$

while if $\tilde{u}(n)=K$ we have:

$$\mu\partial_n\tilde{V}(n) + \frac{1}{2}\sigma^2\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) + K(1-\partial_n\tilde{V}(n)) = 0.$$

Intuitively, given the structure of $\tilde{u}$, there exists a threshold $\tilde{n}$ such that for $n \geq \tilde{n}$ dividends are paid out at maximal possible speed, and $u(n)=0$ for $n<\tilde{n}$.

Hence, the solution features a **free boundary** *$\tilde{n}$ such that:*

$$\mu\partial_n\tilde{V}(n) + \frac{\sigma^2}{2}\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) = 0 \quad \text{if } n<\tilde{n} \tag{*}$$
$$(\mu-K)\partial_n\tilde{V}(n) + \frac{\sigma^2}{2}\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) + K = 0 \quad \text{if } n\geq\tilde{n} \tag{**}$$

Hence, given $\tilde{V}$,

Remark 187. *The optimal control $\tilde{u}(n)$ is given by:*

$$\tilde{u}(n) = \begin{cases} K & \text{if } 1-\partial_n\tilde{V} \geq 0 \\ 0 & \text{if } 1-\partial_n\tilde{V} < 0 \end{cases}$$

For n s.t. $\tilde{u}(n)=0$, it must hold:

$$\mu\partial_n\tilde{V}(n) + \frac{1}{2}\sigma^2\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) = 0,$$

while if $\tilde{u}(n)=K$ we have

$$\mu\partial_n\tilde{V}(n) + \frac{1}{2}\sigma^2\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) + K\left(1-\partial_n\tilde{V}(n)\right) = 0.$$

Intuitively, given the structure of $\tilde{u}$, there exists a $\tilde{n}$ s.t. for $n\geq\tilde{n}$ dividends are paid out at maximal possible speed, and s.t. $u(n)=0$ for $n<\tilde{n}$.

Hence, the solution features a **free boundary** *$\tilde{n}$ s.t.*

$$\mu\partial_n\tilde{V}(n) + \frac{\sigma^2}{2}\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) = 0 \quad \text{if } n<\tilde{n} \tag{*}$$
$$(\mu-K)\partial_n\tilde{V}(n) + \frac{\sigma^2}{2}\partial_{nn}\tilde{V}(n) - \rho\tilde{V}(n) + K = 0 \quad \text{if } n\geq\tilde{n} \tag{**}$$

Remark 188. *For (**), we have*

$$V''(n) + \frac{\mu-\kappa}{\frac{\sigma^2}{2}}V'(n) - \frac{p}{\frac{\sigma^2}{2}}V(n) + \frac{K}{\frac{\sigma^2}{2}} = 0$$

which is a 2^{nd} order ODE. Its general solution is

$$V(n) = C_1 e^{\lambda_1 n} + C_2 e^{\lambda_2 n} + \frac{K}{p}$$

with λ_1, λ_2 solving

$$\lambda^2 + \frac{\mu - \kappa}{\frac{\sigma^2}{2}}\lambda - \frac{p}{\frac{\sigma^2}{2}} = 0$$

$$\implies \lambda_{1,2} = \frac{\frac{\kappa-\mu}{\frac{\sigma^2}{2}} \pm \sqrt{\left(\frac{\mu-\kappa}{\frac{\sigma^2}{2}}\right)^2 + 4\frac{p}{\frac{\sigma^2}{2}}}}{2}$$

Notice that $\lambda_1 > 0, \lambda_2 < 0$ and hence since $\tilde{V}(n) \leq \frac{K}{p}$ because $u(n) \leq K$, $C_1 = 0$!

Remark 189. $\implies V(n) = \frac{K}{p} + C_2 e^{\lambda_2 n}$,

and $C_2 \leq 0$ because $V(n)$ non-decreasing:

$$\frac{K}{p} + \lambda_2 C_2 e^{\lambda_2 n} \geq 0$$

$$\implies \lambda_2, C_2 \leq 0 \text{ both!}$$

Instead, when $n < \check{n}$, the function satisfies

$$(*) \to \frac{1}{2}\sigma^2 \partial_{nn}\tilde{V}(n) + \mu\partial_n \tilde{V}(n) - p\tilde{V}(n) = 0$$

Hence, it is of the form

$$V(n) = A_1 e^{r_1 n} + A_2 e^{r_2 n}$$

with r_1, r_2 solving

$$r^2 + \frac{\mu}{\frac{\sigma^2}{2}} r - \frac{p}{\frac{\sigma^2}{2}} = 0, \quad D = \Delta$$

$$r_{1,2} = \frac{-\frac{\mu}{\frac{\sigma^2}{2}} \pm \sqrt{\left(\frac{\mu}{\frac{\sigma^2}{2}}\right)^2 + \frac{p}{\frac{\sigma^2}{2}}}}{2}$$

28 Appendix

28.1 d-systems and Monotone Class Theorem

Definition 190 (d-system)

[D. Williams, A1.2] A collection $\mathcal{D}$ of subsets of E is a ***d*-system** if:

(i) $E \in \mathcal{D}$

(ii) $A, B \in \mathcal{D}$ and $A \subseteq B \implies B \setminus A \in \mathcal{D}$

(iii) $(A_n)_{n\in\mathbb{N}} \subset \mathcal{D}$ s.t. $A_n \subseteq A_{n+1}\ \forall n \in \mathbb{N} \implies \bigcup_{n\in\mathbb{N}} A_n \in \mathcal{D}$

Proposition 191

A collection of sets Σ on E is a σ-algebra iff it's a π-system and a d-system.

Lemma 192 (Dynkin's lemma)

If $\mathcal{I}$ is a π-system on E then $\sigma(\mathcal{I}) = d(\mathcal{I})$.

Theorem 193 (Monotone class theorem)

[D. Williams, Thm 3.14] Let $\mathcal{H}$ be a class of (bounded) functions $f : E \to \mathbb{R}$ s.t.

(i) $\mathcal{H}$ is a vector space.

(ii) $\mathcal{H}$ contains the constant function $f(x) = 1$.

(iii) If $(f_n)_{n\in\mathbb{N}} \subset \mathcal{H}$ s.t. $0 \le f_n \le f_{n+1}\ \forall n \in \mathbb{N}$ s.t. $f_n \uparrow f$, then $f \in \mathcal{H}$.

Then, if $\mathcal{H}$ contains the indicator functions of all sets from a π-system $\mathcal{I}$ on E, then $\mathcal{H} \supseteq \sigma(\mathcal{I})$.

Remark 194 (Sketch of proof)**.** *The collection $\mathcal{D} := \{F \subseteq E : \mathbb{1}_F \in \mathcal{H}\}$ is a d-system on E that contains a π-system $\mathcal{I}$, then $\sigma(\mathcal{I}) \subseteq \mathcal{D}$. If $f \in m\sigma(\mathcal{I})$ then it can be approximated by simple functions.*

28.2 Uniformly integrable families of r.v.'s

Definition 195

A family $\{X_\alpha, \alpha \in I\}$ of r.v.'s is **uniformly integrable** (u.i.) if

$$\lim_{c\to\infty} \sup_{\alpha\in I} \mathbb{E}\left[|X_\alpha| \mathbb{1}_{\{|X_\alpha|>c\}}\right] = 0.$$

Homework 196

Show that the family formed by a single random variable $X \in \mathcal{L}^1$ is u.i.

Lemma 197

Let $X \in \mathcal{L}^1$. For any $\varepsilon > 0$, $\exists \delta_\varepsilon > 0$ s.t.

$$\mathbb{E}\left[|X| \mathbb{1}_A\right] < \varepsilon \quad \forall A \in \mathcal{F} \text{ s.t. } \mathbb{P}(A) < \delta_\varepsilon.$$

Proof. Fix $\varepsilon > 0$. For arbitrary $\lambda > 0$ we have

$$\begin{aligned}\mathbb{E}\left[|X| \mathbb{1}_A\right] &= \mathbb{E}\left[|X| \left(\mathbb{1}_{A \cap \{|X| > \lambda\}} + \mathbb{1}_{A \cap \{|X| \leq \lambda\}}\right)\right] \\ &\leq \mathbb{E}\left[|X| \mathbb{1}_{\{|X| > \lambda\}}\right] + \lambda \mathbb{P}(A).\end{aligned}$$

By MON $\lim_{\lambda \to \infty} \mathbb{E}\left[|X| \mathbb{1}_{\{|X| > \lambda\}}\right] = 0$ and therefore $\exists \lambda_\varepsilon > 0$ s.t. $\mathbb{E}\left[|X| \mathbb{1}_{\{|X| > \lambda_\varepsilon\}}\right] < \frac{\varepsilon}{2}$. Moreover I can choose $\mathbb{P}(A) < \delta_\varepsilon$ for $\delta_\varepsilon > 0$ s.t. $\lambda_\varepsilon \delta_\varepsilon < \frac{\varepsilon}{2}$. Then $\mathbb{E}\left[|X| \mathbb{1}_A\right] < \varepsilon$ $\forall A \in \mathcal{F}$ s.t. $\mathbb{P}(A) < \delta_\varepsilon$. $\square$

Lemma 198

Let $X \in \mathcal{L}^1$ and $(\mathcal{G}_\beta)_{\beta \in B} \subset \mathcal{F}$ be a family of σ-algebras. Then $\{\mathbb{E}[X|\mathcal{G}_\beta], \beta \in B\}$ is u.i.

Proof. We want to show that $\sup_\beta \mathbb{E}\left[|\mathbb{E}[X|\mathcal{G}_\beta]| \mathbb{1}_{\{|\mathbb{E}[X|\mathcal{G}_\beta]| > c\}}\right] \to 0$ as $c \to \infty$. By Jensen's inequality, $|\mathbb{E}[X|\mathcal{G}_\beta]| \leq \mathbb{E}[|X||\mathcal{G}_\beta]$. Then

$$\begin{aligned}\sup_\beta \mathbb{E}\left[|\mathbb{E}[X|\mathcal{G}_\beta]| \mathbb{1}_{\{|\mathbb{E}[X|\mathcal{G}_\beta]| > c\}}\right] &\leq \sup_\beta \mathbb{E}\left[\mathbb{E}[|X||\mathcal{G}_\beta] \mathbb{1}_{\{\mathbb{E}[|X||\mathcal{G}_\beta] > c\}}\right] \\ &= \sup_\beta \mathbb{E}\left[|X| \mathbb{1}_{\{\mathbb{E}[|X||\mathcal{G}_\beta] > c\}}\right]\end{aligned}$$

where the last equality follows from the definition of conditional expectation since $\{\mathbb{E}[|X||\mathcal{G}_\beta] > c\} \in \mathcal{G}_\beta$.

Remark 199. *The inequality* $\mathbb{E}\left[|\mathbb{E}[X|\mathcal{G}_\beta]| \mathbb{1}_{\{|\mathbb{E}[X|\mathcal{G}_\beta]| > c\}}\right] \leq \mathbb{E}\left[|X| \mathbb{1}_{\{\mathbb{E}[|X||\mathcal{G}_\beta] > c\}}\right]$ *is left as* **HW**.

Let $A_\beta := \{\mathbb{E}[|X||\mathcal{G}_\beta] > c\}$. By Markov's inequality, $\mathbb{P}(A_\beta) \leq \frac{\mathbb{E}[\mathbb{E}[|X||\mathcal{G}_\beta]]}{c} = \frac{\mathbb{E}[|X|]}{c}$ $\forall \beta \in B$. Then I can choose $c_\varepsilon > 0$ s.t. $\mathbb{P}(A_\beta) < \delta_\varepsilon$ $\forall \beta \in B$. By Lemma 197, $\mathbb{E}\left[|X| \mathbb{1}_{A_\beta}\right] < \varepsilon$ $\forall \beta \in B$ and therefore

$$\lim_{c \to \infty} \sup_\beta \mathbb{E}\left[|\mathbb{E}[X|\mathcal{G}_\beta]| \mathbb{1}_{\{|\mathbb{E}[X|\mathcal{G}_\beta]| > c\}}\right] \leq \varepsilon.$$

Since ε is arbitrary, the result follows by letting $\varepsilon \to 0$. $\square$

Lemma 200

Let $X \in \mathcal{L}^1, X \geq 0$. If $(X_n)_{n \in \mathbb{Z}^+} \subset \mathcal{L}^1$ is s.t. $|X_n| \leq \mathbb{E}[X|\mathcal{F}_n]$ $\forall n \in \mathbb{Z}^+$, then $(X_n)_{n \in \mathbb{Z}^+}$ is u.i.

Proof. We have

$$\begin{aligned}\mathbb{E}\left[|X_n| \mathbb{1}_{\{|X_n| > c\}}\right] &\leq \mathbb{E}\left[\mathbb{E}[X|\mathcal{F}_n] \mathbb{1}_{\{\mathbb{E}[X|\mathcal{F}_n] > c\}}\right] \\ &= \mathbb{E}\left[X \mathbb{1}_{\{\mathbb{E}[X|\mathcal{F}_n] > c\}}\right].\end{aligned}$$

As in the proof of Lemma 198, we can choose $c_\varepsilon > 0$ s.t. $\mathbb{P}(\mathbb{E}[X|\mathcal{F}_n] > c_\varepsilon) < \delta_\varepsilon$ $\forall n \in \mathbb{Z}^+$ and therefore, by Lemma 197, $\mathbb{E}\left[X \mathbb{1}_{\{\mathbb{E}[X|\mathcal{F}_n] > c_\varepsilon\}}\right] < \varepsilon$. Thus

$$\lim_{c\to\infty} \sup_{n\in\mathbb{Z}^+} \mathbb{E}\left[|X_n| \mathbb{1}_{\{|X_n|>c\}}\right] \leq \varepsilon.$$

Letting $\varepsilon \to 0$ gives the result. □

28.3 Convergence Theorems for Martingales

Remark 201. *If $\{X_\alpha, \alpha \in I\}$ is u.i. then $\sup_{\alpha\in I} \mathbb{E}[|X_\alpha|] < \infty$, because*

$$\sup_{\alpha\in I} \mathbb{E}[|X_\alpha|(\mathbb{1}_{\{|X_\alpha|\leq c\}} + \mathbb{1}_{\{|X_\alpha|>c\}})] \leq c + \sup_{\alpha\in I} \mathbb{E}[|X_\alpha| \mathbb{1}_{\{|X_\alpha|>c\}}]$$

and for $c > 0$ sufficiently large, this is $\leq c + 1$.

Remark 202. *The following results provide a summary of martingale convergence theorems. For more details, see Chapter 10 in the lecture notes for PFF.*

Lemma 203 (Doob's upcrossing)
Let (X_n) be a super-martingale. Then

$$(b - a)\mathbb{E}[U_N[a, b]] \leq \mathbb{E}[(X_N - a)^+].$$

Corollary 204
Let $(X_n)_{n\in\mathbb{Z}^+}$ be a super-martingale with $\sup_n \mathbb{E}[|X_n|] < \infty$ (i.e., bounded in $\mathcal{L}^1$). Then

$$(b - a)\mathbb{E}[U_\infty[a, b]] \leq |a| + \sup_n \mathbb{E}[|X_n|] < \infty$$

which implies $\mathbb{P}(U_\infty[a, b] = \infty) = 0$.

Theorem 205
Let $(X_n)_{n\in\mathbb{Z}^+}$ be a super-martingale bounded in $\mathcal{L}^1$. Then

$$\lim_{n\to\infty} X_n = X_\infty \quad \mathbb{P}\text{-a.s.}$$

Proposition 206
Let $(M_n)_{n\in\mathbb{Z}^+}$ be a **u.i. martingale**. Then:

(a) $M_\infty := \lim_{n\to\infty} M_n$ exists a.s. and in $\mathcal{L}^1$.

(b) $M_n = \mathbb{E}[M_\infty|\mathcal{F}_n]$.

Proof. Since $(M_n)_{n\in\mathbb{Z}^+}$ is u.i., then it's bounded in $\mathcal{L}^1$ and therefore $M_\infty = \lim_{n\to\infty} M_n$ exists a.s.

Notice that by Fatou's lemma, $\mathbb{E}[|M_\infty|] \leq \liminf_{n\to\infty} \mathbb{E}[|M_n|] < \infty$, so $M_\infty \in \mathcal{L}^1$.

> **Homework 207**
> Show that $\{M_n - M_\infty, n \in \mathbb{Z}^+\}$ is u.i.

We now prove convergence in $\mathcal{L}^1$. We have:

$$\begin{aligned}\mathbb{E}[|M_n - M_\infty|] &= \mathbb{E}[|M_n - M_\infty|(\mathbb{1}_{\{|M_n - M_\infty| \le \varepsilon\}} + \mathbb{1}_{\{|M_n - M_\infty| > \varepsilon\}})] \\ &\le \varepsilon + \mathbb{E}[|M_n - M_\infty|\mathbb{1}_{\{|M_n - M_\infty| > \varepsilon\}}]\end{aligned}$$

We can split the second term as:

$$\begin{aligned}\mathbb{E}[|M_n - M_\infty|\mathbb{1}_{\{|M_n - M_\infty| > \varepsilon\}}] &= \mathbb{E}[|M_n - M_\infty|\mathbb{1}_{\{|M_n - M_\infty| > \varepsilon\} \cap \{|M_n - M_\infty| \le c\}}] \\ &\quad + \mathbb{E}[|M_n - M_\infty|\mathbb{1}_{\{|M_n - M_\infty| > \varepsilon\} \cap \{|M_n - M_\infty| > c\}}] \\ &\le c\mathbb{P}(|M_n - M_\infty| > \varepsilon) + \mathbb{E}[|M_n - M_\infty|\mathbb{1}_{\{|M_n - M_\infty| > c\}}]\end{aligned}$$

This implies:

$$\mathbb{E}[|M_n - M_\infty|] \le \varepsilon + c\mathbb{P}(|M_n - M_\infty| > \varepsilon) + \sup_{n \in \mathbb{Z}^+} \mathbb{E}[|M_n - M_\infty|\mathbb{1}_{\{|M_n - M_\infty| > c\}}]$$

For sufficiently large $c_\varepsilon > 0$, by uniform integrability, we have $\sup_{n \in \mathbb{Z}^+} \mathbb{E}[|M_n - M_\infty|\mathbb{1}_{\{|M_n - M_\infty| > c_\varepsilon\}}] < \varepsilon$. Then:

$$\mathbb{E}[|M_n - M_\infty|] \le 2\varepsilon + c_\varepsilon\mathbb{P}(|M_n - M_\infty| > \varepsilon).$$

Since $M_n \to M_\infty$ a.s., it also converges in probability, so $\lim_{n\to\infty} \mathbb{P}(|M_n - M_\infty| > \varepsilon) = 0$. Thus, $\limsup_{n\to\infty} \mathbb{E}[|M_n - M_\infty|] \le 2\varepsilon$. Letting $\varepsilon \downarrow 0$, we conclude

$$\lim_{n\to\infty} \mathbb{E}[|M_n - M_\infty|] = 0.$$

It remains to prove (b). Fix $n \in \mathbb{Z}^+$ and $A \in \mathcal{F}_n$. Then for all $r \ge n$:

$$\mathbb{E}[M_n\mathbb{1}_A] = \mathbb{E}[M_r\mathbb{1}_A] = \mathbb{E}[M_\infty\mathbb{1}_A] + \mathbb{E}[(M_r - M_\infty)\mathbb{1}_A]$$

> **Homework 208**
> Show that $\mathbb{E}[M_n\mathbb{1}_A] = \mathbb{E}[M_r\mathbb{1}_A]$ for $r \ge n$ and $A \in \mathcal{F}_n$.

We have $|\mathbb{E}[(M_r - M_\infty)\mathbb{1}_A]| \le \mathbb{E}[|M_r - M_\infty|] \xrightarrow{r\to\infty} 0$. Thus, $\mathbb{E}[M_n\mathbb{1}_A] = \mathbb{E}[M_\infty\mathbb{1}_A]$ for all $A \in \mathcal{F}_n$, which implies $M_n = \mathbb{E}[M_\infty|\mathcal{F}_n]$. □

This transcription follows the style and structure of your previous chapters, utilizing the ‘lindrew‘ package and the specific math commands provided in your preamble. Formal statements are placed in their respective environments, and the professor's additional notes are formatted as remarks.

28.4 The Essential Supremum

Remark 209. *Let $(Z_\alpha)_{\alpha\in I}$ be a family of random variables on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.*

- *If I is a countable set (e.g., $I = \mathbb{N}$) then we know that $\bar{Z}(\omega) := \sup_{\alpha\in I} Z_\alpha(\omega)$ is a $\mathcal{F}$-measurable function.*

- *If I is uncountable, then $\bar{Z}$ may not be measurable. We need a different notion:* **essential supremum**.

Lemma 210

There exists a countable subset $J \subset I$ such that $Z^*(\omega) := \sup_{\alpha \in J} Z_\alpha(\omega)$ satisfies:

(i) $\mathbb{P}(Z_\alpha \leq Z^*) = 1$ for all $\alpha \in I$.

(ii) If $\hat{Z} \in m\mathcal{F}$ is such that $\mathbb{P}(Z_\alpha \leq \hat{Z}) = 1$ for all $\alpha \in I$, then $\mathbb{P}(\hat{Z} \geq Z^*) = 1$. (i.e., Z^* is the smallest $\mathcal{F}$-measurable function that satisfies (i)).

Moreover, if the family $(Z_\alpha)_{\alpha \in I}$ is **upwards directed** in the sense that for all $\alpha_1, \alpha_2 \in I$, there exists $\alpha_3 \in I$ such that $Z_{\alpha_3} \geq Z_{\alpha_1} \vee Z_{\alpha_2}$ $\mathbb{P}$-a.s., then there exists a sequence $(\alpha_n)_{n \in \mathbb{N}} \subset I$ such that $Z_{\alpha_n} \leq Z_{\alpha_{n+1}}$ for all n, $\mathbb{P}$-a.s., and $Z^* = \lim_{n \to \infty} Z_{\alpha_n}$ $\mathbb{P}$-a.s.

Proof. We can assume $|Z_\alpha| \leq 1$ because if not we can redefine $Z'_\alpha := \frac{2}{\pi} \arctan(Z_\alpha)$, which is equivalent for our purposes because $\arctan(x)$ is a strictly increasing function. Recall that $\tan x = \frac{\sin x}{\cos x}$.

Let $\mathcal{C}$ be the collection of all countable subsets of I. Define:

$$a := \sup_{C \in \mathcal{C}} \mathbb{E}\left[\sup_{\alpha \in C} Z_\alpha\right].$$

By the definition of supremum, there exists a sequence of countable sets $(C_n)_{n \in \mathbb{N}} \subset \mathcal{C}$ such that $C_n \subseteq C_{n+1}$ for all n and $a = \lim_{n \to \infty} \mathbb{E}\left[\sup_{\alpha \in C_n} Z_\alpha\right]$.

Set $J := \bigcup_{n \in \mathbb{N}} C_n$ so that J is a countable set. Take $Z^* := \sup_{\alpha \in J} Z_\alpha$ and notice that

$$Z^* = \sup_{\alpha \in \bigcup_{n \in \mathbb{N}} C_n} Z_\alpha = \sup_{n \in \mathbb{N}} \sup_{\alpha \in C_n} Z_\alpha = \lim_{n \to \infty} \sup_{\alpha \in C_n} Z_\alpha.$$

By the Monotone Convergence Theorem (MON), $a = \mathbb{E}[Z^*]$.

Now $Z_\alpha \leq Z^*$ for all $\alpha \in J$. For $\bar{\alpha} \notin J$, let us argue by contradiction and assume $\mathbb{P}(Z_{\bar{\alpha}} > Z^*) > 0$. Then $\mathbb{E}[Z_{\bar{\alpha}} \vee Z^*] > \mathbb{E}[Z^*] = a$. However, the set $J' := J \cup \{\bar{\alpha}\}$ is countable, hence $J' \in \mathcal{C}$ and therefore

$$a = \sup_{C \in \mathcal{C}} \mathbb{E}\left[\sup_{\alpha \in C} Z_\alpha\right] \geq \mathbb{E}\left[\sup_{\alpha \in J'} Z_\alpha\right] = \mathbb{E}[Z_{\bar{\alpha}} \vee Z^*] > a,$$

which gives us a contradiction. Thus $\mathbb{P}(Z_\alpha \leq Z^*) = 1$ for all $\alpha \in I$.

Homework 211

Prove property (ii).

Let's prove the final claim: assume for all $\alpha, \beta \in I$, there exists $\gamma \in I$ such that $Z_\gamma \geq Z_\alpha \vee Z_\beta$ $\mathbb{P}$-a.s. Let $Z^* = \sup_{\alpha \in J^0} Z_\alpha$ where $J^0 = \{\alpha_1^0, \alpha_2^0, \ldots, \alpha_n^0, \ldots\}$ is the countable set from the first part.

Take $\alpha_1 = \alpha_1^0$. Let $\alpha_2 \in I$ be such that $Z_{\alpha_2} \geq Z_{\alpha_1} \vee Z^0_{\alpha_2}$. Inductively, take $\alpha_{n+1} \in I$ so that $Z_{\alpha_{n+1}} \geq Z_{\alpha_n} \vee Z_{\alpha^0_{n+1}}$. This defines a countable set $J = \{\alpha_1, \alpha_2, \ldots, \alpha_n, \ldots\}$ and by construction $Z_{\alpha_n} \geq Z_{\alpha_j^0}$ for all $1 \leq j \leq n$. Moreover, $Z_{\alpha_n} \leq Z_{\alpha_{n+1}}$ and so $\check{Z} := \lim_{n \to \infty} Z_{\alpha_n} \geq Z^*$.

However, J is countable and therefore $\mathbb{E}[\check{Z}] \leq \sup_{C \in \mathcal{C}} \mathbb{E}\left[\sup_{\alpha \in C} Z_\alpha\right] = \mathbb{E}[Z^*]$ by the previous part of the proof. We conclude $Z^* = \check{Z}$ $\mathbb{P}$-a.s. $\square$

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